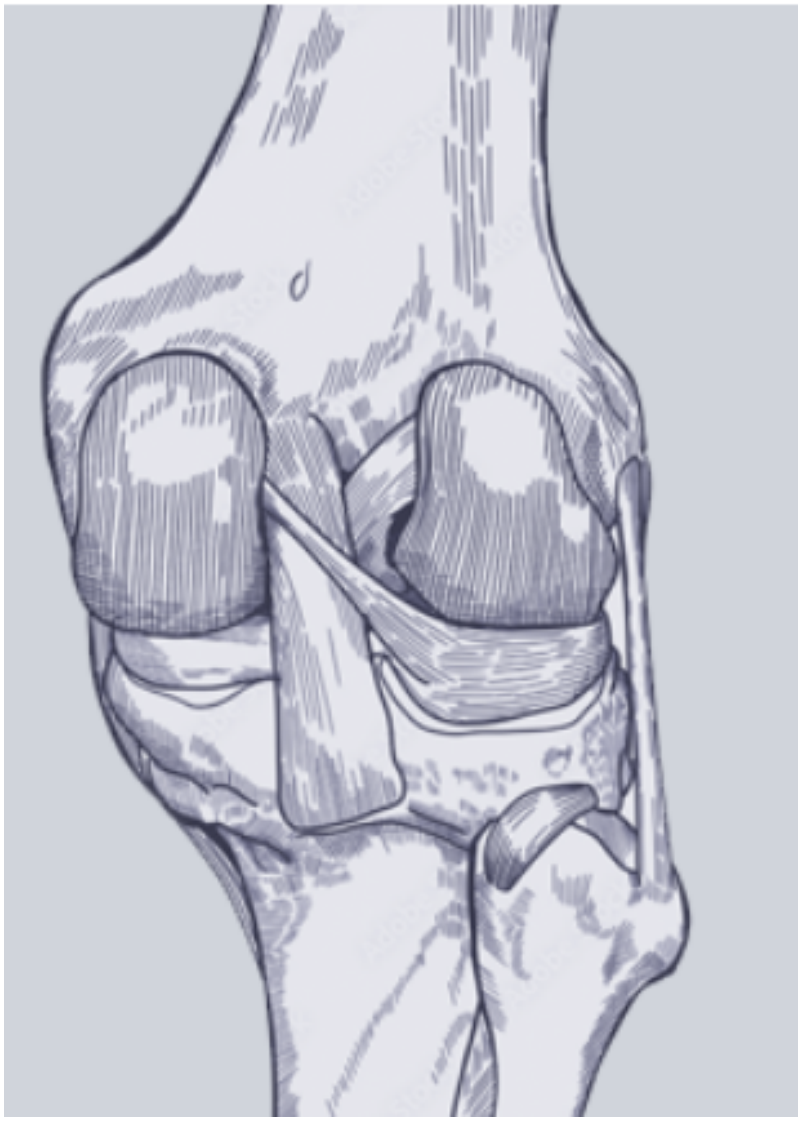
COURSE TRANSCRIPT

Shoulder and Knee Sports

8 lectures



Prof Dr Maged Samy

Watch course

TRANS PRO
Developed by FAM

Lecturers

Prof Dr Maged Samy

IMPORTANT

The world is changing very fast, especially with the rapid development of Artificial Intelligence (AI). We believe we should keep up with these changes and use them to make learning easier and more accessible.

This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

This work is completely non-commercial and was created simply to help those seeking knowledge and make educational resources more accessible.

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How to use this book

- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
- Click any section heading to return to the lecture's contents.
- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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The Painful Stiff Shoulder

(Diagnostic Approach and Management Outlines)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 12) “main restraint of ER in abduction” → “main restraint of ER in adduction”

C2 (p. 13) “isolated subacromial decompression, no tendon tears” → “isolated impingement, no tendon tear”

C3 (p. 13) “rarely need subacromial impingement alone” → “decompression alone rarely needed”

C4 (p. 17) “limited ER in abduction” → “limited ER in adduction”

C5 (p. 18) “limited passive ER and abduction” → “limited passive ER in adduction”

C6 (p. 19) “scapular retroversion” → “glenoid retroversion”

C7 (p. 19) “cemented and cemented stems” → “cemented or cementless stems”

Queried - kept as spoken, but looks wrong

Q1 (p. 16) as spoken; AC joint painful arc is classically the terminal arc, about 140/160–180°, not 110–150°

Presenting Complaints Ask AI

- Shoulder complaints:
  - **Pain** — most common
  - Limited range of motion — "can't reach behind my back", "can't raise my hand"
  - Crepitus or noise on elevation
  - Instability
  - Weakness — can't carry with the arm
- Pain + limitation → two commonest presentations, usually together
  - Almost all shoulder pathologies → painful
  - Any pain → **painful limitation** of range
  - → characterise the limitation in front of you

True Stiffness or Pain? Ask AI

- Key question → **true stiffness** vs limitation due to pain
  - Impingement → every painful range limited by pain
  - Rotator cuff tears → same
  - Adhesive capsulitis → truly stiff
    - Pain abolished → limitation persists; pain not the cause
- Painful limitation also from:
  - Biceps pathology
  - Calcific tendinitis
  - AC joint pathology
- Instability → pain, but no limitation
  - Expected complaint → shoulder dislocates
  - **Occult instability** → overstretching from repeated microtrauma → pain only
    - Pain in overhead activities
    - Painful in impingement arc → misdiagnosed as impingement
    - Real pathology → instability
- Pain may be referred from **cervical spine**
- Painful stiff shoulder → first question: true stiffness or stiffness due to pain?

True Stiffness Ask AI

Definition Ask AI

- **True stiffness** → limitation of both active and passive range
- Active-only limitation, passive free → muscle problem
  - Muscle itself → tendon injury
  - Nerve supply to the muscle
- Passive also limited → two possibilities
  - Truly stiff
  - Passive range painful → patient resists → limitation due to pain
- Passive limitation can still be due to pain → watch closely
- True stiffness → limitation persists with **abolition of pain**

Confirming True Stiffness Ask AI

- Final diagnosis → abolish pain first
  - Local injections
  - **Examination under anesthesia (EUA)**
- Frozen shoulder diagnosed → **MUA** planned
  - Anaesthetised → full range → surgeon pleased
  - Patient wakes → same limitation returns
  - → not truly stiff; limitation was due to pain
- True stiffness → active + passive limitation persisting without pain

Causes of True Stiffness Ask AI

- Painful limited shoulder → differentiate true stiffness vs pain-caused limitation
- Causes of true stiffness:
  - **Primary frozen shoulder** (adhesive capsulitis) — most common
  - Shoulder arthritis — less common
  - **Secondary frozen shoulder** → post-traumatic or post-operative
  - Locked dislocation

Painful Limitation versus True Stiffness Ask AI

- True stiffness → severe passive limitation even without pain
- Painful limitation → pain abolished → normal range
- Causes of painful limitation:
  - **Subacromial impingement**
  - Cuff tears
  - Biceps pathology
  - AC arthritis impinging on the cuff
  - Calcific tendinitis
- Pain abolished or EUA → almost full passive range

Feature	True stiffness	Painful limitation
Active range	Limited	Limited
Passive range	Limited	Limited by pain
After pain abolished / EUA	Still limited	Almost full
Passive ER in adduction	Limited	Almost normal
Causes	Primary frozen shoulder, arthritis, secondary frozen shoulder, locked dislocation	Subacromial impingement, cuff tears, biceps, AC arthritis, calcific tendinitis

Key Test: Passive ER in Adduction Ask AI

- Examine active + passive range → then passive range in a non-painful direction
- Most important test → **passive external rotation in adduction**
- Always start with active range
  - Full active range → passive surely normal → skip

  - Poor active range → check passive
- Why this test:
  - **Non-painful position**
    - Not an impinging position
    - Not painful in cuff tears
    - Not painful in labral pathology
    - Not painful in AC arthritis
  - → limitation here = **true limitation**
    - Frozen shoulder
    - **Locked posterior dislocation**
    - Primary or secondary shoulder osteoarthritis
  - Mandatory before diagnosing frozen shoulder
  - First range lost in frozen shoulder → pathology begins in **rotator interval**
    - Rotator interval = anterosuperior capsule → main restraint of ER in adduction *[C1 · transcript: "main restraint of ER in abduction"]*
    - → frozen shoulder diagnosable in early phases

High Yield
- Passive ER in adduction → non-painful position → limitation = true stiffness
- First range lost in frozen shoulder → rotator interval pathology
- Full range under anesthesia → not truly stiff

Coexisting Pathologies Ask AI

- Impingement and cuff tears → detailed next lecture; focus here → true stiffness
- Different pathologies **coexist** in the same shoulder
  - Diagnose every painful pathology
  - Treat all → one left → pain + painful limitation persist

Subacromial Impingement Ask AI

Mechanism and Sites Ask AI

- **Outlet impingement** → rotator cuff between acromion and humeral head
- AC joint → **non-outlet impingement**
- Acromion shapes → flat, curved, hooked — previously emphasised
  - Subacromial impingement → **clinical diagnosis**
  - Acromion shape → relevance quite limited
  - True impingement possible with flat; none possible with hooked
- Sites:
  - **Anterolateral edge of acromion** — mostly
  - AC joint → non-outlet
  - Greater tuberosity
- Instability → pain **mimics impingement**; not true impingement
- Stages → oedema → fibrosis → **tendon tears**

Clinical Diagnosis Ask AI

Impingement

- Pain at anterolateral corner of acromion → **Codman's point**
  - Radiates down to elbow → normal with impingement
- **Painful arc** → 70°–110°
  - Arm midway between abduction and flexion, internal rotation, moved passively
- **Hawkins sign** → pain at Codman's point, radiating to elbow
- Re-examine passive ER in adduction → exclude frozen shoulder
  - **Frozen shoulder** → painful in all movements
  - → apparent painful arc, if arc even completed
  - → mimics any pathology
- Positive tests ≠ **isolated impingement**
  - Coexisting pathologies in painful limitation:
    - Impingement
    - Supraspinatus or infraspinatus tendon tear
    - Biceps pathology
    - Subscapularis injury
    - AC joint pathology
  - Examine all → treat all
- Not truly stiff + impingement positive → **subacromial impingement plus other things**
- Subacromial injection → repeat test → pain improves? — rarely done nowadays

Treatment Ask AI

Impingement

- **Isolated subacromial decompression** → out of favour
  - Once commonest shoulder operation, at least arthroscopically
- Isolated impingement, no tendon tear → **conservative** *[C2 · transcript: "isolated subacromial decompression, no tendon tears"]*
  - NSAIDs
  - Physiotherapy → **external rotator strengthening** → cuff away from impinging position
  - Local injections
    - **Local steroid** → MRI first, confirming no partial cuff tears
- Conservative fails → widen **subacromial space**
  - Burr anterolateral aspect of acromion — detailed next lecture
- Decompression alone → rarely needed; conservative treatment → good results *[C3 · transcript: "rarely need subacromial impingement alone"]*

Pitfall

- Occult instability → pain in impingement arc → misdiagnosed as impingement
- Frozen shoulder → painful in all movements → mimics any pathology
- Steroid injection only after MRI excludes partial cuff tears

Rotator Cuff Tears Ask AI

Clinical Tests Ask AI

- Painful limitation, not truly stiff, impingement positive → **examine whole cuff**
  - All or most cuff tears → impingement test positive

  - Positive test → impingement ± superadded cuff tears
- **Supraspinatus** → abductor
  - **Empty can test** (Jobe test) → very sensitive
    - Good power elevating, midway abduction/flexion, internal rotation → no supraspinatus tear
    - Weakness may be pain → internal rotation = **impinging position**
  - Repeat in external rotation → **full can test**
- **Infraspinatus** → external rotator → resisted external rotation
- **Subscapularis**
  - **Lift-off test** → hand behind back, held off against push
    - Often position unachievable due to pain
  - **Bear hug test** → easier; hand pressing down in internal rotation against resistance

MRI Ask AI

Cuff

- Space between acromion and cuff; **spur** → not very important
- Look for **small cuff tears**
  - Coronal cuts, fat suppression (bright image) → T1 easily misses them
- Small cuff tear found →
  - No steroid injection
  - **Indicated for surgery**
- No tears → rule: conservative

PASTA Lesion Ask AI

- Bursal side intact, articular side torn → **PASTA lesion**
  - Partial articular-side tear; only small piece intact above
  - Very painful, night pain, severe limitation due to pain
  - Pain and limitation → usually **worse than complete tears**
- Management
  - Confirm not truly stiff → ER in adduction good (non-painful)
  - **No steroid injection** → almost complete; only nominally partial
  - Needs surgery → very painful; not "just a partial tear"

Exam Pearl

- PASTA → partial articular-side tear, bursal side intact
- Worse pain and limitation than complete tears
- No steroid injection → surgery

Arthroscopic Cuff Repair Ask AI

- First → proper **cuff exposure** → clean subacromial space
- **Acromioplasty** → not routine in cuff tears, unlike the past
  - Good space → no acromioplasty; clean bursa well to see cuff
  - Shave bursa **tangentially** → downward shaving cuts the cuff
  - Narrow space → acromioplasty
- View: cuff, exposed humeral head, **greater tuberosity** with residual torn cuff
- Shave greater tuberosity → 1, 2 or 3 anchors by tear size
- Suture-passing instruments → cuff back to greater tuberosity
  - Pass only → second instrument to retrieve suture

  - **Pass and retrieve** → easier; no second instrument
- Sutures **distributed anterior → posterior** — not one central stitch
  - Anchor usually **double-loaded** → 4 sutures
  - 2 anchors, single row → 8 sutures
  - Anterior → middle → posterior empty → pass more
  - Whole cuff grasped, no part left → **tie knots**
- First cause of painful limitation → **impingement**
- Impingement test positive → impingement ± cuff tear → examine cuff
  - Cuff tear → **real indication for surgery**
- **Partial tears** → addressed even when small → debride, or complete and repair

Long Head of Biceps and Subscapularis Ask AI

Biceps Pulley Ask AI

- Third cause of painful limitation → **long head of biceps**
  - May coexist with impingement + tear
- **Biceps pulley system** → keeps long head of biceps in track
  - Proximal → superior glenohumeral ligament, coracohumeral ligament, supraspinatus
  - Distal → subscapularis
- **Partial upper-border subscapularis tear** → long head of biceps almost always subluxed → pain
  - Treat both → long head of biceps tenotomy + subscapularis repair
- Long head of biceps out of groove → **subluxed or dislocated**; subscapularis tear beside it
- Subluxed + frayed long head of biceps → definite **source of pain**
  - Decompression + supraspinatus repair alone → pain persists → limitation due to pain
  - → **biceps tenotomy** if elderly, or tenodesis

Biceps Management Ask AI

- Frayed or subluxed biceps → suspect **upper-border subscapularis tear** → repair
  - All procedures done together
- **Biceps tenotomy** → cut completely until loose
  - Small attached piece → very high tension on fibres → increasing pain
- Painful limited shoulder → may include biceps pathology
- **Ruptured biceps** → no repair
  - Diseased biceps → treatment = cutting → rupture did the job itself
  - Above 50 → not repaired

Subscapularis Tear Ask AI

- Subscapularis lifted off **lesser tuberosity**
- Intra-articular view → **upper border** only
- Upper-border injury → two things:
  - Long head of biceps subluxed → **tenotomy or tenodesis**, even with subscapularis repair
    - Subscapularis repair → doesn't solve the biceps problem
  - Subscapularis repair → anchor in **lesser tuberosity**, not greater
- Anchor → pass through subscapularis substance → suture
- All pathologies together → diagnose all, attack all

Clinical Pearl

- Upper-border subscapularis tear → long head of biceps almost always subluxed
- Subscapularis repair doesn't fix biceps → tenotomy or tenodesis still needed
- Tenotomy → cut completely; partial attachment → increasing pain
- Ruptured diseased biceps above 50 → no repair

AC Joint Pathology Ask AI

- AC joint **osteophyte** → impinges on cuff → pain → limitation
- **O'Brien test**; painful arc higher than classic impingement arc

Painful arc	Range
Subacromial impingement	70°–110°
AC joint pathology	110°–150° *[Q1 · as spoken; AC joint painful arc is classically the terminal arc, about 140/160–180°, not 110–150°]*

Treatment Ask AI

AC joint

- Two approaches → arthroscopic or open
- Choice by pain source:

Pain source	Finding	Treatment
Impingement only	Osteophyte impinging on cuff	Remove osteophytes, lower end of clavicle
Degenerative AC joint itself	Direct tenderness over AC joint	Osteophyte removal insufficient → distal clavicle excision

- **Mumford procedure** → distal (lateral) clavicle excision, 8–10 mm
  - → **resection arthroplasty** → pain relief

Painful Limitation: Summary Ask AI

- Passive **ER in adduction** almost normal → not truly stiff
- Coexisting pathologies → treat all present:
  - Impingement
  - Supraspinatus or infraspinatus tear
  - Biceps pathology
  - Subscapularis injury
  - AC pathology
- **Single pathology** → rare (possible exception: small tears); usually more than one

Primary Frozen Shoulder Ask AI

- **Truly stiff shoulder**:
  - Active + passive limitation

  - Limited ER in adduction *[C4 · transcript: "limited ER in abduction"]*
  - Limited even under anesthesia
- First cause → **primary frozen shoulder** = adhesive capsulitis

Pathology Ask AI

Frozen

- Begins in **rotator interval** → between supraspinatus (above) and subscapularis (below)
- Severe **rotator interval synovitis** → very characteristic; only in primary frozen shoulder
- **Coracohumeral ligament fibrosis** → in rotator interval
  - Capsular thickening there
  - Capsular contracture
- Histology → many cells, marked fibrotic changes

Imaging Ask AI

Frozen

- Frozen shoulder → **clinical diagnosis** ; radiology mainly for other pathologies
  - **X-ray** → exclude shoulder arthritis; arthritis easily diagnosed as frozen shoulder
  - MRI → mainly cuff intact; obliterated axillary pouch, capsular thickening
- Osteoarthritis treated as frozen → **capsular release** → results quite limited

Natural History Ask AI

- Three stages → **self-limiting disease**

Stage	Timing	Features
1	3–4 months	Severe pain, night pain; little limitation
2 — Frozen phase	After pain stage	Pain decreasing; limitation increasing; fibrosis
3 — Thawing phase	After about 9–12 months	Range improves again

- Improves if left alone → usually **not to normal range** → better managed

Arthroscopic Findings Ask AI

Frozen

- Posterior portal → **rotator interval synovitis** beneath long head of biceps
  - Characteristic → seen only in frozen shoulder

Manipulation Under Anesthesia Ask AI

- Release also needed
- **MUA** → limited place
  - Always start with **forward flexion**
  - Sudden popping and release → only sign of success
  - No release in forward flexion → don't continue

Arthroscopic Capsular Release Ask AI

- Right shoulder, posterior view → rotator cuff, biceps, glenoid, humerus, rotator interval
- Step 1 → **rotator interval synovectomy** → remove all tissue
  - Left behind → pain → painful limitation → no felt improvement
- Release whole rotator interval → **coracoid seen from within shoulder**
  - Coracoid seen → **coracohumeral ligament** released

  - Ligament itself → sometimes visible, sometimes not
- **Beach chair** → progressive anterior capsule release → ER improves → continue inferiorly
- Approaching axilla → stop **radiofrequency**
  - Thermal conduction → **axillary nerve** injury — not direct injury
  - Remaining inferior capsule → simple mechanical punch

Pitfall
- Near axilla → no radiofrequency
- Thermal conduction → axillary nerve injury
- Inferior capsule → mechanical instrument instead

Shoulder Arthritis Ask AI

Clinical and Imaging Ask AI

Arthritis

- Next cause of truly stiff shoulder → **shoulder arthritis**
  - Active + passive limitation; limited passive ER in adduction
  - True limitation even under anesthesia
  - Truly stiff and painful → **catching**, grinding
  - Don't miss → don't misdiagnose as frozen shoulder
- Pain, night pain, true limitation, limited passive ER in adduction → looks like frozen shoulder *[C5 · transcript: "limited passive ER and abduction"]*
  - **MRI** → arthritis unclear unless advanced
- X-ray, same patient:
  - Inferior **osteophytes**
  - Joint-space narrowing
  - Subchondral sclerosis
  - → X-ray mandatory

Pitfall
- MRI misses arthritis unless advanced → X-ray mandatory
- Osteoarthritis treated as frozen shoulder → capsular release → limited results

Treatment and Pre-operative Assessment Ask AI

Arthritis

- Treatment → **shoulder replacement**, total or reverse
  - Cautious in **rheumatoid arthritis** → intact cuff required
  - Deficient or torn cuff → no anatomic replacement → reconsider procedure
- Pre-operative assessment:
  - Main complaint → **stiffness or pain**? → prognosis
    - Pain → much better
    - Stiffness → better, but not normal range
    - Stiffness-dominant → patient must understand this
  - **Rotator cuff and deltoid** → functioning
  - X-ray → AP + **axillary view** → glenoid erosion
  - CT or MRI → glenoid; MRI → cuff intact

Glenoid Retroversion Ask AI

- Primary shoulder arthritis → **posterior glenoid wear**
  - → increased glenoid retroversion
  - → posterior subluxation → instability
  - → correct during the procedure
- Measurement on CT or MRI:
  - **Scapular line** → medial end of scapular spine → middle of glenoid
  - **Glenoid line** → anterior to posterior glenoid lips
  - Third line → perpendicular to scapular line
  - Angle perpendicular ↔ glenoid line = **glenoid retroversion** *[C6 · transcript: "scapular retroversion"]*
- Target ≤7°; up to 10° acceptable
- Beyond → too retroverted → **posterior instability** → correct intra-operatively

Shoulder Replacement Implants Ask AI

- Stems → cemented or cementless; **cemented** most used here *[C7 · transcript: "cemented and cemented stems"]*
- Heads → **standard** or offset (one side larger, one smaller)
- **Glenoid component** → main problem of total shoulder replacement
  - Increased loosening
  - Glenoid erosion
- Glenoid component types:

Component	Behaviour	Status
Metal-backed	Severe erosions, early loosening	Abandoned in many places
Modified metal-backed (hydroxyapatite coating, hybrid cage)	Longevity somewhat improved	Best case → not much worse than all-polyethylene
All-polyethylene cemented, pegged or keeled	—	Gold standard

- **Pegged** → biomechanically better ; functionally ≈ keeled
- Trend → all-polyethylene cemented with large **central fluted peg**
  - Bone graft from humeral head into peg → **bony ingrowth**
  - → partially cemented, partially bony ingrowth

Total Shoulder Replacement Technique Ask AI

Approach and Humeral Preparation Ask AI

- Standard **deltopectoral approach**
  - Release part of pectoralis major
  - Release **subscapularis** just medial to bicipital groove → tag with sutures
- Dislocate → define **anatomical neck** = cut level; retractors protect surroundings
- Reamer entry point → lateral and anterior
  - **Reamers** in increasing diameters
- **Humeral head cut** → ~30° retroversion

  - 2 rods → 20° and 40°
  - Forearm between rods → ~30° ER = humeral head retroversion
- **Provisional stem** + head
  - Head → reaches cuff attachment laterally, overlies **calcar** medially
  - Standard head achieves this → fine
  - Falls short → **offset head**, rotated to reach both

Glenoid Preparation Ask AI

- **Hohmann retractors** anterior and posterior to glenoid → mark glenoid midpoint
- Remove glenoid cartilage → **glenoid size**
  - Sizer exactly on glenoid borders, not outside
  - Central hole at centre
- **Glenoid reamers** over central hole
- Exaggerated retroversion → **correct at this stage**:
  - Before reaming → burr → slight anterior reaming
  - During reaming → handle pushed anteriorly → anterior reamed more than posterior
- Holes for pegs or **keel**; here keeled → holes connected
- Impact bone into keel **slot**

Cementing the All-Polyethylene Glenoid Ask AI

- **No cement on surface** → slot only — very important
- Slot preparation:
  - Impact cancellous bone with **pressurizer**
  - Repeat reaming with **keel locator** → good fit
- **Trial** → very well seated: keel inside, component lying on glenoid
- Cement keel only:
  - Dry slot very well
  - Inject cement → pressurizer with gauze → press while soft
    - → cement **interdigitates** within cancellous bone
  - Pressurizer removal pulls cement → inject more → re-pressurize
  - Insert **final glenoid component** → press until cement sets

Head Sizing and Closure Ask AI

- Provisional stem carries **counterbore** → countersink → stem well seated
  - Insert stem and head
- **Head diameter** unrelated to removed head — unlike hip
  - Removed 46 head ≠ insert 46 head
- Diameter check → translate head relative to glenoid
  - Push posteriorly or inferiorly → ~50% overlaps glenoid → correct
  - Moves less → **too large**
  - More than half behind or below glenoid → too small
- Fixed sizes → 46, 50, 54; usually 3–4 sizes by company
- Each diameter → several **thicknesses**; 54 mm head → 15, 18, 21, 24
- Thickness → subscapularis back in good tension at 30° ER
  - Won't reach → **too thick** → reduce thickness

Head parameter	Options	Determined by
Diameter	46, 50, 54 (3–4 sizes by company)	~50% translation over glenoid posteriorly or inferiorly
Thickness	15, 18, 21, 24 (54 mm head)	Subscapularis back in good tension at 30° ER
Type	Standard or offset	Reaches cuff attachment laterally, overlies calcar medially

- Drill holes in **anterior cortex** → non-absorbable sutures → later subscapularis repair
- Canal **plug** → cement → stem → chosen head
  - Diameter, thickness, standard or offset
  - → **subscapularis repair**

Outcome and Rehabilitation Ask AI

TSR

- **Modern cementation techniques** → very good longevity up to 15 years
- Early passive and active-assisted motion
- Avoid **active internal rotation** before 6 weeks → protect subscapularis

Must-Know

- Gold standard glenoid → all-polyethylene cemented, pegged or keeled
- Cement keel slot only → never the glenoid surface
- Humeral head cut → ~30° retroversion
- Glenoid retroversion → target ≤7°, up to 10° acceptable
- No active internal rotation before 6 weeks

Calcific Tendinitis Ask AI

- Last cause in painful stiff shoulder
  - Not true stiffness → but **simulates frozen shoulder**
  - Severe pain; night pain in many
  - Women, 40–60 years; can be **bilateral**
  - Pain may block even ER in adduction
  - Under anesthesia → **full range**
- Calcifications → poorly seen on **MRI**, may be missed; obvious on X-ray
  - Calcific tendinitis + arthritis → two conditions missed on MRI, especially early
  - Main diagnosis → **X-ray**

Treatment Ask AI

Calcific

- **Shock waves**
- **Simple needling** → puncture cuff → hydroxyapatite deposits escape
  - Arthroscopic or ultrasound-guided
- **Arthroscopic shaving**
  - Whole deposit removal unnecessary → open space in cuff → deposits come out
  - After shaving → reassess cuff → repair if too much injury
  - Repair often not needed

Take-Home Messages Ask AI

- Stiff shoulder → **true stiffness** vs limitation due to pain
  - Differentiated by passive ER in adduction
  - Frozen shoulder for release → EUA before release
  - Full range under anesthesia → **wrong diagnosis**
- Painful limited range → identify the pathology
  - Different pathologies → usually **coexist**
  - Treat all present → pain-free shoulder, good range
- **Shoulder arthritis** + calcific tendinitis → X-ray better than MRI

Rotator Cuff Tear

(Anatomy, Diagnosis and Management)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 32) “passive ER in abduction” → “passive ER in adduction”

C2 (p. 33) “PASTA tends to heal spontaneously” → “PASTA does not heal spontaneously”

C3 (p. 46) “passive ER in abduction not limited” → “passive ER in adduction not limited”

C4 (p. 46) “limited passive ER in abduction” → “limited passive ER in adduction”

C5 (p. 47) “repairable cuff tears” → “irreparable cuff tears”

Queried - kept as spoken, but looks wrong

Q1 (p. 34) As spoken; partly uncovered head is later called type 2. Patte stage 1 means minimal retraction.

Q2 (p. 37) As spoken; duloxetine is a serotonin-noradrenaline reuptake inhibitor (SNRI), not an SSRI.

Q3 (p. 44) As spoken; in Hamada grading, grade 1 is AHD of 6 mm or more and grade 2 is 5 mm or less.

Webinar Opening and Introduction Ask AI

- Weekly meeting → **Monday 10:30 pm Cairo time**, since February
  - Fridays crowded with events → Monday allows following more webinars across specialties
- Lecture focus:
  - Topic-based teaching
  - Exam preparation
  - Practical applications of previous lectures; applied daily work
  - Important topics missed or under-covered in the past period
- Speaker → **Prof. Dr. Maged Samy**, shoulder surgery, Egypt and Middle East
- Organizers → Nile Hospital: Dr. Ahmed Soliman, Dr. Seif Mohamed, Dr. Omar El-Hanafy
  - Sponsor → Aspen, anticoagulant product
- Chair → **Dr. Atef El-Beltagy**, Professor of Orthopaedic Surgery, Ain Shams
  - One day per week; topic deserving two sessions → held over two days
- Thanks → Orthopaedic Department of Health Insurance, sponsors, speakers
- COVID-19 benefit → online teaching; experienced speakers discovered
- Topic → **rotator cuff tears: assessment and lines of treatment**
  - Estimate: perhaps 5% of colleagues still unfamiliar with it
- Sporting aside → Al Ahly defeat by Bayern Munich; 15 minutes without touching the ball
- Speaker's title → **Rotator cuff tears: assessment and treatment options**

Rotator Cuff Anatomy and Function Ask AI

Components Ask AI

- Four muscles → supraspinatus, infraspinatus, teres minor, subscapularis
  - Cuff surrounding the **glenohumeral joint**
- Main role → **compress the humeral head against the glenoid**
  - Maintains stability; keeps joint congruent
  - Stability, not movement

Insertions Ask AI

- Anterior view → **subscapularis** on lesser tuberosity; long head of biceps alongside
- Lateral view, anterior → posterior:
  - **Lesser tuberosity** → subscapularis
  - **Greater tuberosity** → supraspinatus → infraspinatus → teres minor

Sheet, Not Tendons Ask AI

- Cuff = **sheets**, not rope-like tendons
  - Supra + infra (+ subscapularis, slightly further) interconnected → one sheet at insertion
- Supra + infra → **common sheet** at insertion
  - Teres minor → separate, from behind
- Supra–infra connection → **rotator cable**
  - Fibers from cable → insertion = **rotator crescent**

Clinical Significance of the Cable Ask AI

- Complete **full-thickness supraspinatus tear** → patient may still abduct
- Intact rotator cable + crescent → one muscle compensates → action possible
  - Result → **weakness**, not loss of action
- Tear ≠ complete loss of function → cuff, not separate tendons

Function Ask AI

- Very vascular → **blood supply** to humeral head
- Primary → **stabilizer of the glenohumeral joint**
  - Head compressed against glenoid → **deltoid** gains biomechanical advantage
  - → flexion, extension, abduction, rotation incl. external rotation
- Secondary → individual muscle actions, not the prime function:

Muscle	Action
Subscapularis	Internal rotator
Supraspinatus	Abductor
Infraspinatus + teres minor (posterior cuff)	External rotators

- Third → blood supply reaches humeral head through cuff

High Yield
- Cuff main role → compress humeral head against glenoid
- Compression → deltoid biomechanical advantage → all movements
- Supra + infra → one sheet; rotator cable → rotator crescent
- Intact cable → abduction possible despite full-thickness supraspinatus tear

Rotator Cuff Tears: Categories and Presentation Ask AI

Categories Ask AI

- Two categories:
  - **Traumatic** → unrelated to age; young or old
  - **Degenerative** → much more common; above 40 years
    - With or without preceding subacromial impingement
- Associated injury → **shoulder dislocation**
- First-time dislocation above age 40 → suspect **associated cuff tear**
  - Exclude by examination and radiology

Clinical Presentation Ask AI

- Main complaint → **pain**, including night pain
- Limitation in range of motion
- **Weakness** → unable to lift heavy objects
- Longstanding → **atrophy** or thinning of visible muscles
  - Visible → supraspinatus, infraspinatus
  - Not visible → subscapularis

- Pain with certain movements
- Weakness compared with other side
- **Crepitus** or crackling sensation
- Summary → pain incl. night pain, limited ROM, crepitus and noise, weakness
- Most tears → **degenerative**, not traumatic
  - Usually preceded by subacromial impingement

Subacromial Impingement Ask AI

Mechanism Ask AI

- Supraspinatus (anterior) + infraspinatus (posterior) → pass between humeral head and acromion
- Coracoid–acromion ligament → **coraco-acromial arch** over cuff
- Acromial **bony spurs** → narrow tendon space
- Abduction + internal rotation → friction → tendon tears
- Impingement tests → also positive in degenerative cuff tears
  - Positive test = impingement **with or without** associated cuff tear
- Sequence: friction → edema → fibrosis, tendinitis → **cuff tears**

Types Ask AI

- Three types:
  - **Outlet** → lateral end of acromion; most famous
  - **Non-outlet** → AC joint, or greater tuberosity prominent post-fracture
  - **Instability** → may present with impingement pain

Outlet Impingement Ask AI

Outlet

- Tenderness at anterolateral corner of acromion → **Codman's point**
  - Pain radiates down upper arm
- **Neer test** → passive elevation, internal rotation, midway between flexion and abduction
  - Painful arc → 70° to 110°
- **Hawkins test** → arm internally rotated, pushed down
  - Pain from shoulder radiating down upper arm
- Positive tests → impingement present; cuff tear status still unknown
- Subacromial **local anesthetic** injection → pain gone → subacromial impingement confirmed
  - State of cuff still unknown
- Treatment → mainly **conservative**
  - Resistant → acromial spurs removed = **subacromial decompression**
  - → wider space, less tendon friction

AC (Non-Outlet) Impingement Ask AI

Non-outlet

- Site → AC joint; outlet lies lateral, at acromion
- Bony prominence from **lateral end of clavicle** → impinges on cuff beneath
- **O'Brien test** → adduction + internal rotation
  - Patient raises arm against downward push → pain over AC joint → AC pathology
- Local tenderness over AC joint
- **Painful arc** → 110° to 140°
- Injection → pain improves

Feature	Outlet	Non-outlet (AC)
Site	Lateral end of acromion	AC joint; lateral clavicle prominence
Tenderness	Codman's point, anterolateral acromion	Over AC joint
Tests	Neer, Hawkins	O'Brien
Painful arc	70° to 110°	110° to 140°
Local anesthetic	Pain abolished	Pain improves

Pitfall

- Positive impingement tests → cuff tear neither confirmed nor excluded
- Local anesthetic relief → confirms impingement, not cuff integrity
- Instability → may mimic impingement pain

Clinical Examination for Cuff Tear Ask AI

Inspection Ask AI

- Orthopaedic examination → inspection, palpation, range of motion, special tests
- Inspection → look for **muscle wasting**
  - Scapular spine → supraspinatus above, infraspinatus below
- Wasting → **chronic** tear, usually retracted

Range of Motion Ask AI

- Order → **active first**, then passive
- Full active range → full passive range by definition → no need to test
  - No disease → full active with limited passive
- Limited active → test passive
  - Example → forward flexion 50° instead of 180°
- Cannot raise arm actively → two possibilities:
  - **Muscle** problem → muscle tear or nerve injury → passive range normal
  - **Stiffness** → active and passive both limited
- Key discriminator → **passive external rotation in adduction**
  - Limited → **true shoulder stiffness**, unrelated to muscle
- True stiffness = limitation in both active and passive ROM
- Cuff tear → active limited; passive **near normal**
  - Pain limits passive range
  - Local anesthetic or **examination under anesthesia** → almost full passive range

Feature	True stiffness	Cuff tear
Active ROM	Limited	Limited
Passive ROM	Limited	Near normal, pain-limited

Feature	True stiffness	Cuff tear
Passive ER in adduction	Limited	Almost normal
Pain abolished (LA / EUA)	Still limited	Almost full

- Causes of true stiffness → passive limited even without pain, even under EUA:
  - **Primary shoulder stiffness** → frozen shoulder, adhesive capsulitis
  - Shoulder arthritis
  - Secondary frozen shoulder
  - Locked dislocation
- Cuff tear passive limitation → due to pain → almost normal once pain abolished
- Causes of ROM limitation due to pain:
  - Impingement
  - Cuff tears
  - Biceps pathology
  - AC arthritis
  - **Calcific tendinitis**

Individual Muscle Testing Ask AI

- Setting → pain, weakness, positive impingement tests, near-normal passive ER in adduction *[C1 · transcript: "passive ER in abduction"]*
  - Test **cuff muscle** power vs other side → impingement alone or associated tear
- Supraspinatus → **Jobe test**
  - Arm in **scapular plane**, midway between abduction and flexion, internal rotation
  - Examiner pushes down; patient pushes up; both sides compared
  - Weakness may be pain-induced → internal rotation = impingement position
  - Repeat in external rotation → supraspinatus moves posteriorly → no impingement on lateral acromion
  - = **full-can test** → still weak → true supraspinatus weakness
- Subscapularis (internal rotator):
  - **Belly-press test** → hand pressed on belly
    - Elbow anterior to coronal plane of trunk → subscapularis, not pectoralis major
  - **Bear-hug test** → easier
    - Patient pushes down against resistance → internal rotation
  - **Lift-off** → hand behind back, lifted off against resistance
    - Many cannot reach **hand-behind-back** → pain
    - Can reach, cannot lift off or very weak → examiner holds hand away; patient only maintains
    - Hand falls back onto back → **lag test** positive
    - Maintenance, not resistance → residual function vs complete loss
- Infraspinatus → **external rotator**
  - External rotation against resistance
- Teres minor → external rotator of **abducted shoulder**
  - Shoulder abducted, then externally rotated
- Abducted arm placed in ER, released → falls → **positive lag test for external rotation**

Muscle	Action	Tests
Supraspinatus	Abduction	Jobe (internal rotation); full-can (external rotation)
Subscapularis	Internal rotation	Belly-press, bear-hug, lift-off, lag
Infraspinatus	External rotation	ER against resistance
Teres minor	ER of abducted shoulder	ER in abduction; ER lag

Pseudoparalysis Ask AI

- Lag test → residual function; detects **pseudoparalysis**
- Passively placed, cannot maintain position → **shoulder pseudoparalysis**
  - → no residual cuff function
  - Elderly with pseudoparalysis → chances of repair quite limited

Exam Pearl

- Passive ER in adduction limited → true stiffness, not cuff tear
- Full active range → passive need not be tested
- Jobe weak, full-can weak → true supraspinatus weakness
- Belly-press → elbow anterior to trunk's coronal plane
- Lag sign → cannot maintain position → no residual function

Classification of Cuff Tears Ask AI

By Thickness Ask AI

- Two sides → **bursal side** facing acromion; articular side facing humeral head
- Full-thickness → whole thickness torn
- Partial, head side → **PASTA lesion** = partial articular-sided supraspinatus tear
  - Joint side only, not whole thickness
- Partial, acromion side → **bursal-sided tears**
- Comparison:

Feature	PASTA (articular-sided)	Bursal-sided
Frequency	More common	Less common
Healing tendency	None spontaneously	Tends to heal *[C2 · transcript: "PASTA tends to heal spontaneously"]*

- PASTA more common → impingement as main cause of cuff tears → not very convincing; view changing
- Impingement-caused tears → logically start bursal side, not articular
  - Impingement frequently associated with tears; causal role **debatable**

Full-Thickness Tears: Size vs Retraction Ask AI

- Full-thickness → bursal to articular, whole thickness
- Size → small, medium, large, massive

- **Tear size ≠ tear retraction**
- Size = anteroposterior extent along greater tuberosity
  - Cuff attachment → about 6 cm, anterior → posterior: supra → infra → teres minor
  - Tears always start anteriorly → extend posteriorly
  - Extent: most anterior supraspinatus → whole supraspinatus → infraspinatus → teres minor
- Retraction → how far torn tendon moved from its place
- **Type 1** → part of humeral head covered, part uncovered; larger tear *[Q1 · As spoken; partly uncovered head is later called type 2. Patte stage 1 means minimal retraction.]*
- Massive by retraction → **whole head exposed**; tendon retracted level with glenoid
- Chronic or degenerative + massive retraction → most likely irreparable
  - Distance > 5 cm → cannot be pulled back despite releases
  - Acute traumatic → different; pulled easily
- Two dimensions → **tear size** and tear retraction

Radiological Progression Ask AI

- Early → normal, or **sclerosis of greater tuberosity**
- Large retracted tear → head uncovered → **upward migration**
  - Cuff → presses head downwards; deltoid → pulls head upwards
  - **Deltoid atony** (e.g. post-fracture) → inferior subluxation
  - No functioning cuff + functioning deltoid → **proximal migration**
- Proximal migration → definitely large tear; no cuff function
- End stage → **cuff tear arthropathy**
  - Acetabularization of acromion
  - Glenohumeral arthritis

High Yield

- PASTA → more common than bursal-sided; no tendency to heal
- Bursal-sided → tends to heal
- Tear size (anteroposterior) ≠ tear retraction (mediolateral)
- Degenerative tear retracted to glenoid → most likely irreparable
- Proximal migration → large tear; end stage → cuff tear arthropathy

MRI Assessment Ask AI

Normal Anatomy by Plane Ask AI

- Coronal → cuff intact all the way to greater tuberosity
- Axial → subscapularis anteriorly → bicipital groove; infraspinatus posteriorly
  - Intact subscapularis → **main stabilizer of long head of biceps** in groove
  - Subscapularis tear → LHB subluxates → **empty bicipital groove**
- Sagittal landmarks → **coracoid** (anterior), clavicle, acromion
  - Anterior → posterior: subscapularis, supraspinatus (above), infraspinatus, teres minor (below)
  - Posterior cuff → infraspinatus + teres minor
  - Small **acromial spur** pressing on supraspinatus → visible in sagittal

Partial Tears Ask AI

- Coronal → cuff thins at greater tuberosity
  - Articular part torn, bursal continuous → **PASTA injury**
  - Articular continuous, bursal torn → **partial bursal-sided tear**
- Bursal-sided → usually associated with impingement
  - Depth < 50% of tendon thickness → **healing potential**
  - Depth > 50% → repair

Full-Thickness Tears Ask AI

- Full-thickness, bursal to articular; head completely covered → **minimal retraction**
  - Further → large; reaching glenoid → massive
- Tendon stump may remain attached at footprint
  - Rule → **no tendon-to-tendon suture**
  - Stump removed → tendon sutured to bone → only route to healing
- Thin line to greater tuberosity → bursal side continuous
  - Empty area beneath → **articular-side tear**
- Most images → **fat suppression** (dark)

Retracted and Massive Tears Ask AI

- Retracted to middle of head → **type 2**, or medium by retraction
- Sagittal → subscapularis, infraspinatus present; supraspinatus almost absent → retracted out of cut
- Degenerated, stopped near glenoid → whole head uncovered → proximal migration
  - → **massive tear**

Subscapularis Tear and Long Head of Biceps Ask AI

- Axial → subscapularis torn from **lesser tuberosity**, retracted
- → long head of biceps subluxed; not seen in groove
- Always assess LHB during cuff examination and arthroscopy
  - Degenerated LHB → **source of pain**
- Cuff repair must not leave pathological LHB → **biceps tenotomy** for pain relief
- Tenodesis site → **bicipital groove, suprapectoral**; not supraglenoid origin

Patient	Procedure	Effect
Young	Tenotomy + tenodesis, suprapectoral	Refixed lower in groove
Above 50	Tenotomy alone; left to retract	Pain relief; no functional difference

- Cuff assessment → supra, infra, plus **subscapularis** and LHB
- Degenerated or subluxed LHB → **very low threshold** for biceps tenotomy
  - Left in place → persistent source of pain

Fatty Infiltration and Atrophy Ask AI

- **Fatty infiltration** → sagittal view: scapular spine, coracoid, cuff
- Progression (white dots = fat):
  - Whole cuff muscle
  - Fatty streaks
  - More white dots; muscle still predominant
  - Fat ≈ muscle

  - Fat great majority
- → **Goutallier classification**, 0 to 4
- More fatty degeneration → weaker cuff → poor function even after repair
  - Repair aim → cover head; prevent superior migration and cuff tear arthropathy
  - Prognosis → **poor function** expected
- **Atrophy** → same view; line from scapular spine to coracoid
  - Normal → muscle above line
  - Muscle shrinking well below line → **severe muscle atrophy**

Pitfall

- Small–medium or partial tears → easily missed on T1
- Fat-suppression coronal images → tear spotted; train on these by default
- Empty bicipital groove → look for subscapularis tear
- Fatty degeneration → sagittal view

Treatment Options and Conservative Management Ask AI

Options Overview Ask AI

- Options:
  - **Non-operative**
  - Repair
  - Limited procedure without repair → "cannot repair it"
  - **Partial cuff repair** → part of tendon restored, rest not
    - Example → most of posterior cuff back; most anterior part won't reach despite release
  - **Patch grafts** → bridge unreachable part
  - Tendon transfers
  - Biodegradable spacers
  - Shoulder arthroplasty

Rationale and Outcomes Ask AI

- Conservative treatment → **no consensus**
- Rationale → large and massive tears → very high re-tear, about 60%
  - Even in good hands → re-tear rate 50% to 60%
- Re-tear on MRI ≠ return to **pre-operative level of complaint**
  - Many remain satisfied after re-tear
  - Satisfaction attributed to long post-operative **physiotherapy programme** → why not without surgery
- Success rates in literature → 30% to 82%

Indications Ask AI

- Main indications:
  - Patient does not want surgery
  - Too elderly
  - Unable to perform surgery
- Relative indication → **partial**, not full-thickness, tear
  - Natural history: full-thickness → increasing retraction → proximal migration → **cuff tear arthropathy**

  - Partial → time for follow-up; if no progression → depends on patient tolerance
- Known irreparable tears; patients already decided for arthroplasty → also managed conservatively meanwhile

Components Ask AI

- Goals → improve function, reduce pain, some power via **physiotherapy** and strengthening
- Measures:
  - Preserve **residual muscle function**
  - Strengthen **deltoid** and scapular stabilizers
  - NSAIDs, pain killers
  - **Duloxetine** (described as selective serotonin reuptake inhibitor) → night pain *[Q2 · As spoken; duloxetine is a serotonin-noradrenaline reuptake inhibitor (SNRI), not an SSRI.]*
  - **Steroid injections** → only once full-thickness
    - Avoided in partial tear → may convert to full-thickness
    - Full-thickness → aim only to abolish pain
  - **PRP and stem cells** → no evidence in established cuff tears
- Ideal candidate:
  - **Congruent joint**, no proximal migration
  - Partial tear preferred
  - Improving ROM and pain
- Full-thickness + insisted conservative + proximal migration → cuff tear arthropathy
  - Then only → **reverse shoulder arthroplasty**
- Cautious selection → very elderly, or partial tear improving with physiotherapy

Pitfall

- Steroid injection in partial tear → may convert to full-thickness
- Full-thickness tear → retracts with time → migration → arthropathy
- Conservative treatment needs congruent joint without proximal migration
- PRP, stem cells → no evidence in established tears

Rotator Cuff Repair: Principles and Technique Ask AI

Steps Ask AI

- Goal → clear view of tear → clean **subacromial space**:
  - Subacromial **bursectomy**
  - Some release of **coraco-acromial ligament**
  - Acromial spurs → removed
    - Routine **acromioplasty** → no longer routine; impingement role debatable
    - Clear bony prominences from acromion or AC joint → removed
  - Cuff mobilized → reaches greater tuberosity **footprint**
  - Repair → anchors, or **transosseous sutures** through bone

Fixation Ask AI

- **Anchor** → enters greater tuberosity bone; sutures passed through tendon
  - → tendon back to normal place on greater tuberosity

- Tendon → **sutured to bone**
  - Suture to residual lateral stump → fails very rapidly
- Transosseous → wide suture through **large bone bridge**
  - Small bone bridge → breaks quickly
- Footprint preparation → thin cortical layer removed until **bleeding spots** appear
  - Not down to **cancellous bone** → weak bone → anchor pull-out, poor purchase
- Anchor inserted at about 45° → **Dead Man's angle**
- Sutures passed through cuff, various **stitch configurations** → tied → cuff back on greater tuberosity

Mini-Open vs Arthroscopic Repair Ask AI

- **Mini-open** → deltoid split incision, 4 or 5 cm
  - Limit → 5 cm below lateral edge of acromion → protects **axillary nerve**
  - Exposes greater tuberosity, head, cuff
- Sequence → bursa removed → acromial spurs removed with **burr**
  - → footprint prepared with **shaver** → anchor → sutures passed through cuff
  - → cuff brought back over humeral head, sutured
- Other routes:
  - Arthroscopic
  - Transosseous (trans-humeral) with **giant needle**
  - Combined arthroscopic + mini-open

Feature	Mini-open	Arthroscopic
Glenohumeral joint	Not seen	Inspected
Subacromial space: acromion, spurs, cuff, greater tuberosity	Seen	Seen
Subscapularis	Not seen	Seen
LHB pathology	Not seen	Seen; tenotomy possible

- Mini-open drawback → missed subscapularis and LHB pathology (quite common, needing tenotomy)
- Compromise → surgeon doing shoulder arthroscopy but not **arthroscopic cuff repair**; outcome unchanged:
  - **Glenohumeral arthroscopy** → check subscapularis, biceps; tenotomy if needed
  - Subacromial **bursoscopy** → bursa removed → cuff tear confirmed
  - Decision to repair → mini-open suture

Tear Configuration Ask AI

- Ideal → **crescent**-shaped; greater tuberosity lateral, head medial
- **U-shaped** → apex cannot be pulled to greater tuberosity
  - First side-to-side closure → **margin convergence**
  - → U converted to crescent → crescent repaired
- **L** or reverse L
  - Long limb posterior → reverse L; opposite → L
  - Limb closed first → residual tear sutured to greater tuberosity

Arthroscopic Repair Sequence Ask AI

- **Glenohumeral arthroscopy** first → articular-side view
  - PASTA → subacromial view shows intact cuff; tear only articular
  - Inspect biceps tendon and subscapularis tendon
  - Early shoulder stiffness → **rotator interval release**
- Then subacromial **bursoscopy** → bursa, not joint → not "arthroscopy"
- Bursectomy → bony spurs removed (**acromioplasty not routine**) → tear, footprint identified
  - → anchor in greater tuberosity → sutures through cuff → tied
- Video (mini-open) → burr; cuff lifted off exposed head; greater tuberosity site
- Greater tuberosity **shaved** → anchor inserted → sutures passed through cuff
- Suture-passing instruments → two types:
  - Pass and retrieve in one pass
  - Pass only → second instrument retrieves suture
- Goal → cuff **covers whole head**, reaches anchor
- Anchors → one, two or more, per **extension of the tear**; stitch types vary
- Anchors usually **double-loaded** → four suture strands
  - One strand of each pair → **simple suture**
  - Both strands → **mattress suture**
  - Choice → preference, tear size, tension; marked retraction → tighter closure
- **Knot pusher** → knots tied
  - → retracted tear back to greater tuberosity; whole head covered (subacromial view)

Subscapularis Repair Ask AI

Subscap

- Glenohumeral view → damaged subscapularis at lesser tuberosity
  - → suspect **biceps pathology**, LHB subluxation
- LHB pulled in → marked **fraying**
- Plan → LHB tenotomy → subscapularis repair to lesser tuberosity → inspect supra, infra
  - Anchor in **lesser tuberosity**, glenohumeral side → sutures through subscapularis → tied → tenotomy
- **Isolated subscapularis tendon injury** → rare (this case isolated)
  - Usually part of tear with supra, infra; at least supra
- Stitch passed and retrieved repeatedly → knot pusher → tendon back on subscapularis footprint
- Management by patient:

Patient	Action	Reason
Instability / dislocation	Repair	Main anterior stabilizer
Elderly, small tear	Biceps tenotomy alone may suffice	Subluxation painful
Young, active	Repair	—

Clinical Pearl

- Tendon → bone only; never tendon to residual stump
- Footprint → bleeding spots, not cancellous bone
- Anchor at about 45° → Dead Man's angle
- U-shaped tear → margin convergence first
- Mini-open → 5 cm below acromion limit; axillary nerve

Massive and Irreparable Tears Ask AI

Large Retracted Tears Ask AI

- Coronal → whole head exposed; retraction at least 4 cm, not yet to glenoid
- Bringing it back → **difficult**
- Marked retraction → **not supraspinatus alone**
  - Sheet → anterior part cannot retract this far with intact posterior part
  - → posterior extension → supra + infra
- Sagittal → supraspinatus space empty; retracted
- Arthroscopy → very thick bursa (as usual locally); head exposed; cuff fails to reach on pull

Mobilization and Release Ask AI

- Release from above and below:
  - **Traction suture** through tendon → pulled
  - **Tissue liberator** (smaller periosteal elevator) → frees **adhesions** above and below → gains length
  - **Extra portal** for instrument
- Posterior part may come; adherent anterior part blocks
- Footprint → **prepared with a shaver** → tendon pulled toward footprint
- Part not coming → **interval release**
  - Cut between posterior cuff and most anterior part → each handled separately
  - One blocked part may block the other

Partial Cuff Repair Ask AI

Partial

- Whole posterior cuff restored; part of most anterior portion not → **partial cuff repair**
- Satisfactory:
  - Good function
  - Prevents proximal migration
  - Prevents **cuff tear arthropathy**
- Releases incl. interval release → bring back what comes; unreached anterior part left as anterior tear
- Indication → **massive tears** not completely repairable

Irreparable Tears Ask AI

- Proximal migration; cuff retracted **medial to the glenoid**
- Medial to glenoid + degenerative (not acute traumatic) → **irreparable**
  - Releases cannot cover the distance
- Such retraction → supra + infra; may reach teres minor
  - Plus at least **upper border of subscapularis**
- Left untreated → cuff tear arthropathy eventually
- Case, same MRI:
  - Glenohumeral → subacromial space passage easy → communicate
  - LHB subluxed, degenerated (upper border subscapularis tear) → pain → **biceps tenotomy** → first step
  - Subscapularis tear: small or elderly → repair may be omitted; larger, younger → repair
  - Subacromial → bony prominence on **undersurface of the acromion** → removed (acromioplasty)
    - Not routine → only tight space or bony prominence

        - Wide space, no prominence → no acromioplasty
        - **Bursectomy** always → cuff lies under bursa
    - Acromioplasty → **cauterization** so cancellous bone does not bleed
    - Undersurface + upper surface released → cuff moves only from medial to lateral of glenoid
    - Still about 5 cm short across exposed head
    - Tenotomy + decompression done → enough to improve patient?
    - Irreparable because:
        - Degenerated → fat whiter than muscle
        - Muscle below spine–coracoid line → **completely atrophied**
        - Completely retracted
        - Proximal migration
- Pattern → most commonly **posterosuperior**: supra + infra ± upper border subscapularis
    - Rarely anterosuperior → subscapularis + supraspinatus

Clinical Signs of Large or Irreparable Tear Ask AI

- No external rotation at all → tear extended anterior → posterior → **large tear**
    - Tears start in supra → extend to infra; rarely start in infra
- Plus positive **lag signs** → ER position not maintained, falls back onto belly
    - → irreparable, or at least large
- **Acromiohumeral distance** < 6 mm → proximal migration started
    - = first stage of shoulder arthritis secondary to cuff tear

Choosing Among Options Ask AI

- **Physiotherapy alone** → fails once migrated
    - Needs **congruent glenohumeral joint**
    - Proximal migration = **superior escape** → deltoid cannot work
    - Deltoid needs stable joint, head compressed against glenoid
    - Migration increases → arthropathy
- **Subacromial decompression + biceps tenotomy** alone → temporary relief, usually not more than a year
    - Then migration increases → arthritis
- Partial repair possible → good option
    - Part of cuff covers head; most anterior part need not reach
- Not possible → key options: **tendon transfers**, reverse arthroplasty
- All **soft-tissue options** (repair, transfers, others) → non-arthritic or mildly arthritic shoulders
    - Advanced arthritis → arthroplasty only
- Priority → **cuff repair**, complete or even partial
    - Partial repair fails → tendon transfers or reverse arthroplasty
- Physical therapy → only with congruent joint; proximal migration → definite failure
- Decompression + tenotomy → very temporary; **partial cuff repair** → excellent if reachable

High Yield

- Cuff medial to glenoid + degenerative → irreparable
- No ER + positive lag signs → irreparable or at least large
- Acromiohumeral distance < 6 mm → proximal migration started
- Superior escape → physiotherapy fails
- Not all massive tears irreparable; all irreparable tears massive

Salvage Procedures for Irreparable Tears Ask AI

Biceps Augmentation Ask AI

- LHB tenotomy needed anyway → **strong tendon** available
- Shown mini-open; possible arthroscopically
- Cuff cannot reach greater tuberosity
- LHB as graft instead of **synthetic material**
  - One side → greater tuberosity; other side → retracted cuff
- = **biceps augmentation**

Tendon Transfers: Patient Selection Ask AI

- Too retracted for augmentation → **tendon transfer**
- Requirements:
  - **No advanced arthritis** → soft tissue cannot address it
  - Fit and active
- Best → patients in their 50s with irreparable cuff tear
- Not for patients in their seventies:
  - Any transferred muscle → **weaker**
  - **Re-education** for new function → prolonged rehabilitation → unsuited to elderly
- Best candidates → active patients in their 50s, irreparable tears

Latissimus Dorsi Transfer Ask AI

Lat dorsi

- Most famous for irreparable **posterosuperior** tears (supra + infra) → described by Gerber
- Detached from **insertion in bicipital groove** → moved to greater tuberosity
- Two actions:
  - **External rotator**
  - Compresses humeral head against glenoid
- Stable compressed joint → **deltoid** works → flexion, abduction
  - Not by latissimus → surprisingly, by deltoid
- Latissimus → external rotation only; gives deltoid advantage
- Technique:
  - Greater tuberosity side → arthroscopic or **mini-open approach**
  - Harvest → incision on **posterior axillary fold**
  - Two muscles: posterior → **teres major**; anterior → latissimus dorsi
  - Cut at bicipital groove attachment → very thin tendon
  - Mobilized **posterior → anterior**, at teres major side
    - **Neurovascular bundle** enters anteriorly → anterior mobilization cuts it → muscle worthless
  - Photo → teres major; elevated latissimus with sutured tendon; bundle entering from front
  - Mobilized until length reaches greater tuberosity; reaches with difficulty
  - Passed **under the deltoid**, upper incision → lower
  - Sutures pulled out at upper wound → **anchors in greater tuberosity** → tendon sutured
- Case:
  - Pre-op → no active ER, no abduction
  - Post-op → **full external rotation**, forward flexion, abduction

  - Abduction by deltoid; latissimus → **stable congruent joint**
- **Less invasive** transfers → greater tuberosity part arthroscopic
  - Harvest open; **all-arthroscopic** also described
- Mechanism:
  - **Tenodesis effect** → compresses head → deltoid biomechanical advantage
  - Active external rotator
- Results → good in irreparable tears
  - Not **normal function** → better function, better pain
- With **arthroplasty**: no intact posterior cuff (infraspinatus + teres minor torn) → add latissimus transfer for ER
  - **Teres minor** present → not needed
  - Complete teres minor tear → not common

Trapezius Transfer Ask AI

Trapezius

- Recent alternative to latissimus dorsi
- **Biomechanically better** in posterosuperior tears
- Latissimus → stood test of time; trapezius → still new
- Drawback → **lacks length** when detached
  - → lengthened with **graft** (abroad or locally in Egypt); selected cases

Pectoralis Major Transfer Ask AI

Pec major

- Subscapularis torn, too retracted → **irreparable anterior cuff tear**
- **Pectoralis major**, sternal head:
  - Cut at bicipital groove
  - Passed **behind the conjoint tendon**
  - Fixed to lesser tuberosity

Transfer	Indication	Route / fixation	Status
Latissimus dorsi (Gerber)	Irreparable posterosuperior	Bicipital groove → under deltoid → greater tuberosity	Stood test of time
Trapezius	Posterosuperior	Graft-lengthened → greater tuberosity	Biomechanically better; still new
Pectoralis major, sternal head	Irreparable subscapularis	Behind conjoint tendon → lesser tuberosity	—

Superior Capsular Reconstruction Ask AI

SCR

- **Superior capsular reconstruction** → appeared 2007; Mihata (Japan)
  - Indication → posterosuperior irreparable tears (supra + infra)
- Goal → prevent proximal migration; compress head → deltoid works
  - Without tendon transfer
- Graft → **fascia lata** or synthetic
  - Fixed glenoid side + greater tuberosity side → head pressed downward
- **Tenodesis effect** → not dynamic, not muscle
  - Prevents proximal migration; head pushed slightly down
- Anchors → glenoid and greater tuberosity

- Durability → how long before tearing and re-migration
  - Still debated; **not yet stood the test of time**
- Patch graft sources → **iliotibial band** autograft, or commercial dermal graft
- Superior migration case → tissue spans glenoid → greater tuberosity; neither muscle nor tendon

Biodegradable Spacer (Balloon) Ask AI

Balloon

- Simpler aim → press humeral head down; prevent proximal migration
- **Biodegradable polymer** sac (balloon) → subacromial space
  - Inflated with 25 cc saline
  - Wedged between acromion and humeral head → presses head down
- Eventually ruptures, leaks saline → polymer dissolves
- Reported improvement → one to three years in **irreparable cuff tears**
- Prerequisite → no arthritis

Cuff Tear Arthropathy and Reverse Shoulder Arthroplasty Ask AI

Arthritis Grading in Cuff Tears Ask AI

- Arthritis → severe pain; all **soft-tissue procedures** fail
- Classification of shoulder arthritis in cuff tears:

Stage	Finding
Grade 1	Acromiohumeral distance < 6 mm *[Q3 · As spoken; in Hamada grading, grade 1 is AHD of 6 mm or more and grade 2 is 5 mm or less.]*
Grade 2	Acromiohumeral distance < 5 mm
Grade 3	Acetabularization; head abutting against acromion
Next	Glenohumeral arthritis evident
Then	Collapse of head

- Grade 3 → any interposed soft tissue tears immediately → soft-tissue procedures fail
- Grade 3 onwards → only **reverse shoulder arthroplasty**

Principle of Reverse Arthroplasty Ask AI

RSA

- **Glenosphere** (head) on glenoid; concavity on humerus
- **Center of rotation** → more medial, more inferior
  - → deltoid tensioned → biomechanical advantage → stronger deltoid
- Result → stable joint + functioning deltoid → movement

Case: Failed Repair Ask AI

- 65-year-old woman; previously repaired cuff → **re-torn**
  - Acromiohumeral distance → 4 mm
  - Re-torn tendon retracted, **just lateral to the glenoid**
- Why not re-repair:
  - Previous repair failed

  - Marked **proximal migration**
  - Very poor tendon quality expected → immediate re-tear; likely unreachable
- Older, weak muscle → **tendon transfer** not ideal
- SCR in very tight space → tears soon → high **friction**
- Decision → reverse shoulder arthroplasty, **deltopectoral approach**

Surgical Steps Ask AI

- **Biceps tendon** → always tenotomy → pain source; no function after arthroplasty
- **Subscapularis** → anterior stability → cut, tagged with tension sutures for later repair
- **Teres minor** intact → posterior cuff present
  - Posterosuperior cuff (supra + infra) torn → head uncovered
- Humeral cut → **just superior to bicipital groove**
  - **Conical reamer** → humeral stem preparation
- Glenoid **base plate** → reach most inferior glenoid margin → avoids **scapular notching**
- Covers mainly **lower half** of glenoid; top not a concern
- Base plate → fixed size 28 mm
  - Entry → **just inferior to the middle** of glenoid
  - Inclined → **inferior tilt** → avoids scapular notching
- Ream → base plate (**peg** + screws) → glenosphere mounted → stem preparation completed
- Stem **liners** → 0, 3, 6 mm; increased with spacer
- **Stability testing** → extension, external rotation, internal rotation
  - Decides polyethylene thickness
- Stable → trial replaced by **definitive polyethylene**
  - Subscapularis sutured to humerus transosseously
- Post-op film → **inferior tilt** → prevents impingement causing scapular notching

Outcome Ask AI

- Pre-op → **pseudoparalysis**; positive lag sign
- Six weeks, before **active strengthening**:
  - Hand to mouth
  - Reasonable abduction, reasonable forward flexion
- Continued improvement with physiotherapy

Take-Home: Irreparable Tears Ask AI

- Irreparable tear → **tendon retraction and thinning**
- Grade 3+ arthritis → marked proximal migration → glenohumeral narrowing → osteophytes

Must-Know

- Decompression + biceps tenotomy alone → limited benefit, rapid deterioration
- Partial repair → excellent option if reachable
- Superior capsular reconstruction → before onset of arthritis
- Tendon transfer → before arthritis, active patients
- Arthritis grade 3+, especially pseudoparalysis → reverse shoulder arthroplasty only

Review MCQs Ask AI

Q1: Biceps Stabilizer Ask AI

- Most important LHB stabilizer in bicipital groove → **subscapularis**
  - Options → supraspinatus, subscapularis, superior glenohumeral ligament, pectoralis major
- Subscapularis injury → suspect **subluxation of the long head of biceps**
- LHB subluxed out of groove → inspect **upper border of subscapularis** arthroscopically
  - Both usually occur together

Q2: Fatty Degeneration on MRI Ask AI

- Best MRI cut for **fatty degeneration**?
  - Options → coronal T1, coronal T2, axial, sagittal
- Answer → **sagittal**; fat amount vs muscle

Q3: Arthroscopic vs Mini-Open Ask AI

- Main advantage of arthroscopic over mini-open → **glenohumeral inspection**
  - Options → smaller scars, earlier active motion, stronger repair, glenohumeral inspection

Q4: Night Pain, No Stiffness Ask AI

- 65-year-old female; pain incl. night pain; limited ROM
  - Passive ER in adduction **not limited** *[C3 · transcript: "passive ER in abduction not limited"]*
  - Options → physiotherapy, RC repair, SCR, reverse arthroplasty
  - Audience split → 50% A, 50% B
- Night pain → cannot be left untreated
- If passive ER in adduction limited → **shoulder stiffness** *[C4 · transcript: "limited passive ER in abduction"]*
  - → physiotherapy first to regain good passive ROM
  - Repair on stiff shoulder + post-op immobilization → remains stiff
- Here → no true stiffness; good passive range
  - Main problem → weakness, limited range, night pain → cuff
- Not a large tear → best option **cuff repair**

Q5: Pseudoparalysis Ask AI

- 65-year-old female; pain + **pseudoparalysis**
  - No active motion; arm passively placed → falls back → cannot maintain arm in space
- About 90% answered D → **reverse shoulder arthroplasty**
- Physiotherapy → fails; **superior escape**, proximal migration
- Cuff repair → cuff too medial; marked migration → would tear immediately
- SCR → not advised when **head abutting against acromion** → cannot push head down

Q	Answer	Key reason
1	Subscapularis	Main LHB stabilizer in groove
2	Sagittal	Fat vs muscle assessed
3	Glenohumeral inspection	Subscapularis, LHB seen
4	Cuff repair	No true stiffness; not large tear

Q	Answer	Key reason
5	Reverse shoulder arthroplasty	Superior escape; cuff too medial

Questions and Answers Ask AI

Grafts vs Transfer; Post-Operative Imaging Ask AI

- Session end → 12 o'clock
- Irreparable tear, middle-aged: allograft, xenograft or muscle transfer? *[C5 · transcript: "repairable cuff tears"]*
  - **Allografts** unavailable locally
  - SCR ≈ allograft equivalent → graft between glenoid and greater tuberosity, not cuff extension
  - Middle-aged → **tendon transfer** definitely best
- Dr. Mohamed Kamal: post-operative imaging → when, what, why?
  - **No routine post-operative imaging**
  - Only with significant complaint → pain or weakness
  - Large and massive tears → re-tear 50% to 60%
  - Re-tear → not necessarily loss of **symptomatic improvement**
  - Imaging → likely finds re-tear → patient **psychologically** worse

Indications by Tear Type Ask AI

- Dr. Wael: 50-year-old, **full-thickness tear** → repair
  - Only partial thickness → would defer repair
  - Full-thickness → retracts by natural history → proximal migration → arthropathy
  - No complaint at all → no surgery; no benefit
  - **Very low threshold** → any complaint, even insignificant → repair advised
- Dr. Mohamed: **partial cuff tears** → act according to complaint
  - No fear of proximal migration or cuff tear arthropathy
  - Experience → **PASTA lesions** more painful than complete tears; no tendency to heal
  - Severe pain → surgery; insignificant pain → wait
- Surgical options for PASTA → two ways:
  - Convert to complete tear → sutured as complete → most surgeons
  - Suture lower articular fibers only → **technically demanding**
    - Tension matched to proximal bursal-sided fibers
- > 50% thickness → **convert into a complete tear** → repair
- Middle-aged, small complete tear, 1 cm → **repair**

Longitudinal and Interstitial Tears Ask AI

- Dr. Ibrahim Fawzy: split **along the line of the fibers** → side-to-side suture? visible on MRI?
  - **Longitudinal tear** alone → should not affect function → no repair
  - Not described among cuff tear types; not common; may be **iatrogenic**
  - With **avulsion** off greater tuberosity → side-to-side first (margin convergence) → fix to greater tuberosity
- **Interstitial tear** → same principles as partial tears → thickness and complaint
  - < 50% → usually **conservative**
  - > 50% + significant pain, weakness → convert to complete tear → repair

  - Location → middle; intact bursal and articular layers
  - Least common type

Cuff Tear with Frozen Shoulder Ask AI

- **No complete consensus**; two ways:
  - Speaker's preference → **good passive range of motion** first, then cuff repair
    - **Arthroscopic release**, not manipulation
  - Same session → rotator interval release + **anterior capsulotomy** + cuff repair
    - Post-repair immobilization → stiffness returns
    - Forced **early range of motion** → may compromise repair
- Experience-based; nothing solid

Ultrasound Ask AI

- **Ultrasound** → has a role; not in common local practice
- **Operator dependent**:
  - Done by surgeon himself (practice elsewhere)
  - Or written report without reviewable image → limited reliance

Massive vs Irreparable Ask AI

- Dr. Adnan: massive = irreparable?
  - Some massive tears → **direct repair** achievable
  - Direct repair not achievable → irreparable
- **Not all massive tears irreparable**; all irreparable tears massive

Biceps and Fixation Choices Ask AI

- Degenerative tears → **low threshold for biceps tenotomy**
  - Subluxation or degeneration of LHB → tenotomy ± tenodesis; little difference
  - Biceps non-functioning; definite **source of pain** → no reason to leave
- Dr. Amin: double-row vs single fixation → no clear consensus
  - Small-to-medium tears → **single row**
  - Large or massive tears → **double row** → secure repair

Post-Operative Rehabilitation Ask AI

- As any **tendon repair**:
  - Passive ROM started; **no active ROM for 6 weeks**
  - Simple **arm sling**; abduction splint not used
    - Abduction splint → very annoying for patient
    - Tendon reaching only in abduction → fails; must reach with hand by side
  - Simple passive exercise → elbow straightened, other hand raises arm in **forward flexion**
  - **External rotation** allowed only without subscapularis repair → stretches subscapularis
  - Active-assisted ROM → not before 6 weeks
  - **Strengthening** → not before 3 months

Phase	Timing
Passive ROM (forward flexion; ER if no subscapularis repair)	Immediately
Active or active-assisted ROM	Not before 6 weeks

Phase	Timing
Strengthening	Not before 3 months

Professional Athlete Ask AI

- Return to previous level after cuff tear?
- **Competitive athlete** → acute traumatic, not degenerative
  - Healthy tendon
  - Good bone stock in humeral head
  - Wide subacromial space
  - → **excellent results for repair**
- Easily pulled back regardless of retraction → no degenerative **adhesions**
- Acute traumatic tear → best outcome; **return to play**
- Session closed; thanks to speaker and attendees

Clinical Pearl

- No routine post-op imaging → re-tear common, symptoms often preserved
- PASTA > 50% → convert to complete tear, then repair
- Frozen shoulder + cuff tear → passive range first, then repair
- No active ROM for 6 weeks; strengthening not before 3 months
- Small-to-medium → single row; large or massive → double row

Arthroscopic Supraspinatus Repair

(Step by Step to a Successful Repair)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 58) “anterolateral subacromial viewing portal” → “anterolateral working cannula”

Case Scenario Ask AI

- Topic: **arthroscopic technique** for small and medium-sized supraspinatus tendon tears
  - Common scenario in practice
- Case: **60-year-old female**
  - Shoulder pain, weakness, limited range of motion
  - **Night pain**
  - **Coronal MRI** → full-thickness tear with retraction

Arthroscopic Portals Ask AI

- Two sets: **glenohumeral joint** portals + subacromial bursoscopy portals

Glenohumeral Portals Ask AI

- **Standard posterior viewing portal**
  - 2 cm distal + 2 cm medial to posterolateral acromial edge
  - Role: viewing
- **Anterior working portal**
  - Just lateral to the **coracoid process**
  - Usually **outside-in technique**
- Glenohumeral joint: posterior = viewing; **anterior** = working

Subacromial Portals Ask AI

- Most rotator cuff repair work → **subacromial space** via subacromial bursoscopy
  - Two standard portals
- **Posterolateral portal**
  - ~2 cm distal to lateral acromial edge, posteriorly
- **Anterolateral portal**
  - More anterior; ~2–2.5 cm distal to anterolateral acromion
  - ~5 mm distal to viewing portal → better instrument view with **30° lens**
- **Anchor portal**
  - More proximal; just lateral to lateral acromial edge
  - Localized with **needle** → best trajectory to footprint
  - Not marked on skin
- Subacromial: **posterolateral** = viewing; anterolateral = working

High Yield

- Glenohumeral: posterior viewing, anterior working
- Subacromial: posterolateral viewing, anterolateral working
- Anchor portal: separate, proximal, needle-localized

Examination Under Anesthesia and Glenohumeral Arthroscopy Ask AI

Examination Under Anesthesia Ask AI

- Preoperative **stiff shoulder** → examination under anesthesia mandatory
- This patient: good external rotation + abduction → not stiff
- Stiffness under anesthesia + cuff tear → add **release of rotator interval and capsule**
  - Otherwise → stiff shoulder postoperatively

Why Start With Glenohumeral Arthroscopy Ask AI

- Most supraspinatus work subacromial, yet procedure **starts with glenohumeral arthroscopy**
- Reasons
  - Cuff seen from **articular side**
    - Head + cuff visible, footprint fully covered → no full-thickness tear
    - No partial articular-sided tear (**PASTA lesion**)
    - Cuff intact from inside, or small **bursal-sided tear**
  - **Subscapularis tendon**: associated upper border tear common
    - Small tear (**Lafosse type I**) → neglect; deal with biceps tendon instead
    - Type II or more → must be repaired
  - **Biceps tendon**: degenerated or subluxed → source of pain
    - Add **biceps tenotomy** → else residual postoperative pain
  - Clear **rotator interval synovitis** → secondary frozen shoulder starting
    - Synovectomy + interval release together with cuff repair
  - Above 40 years + **first-time dislocation** → often Bankart lesion + cuff tear
    - Glenohumeral arthroscopy needed for **Bankart repair**

Arthroscopic Findings in This Case Ask AI

- **Subscapularis**: small upper border tear
- **Long head of biceps**: degeneration, thinning → pain source
- Cuff fully separated; footprint fully exposed → **full-thickness tear**

Biceps Tenotomy Ask AI

- Upper border subscapularis tear + subluxed, degenerated biceps → **biceps tenotomy**
  - Instrument: **basket forceps**
- Tenotomy must be **complete** → else residual pain after cuff repair
- Portals: **posterior glenohumeral viewing** + anterior working
- No biceps fibers left intact → else **source of pain**

Pitfall

- Stiffness under anesthesia ignored → stiff shoulder postoperatively
- Incomplete biceps tenotomy → residual pain after cuff repair

Subacromial Bursectomy and Footprint Preparation Ask AI

Bursectomy Ask AI

- Shift to **subacromial bursoscopy** → start with bursa removal
- Reasons
  - **Visualization**: cuff, tear, footprint
  - Inflamed bursa (**bursitis**) → common cause of pain
- **Acromioplasty** / lateral acromial bone removal: not routine
  - Only with clear **bony impingement** (bony knob to remove)
- Tools: **radiofrequency** or bipolar cautery
- Target: cuff, tear, footprint (**greater tuberosity**) → anchor site
- Rule: bursectomy for visualization; no acromioplasty without clear **bony impingement**

Footprint Preparation Ask AI

Footprint

- Enough bursa removed → cuff, tear, **greater tuberosity** visible
- Goal: **bleeding points** on footprint
  - Shaver, curette, file, or **burr** — cautiously
- Do not remove the whole **cortical layer**
  - Pure cancellous bone, esp. **osteoporotic** → poor anchor purchase → pull-out
- Sequence: bursectomy → tear + footprint visualization → **footprint preparation** (shaving)
- Little **shaving of the cuff undersurface** → better healing

Pitfall

- Whole cortical layer removed → poor purchase → anchor pull-out
- Highest risk: osteoporotic bone

Anchor Portal and Medial-Row Anchors Ask AI

Anchor Portal Ask AI

- Localized with **needle** → good trajectory
  - Stab incision with scalpel at that point
- **Separate portal**, proximal to posterolateral + anterolateral subacromial portals

Anchor Insertion Ask AI

- Insert until **laser mark** (black line) fully under bone
  - Else **prominent anchor** → impingement

Number of Anchors Ask AI

- Assess: cuff → remove bursa → footprint → **tear size**
- Enough space for **two anchors**?
  - Tunnels too close → **purchase lost**
  - This case: small tear → only one anchor

Cannula Timing and Portal Roles Ask AI

- Bursectomy: posterolateral viewing + **anterolateral working**
- Lecturer delays **cannula** insertion
  - Without it → freer RF probe + instruments: lateral, medial, posterior
- Three portals: posterolateral viewing; anterolateral working; separate **anchor portal**
  - Anchor portal defined only by needle → good trajectory
- Anchor via separate portal → its threads exit there
- **Cannula** now → anterolateral working portal
  - Separate portal = anchor placement + **suture handling**
- Threads retrieved through cannula **one at a time**
  - >1 thread in cannula while passing → **entanglement**
  - Rest stay in anchor portal; pull only the one being passed

Medium-Sized Tear: Two Medial-Row Anchors Ask AI

- After needle: **artery forceps** in, withdrawn open → space for anchor
- **Anterior anchor** first → shoulder in external rotation
- Second anchor if space → **more anchors** = stresses distributed
- **Posterior anchor** → shoulder in internal rotation
- **Medial row** (first row): to laser mark; both via same portal

Suture Management and Passing Ask AI

Threads Across Portals Ask AI

- Two anchors → too many sutures
- **Ring forceps**: posterior glenohumeral portal → subacromial space
  - Four threads of posterior anchor → out via **posterior portal**
  - Fewer threads at anchor portal → easier handling
- Rule: two anchors via one anchor portal → not all threads exit there
  - Posterior anchor threads → posterior portal, or anterior anchor threads → **anterior portal**
  - Too many threads from one portal → **entanglement**
- Lecturer routine: posterior anchor's four threads → **posterior portal**
  - All work via cannula in anterolateral portal, thread by thread

Single Anchor: Suture Passing Ask AI

Small tear

- One anchor (small tear) → **four threads**
- Cannula insertion: **blunt obturator** through cannula
  - Obturator directs cannula in
- **Ring forceps**: one thread into cannula; other three stay in anchor portal
  - Only one thread through cannula in anterolateral portal
- Passed with **suture-passing instrument**
- Passed thread exits cannula → retrieved via **anchor portal** → next suture
  - One suture in cannula at a time → else entanglement

Mattress Sutures Ask AI

- **Mattress sutures** preferred
- **External rotation** → pass two anterior sutures

- **Internal rotation** → pass two posterior sutures
- Each passed suture → **ring forceps** retrieval via anchor portal
- Passed sutures marked with **artery forceps**
  - Held near ends → threads not lost, not damaged by forceps
- Internal rotation → further posterior for next suture
- **Good, deep bite**; not near edge
  - Near-edge sutures → **cut through** on tying or pulling
- Watch anchor while pulling: suture **runs through anchor** → wrong limb
- Last suture → **maximum internal rotation**
- Four sutures = **two mattress sutures**: one anterior, one posterior
  - Two sutures form one mattress
  - Sufficient for small tear
- Per thread: anchor portal → out **anterolateral portal** → pass → back to anchor portal

Two Anchors: Up to Eight Sutures Ask AI

- Two anchors → up to **eight sutures**
  - Two mattress knots (four sutures) **anteriorly** + two (four sutures) posteriorly
- Sequence
  - Posterior anchor sutures → out **posterior portal**
  - **Anterior anchor** first: threads one by one; all four passed
  - Then **posterior anchor**: threads one by one via anterolateral subacromial cannula
  - Up to eight sutures → **good repair**
- Posterior sutures (posterior portal) → anterolateral working cannula → passed in **internal rotation** *[C1 · transcript: "anterolateral subacromial viewing portal"]*

Distributing Sutures Ask AI

Sutures

- Many sutures → not all **on the same line**
  - Some a bit **medial** (deep in tendon); some lateral (nearer edge)
- Tied → whole area between mattress limbs contacts footprint → larger **contact surface area**
- Different depths lateral → medial → better **footprint contact**

Knot Tying Rule Ask AI

Knots

- Only **two matching sutures** in cannula at a time → tie
  - Others stay in other portals
  - Always two by two → else **entanglement**

Pitfall

- More than one thread in cannula while passing → entanglement
- More than two sutures in cannula while tying → entanglement
- All threads from one portal → entanglement
- Bite near cuff edge → cut-through

Lateral-Row Augmentation Ask AI

Assessing the Repair: Small Tear, Single Anchor Ask AI

- Small cuff tear, good repair: **four sutures** passed
- **Anterior mattress knot** + posterior mattress knot; footprint covered
- **Elevating the cuff** → compressed in one area only
  - **Line contact**, not surface contact
  - Part of cuff not compressed; middle not properly repaired

Accessory Sutures Ask AI

- Options: cut one existing suture, or separate **FiberWire**
  - **Accessory suture**: not related to any anchor; passed in poorly fixed area
- Combined with other sutures → **sutureless (knotless) anchor** as lateral row

Loading the Lateral-Row Anchor Ask AI

Lateral row

- Footprint not truly covered → take sutures (more addable with suture passer)
- Down onto **lateral cortex** of humerus → anchor well down
- Sutureless anchor: **PushLock**, SwiveLock, or equivalent
- Pilot hole: tap → hammer to **laser mark**
  - All via cannula in **anterolateral working subacromial portal**
- All sutures loaded into **sutureless anchor**

Anchor tip	Insertion	Note
Metal tip at front	Hammered (tapped)	—
PEEK only, no metal	Screwed	Hammering → PEEK breaks

Lateral-Row Anchor Insertion Ask AI

Lateral row

- Hammer in; hand raised slightly (proximal) → anchor directed slightly **distal**
  - Prevents **avulsion of the greater tuberosity**
- One medial anchor → add second, **lateral-row anchor**
- **Two medial anchors** → lateral row still possible
  - Clear lateral humerus
  - Two sutures from anterior anchor + two threads from **posterior anchor** → load, fix
  - Whole cuff compressed on **greater tuberosity** → better footprint coverage
- All work via cannula in **anterolateral portal**
  - Common difficulty: imagining viewing vs working portal
  - Viewing = **posterolateral subacromial portal**; working = anterolateral
- Preference: **hammered anchors**
  - **Clear view** → prepared hole not lost
  - Hand raised → anchor slightly down → no tuberosity avulsion; stay **distal**
- Result: whole footprint covered; **whole cuff compressed**

Single Row vs Double Row Ask AI

Line Contact After a Single Row Ask AI

- Small and medium-sized tears → **double row** usually not needed
- Risk: cuff compressed on greater tuberosity **only along one line** → undesirable
- Elevating cuff after seemingly good repair → compressed only along **medial line** (knots)
  - Hence: threads a bit medial + a bit lateral (not near edge) → better coverage
  - Such repair **might fail**
- Acceptable in a **small tear**
- **Larger tear** → high incidence of failure
  - Uncovered footprint; cuff not pressed against it

Rationale for the Lateral Row Ask AI

- Threads → additional **lateral row** → whole cuff over whole footprint
  - Down to **lateral surface of the humerus**
  - Much better footprint coverage
  - Larger **area of contact** between cuff and footprint
- Sutureless anchor here: **PushLock** — threads loaded, hammered in
- Whole footprint covered; no greater tuberosity visible; cuff pressed → **better healing**
- Threads: medial anchor → **lateral surface of the humerus** → cuff compressed on footprint

Technique Summary (Video Review) Ask AI

Setup and Preparation Ask AI

- Start: **glenohumeral arthroscopy**
- **Subacromial bursectomy**: posterolateral viewing + anterolateral working
  - Bursa removed → see footprint, see cuff, remove **inflamed bursa**
- **Footprint preparation**: shaver, curette, file → bleeding points
  - Whole cortical bone kept → better **anchor purchase** + good healing bed
- Little shaving of the **undersurface of the cuff**

Anchor Portal and Position Ask AI

- **Needle** localization → good trajectory → anchor portal made there
  - After scalpel: **artery forceps** in, opened, withdrawn open
  - → anchor not entangled by soft tissue and bursa
- **Anchor position** by tear type

Tear	Anchor position	Purpose
Small–medium, easily pulled over footprint	A little lateral; not all the way medial to articular cartilage margin	Large cuff part on greater tuberosity → larger healing surface
Large, retracted	Medial, near cartilage edge	Repair with less tension

- **Laser mark** (black ring on handle) fully under bone → no protruding anchor

Cannula and Suture Passing Ask AI

Summary

- **Plastic shoulder cannula** → anterolateral subacromial working portal
  - **Blunt obturator** inside cannula directs it
- Threads **one by one**: one in cannula; others in anchor-placement portal
- **Suture-passing instrument**: any company you are comfortable with
- Passed thread returns into cannula → must not stay there
  - **Ring forceps** retrieval via anchor-placement portal
  - Marked with **artery forceps** → passed vs waiting threads
- All sutures passed → always **mattress sutures**

Suture Configurations, Distribution and Knot Tying Ask AI

Suture Configurations Ask AI

- Mattress configuration = **simplest** form of suture passage
- Complex: **Mason-Allen**, modified Mason-Allen stitches
  - For **really degenerated, retracted cuffs** → large pull expected
  - Sutures must not rip out of cuff

Suture Distribution Ask AI

Sutures

- Spread between anterior and posterior tear — **same portal**
  - **External rotation** → reach anteriorly
  - **Internal rotation** → reach posteriorly
- This case: two anterior + one posterior done; last posterior now — **single anchor**
- **Good bites**, not near edge
- Two anchors, eight sutures: within same knot, some sutures **medial** to others
  - Mattress knot → **larger surface area** on footprint
  - All on same line → one line of greater tuberosity → **line fixation**
  - Goal: **surface fixation** → better healing

Knot Tying Ask AI

Knots

- All passed → **two matching sutures** at a time into cannula
  - Others outside in anchor portal
  - Only those two while tying → else **entanglement**
- Check whether sutures **running (sliding)**

Sutures	Knot
Running	Sliding knot, e.g. fisherman's knot
Not running (any reason)	No sliding knot; simple knots (half hitches) alternately

- Sequence: sliding knot → 3–4 **half hitches** on top

Managing Suture Limbs Ask AI

Knots

- **Double row** planned → limbs not cut; retrieved via anchor portal for lateral-row anchor
- No lateral row → cut each suture **once tied**
- Lateral row planned → limbs left; one end may be cut, other kept
  - Depends on number of sutures loaded into lateral row

  - Depends on **one or two lateral-row anchors**
- Usually **four threads** → lateral row: two from anterior + two from posterior anchor

Double-Row Repair: Indications and Placement Ask AI

When to Add a Lateral Row Ask AI

- After good repair: cuff fixed **only along a medial line**; no whole-footprint compression
  - Small–medium tear → should do well
  - **Degenerated**, under tension, or somewhat large → bigger chance of failure
- Uncertain: cuff elevatable, fixed medially only
  - No compression over large footprint part → lower chances of **healing**
  - → consider **lateral-row sutureless anchor**
  - Use same threads just tied
  - **Free threads** (no anchor) passed untied if part of cuff floating
  - Goal: compression as demonstrated with **blunt obturator**
- Sutures loaded → **lateral-row sutureless anchor**
- Same portals: **posterolateral subacromial viewing** + anterolateral working
- **Lateral surface of the humerus** cleared → clear bone visualization

Adding an Extra Suture Ask AI

- One more suture = **risky**
  - Cuff well elevated → suture passer passes easily
  - Not elevated → **previous sutures injured**
  - Not advised unless area clearly elevated where suture really wanted
- Source: free **FiberWire**, or cut one limb of a tied suture
  - Other limb → through sutureless anchor

Lateral-Row Anchor Placement Ask AI

Lateral row

- Clear **lateral cortex** well down; not near top of greater tuberosity
  - Else fragment **breaks** on anchor insertion
- Each anchor: **specific introducer**; tap to laser mark
- Sutures through anchor **hole (eyelet)**
  - Usually a shuttle loop pulls them through → then hammered in
  - Shuttling easy → anchor always has a **shuttle loop**
- Hammered in; **not prominent**
- Sutures compress cuff against **whole footprint**

Indications Ask AI

Lateral row

- **Lateral row** by tear size

Tear size	Lateral row
Small or medium	Not necessary routinely; add if large cuff part clearly not compressed after repair
Large	Definitely advised

- Final: anchor down on lateral cortex → whole upper greater tuberosity covered by cuff

- Whole cuff over head + greater tuberosity, firmly compressed → **good outcome**
  - Especially small and medium-sized tears

High Yield
- Laser mark fully under bone → no prominent anchor
- Lateral-row anchor well down on lateral cortex → no tuberosity fracture
- Hand raised → anchor directed distal → no tuberosity avulsion
- Metal tip → hammered; PEEK only → screwed
- Large tears → lateral row definitely advised

Key Steps for a Successful Repair Ask AI

- Small to medium-sized **supraspinatus tendon tear**: key steps
  - Start with **glenohumeral arthroscopy**
    - Look for **upper border subscapularis tears**
    - Decide need for **biceps tenotomy**
    - Confirm cuff tear from **articular side**
    - **Rotator interval synovitis** → synovectomy and release
  - **Subacromial bursectomy**: see cuff, mobilize, confirm whole-footprint coverage when pulled
  - See footprint; prepare **greater tuberosity**; cuff release
  - Insert **anchor**
  - Sutures **well distributed** across tear, anterior → posterior
  - Tie: preferably **mattress sutures**, all threads passed
  - **Double-row repair** when needed

Must-Know
- Always start with glenohumeral arthroscopy
- Bursectomy routine; acromioplasty only for bony impingement
- Bleeding footprint, cortex preserved
- Goal: surface contact, not line contact

Anterior Shoulder Instability

(Classification, Assessment and Treatment Algorithm)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 86) “83% of 30 mm = 26 mm” → “83% of 30 mm ≈ 25 mm”

C2 (p. 86) “glenoid track 26 mm” → “glenoid track ≈ 25 mm”

Session Opening and Objectives Ask AI

Opening of the Session Ask AI

- Participants admitted; addressed to Dr. Osama
- Second lecture in the series of **Nile Hospital for Health Insurance**
- Chair: **Prof. Dr. Atef El-Beltagy**
  - Professor of Orthopaedic Surgery, Ain Shams University
  - Present in all lectures
- Thanked for the momentum given to the course
- Lecturer: **Prof. Dr. Maged Samy**, Professor of Orthopaedic Surgery
- Requests to attendees
  - Share the published link and lecture schedule
  - Lectures recorded → availability decided later
  - Suggestions → Facebook page, **Orthopaedic Department**, Nile Hospital for Health Insurance
  - Mid-lecture break → chair may stress a point
  - End of lecture → chat and questions to Dr. Maged and Dr. Atef
  - Cameras off → no server overload
- Dr. Atef: happy new year, blessed Ramadan
- Second lecture in the **online teaching** series
  - Sponsor: Orthopaedic Department, Nile Hospital — headed by **Dr. Osama Abdel-Rashid**
  - Participation: **Elixir** company, represented by Dr. Mohamed
  - Aim: vital, debated topics — never absent from **Master's, MD or Fellowship** exams
- Prof. Dr. Maged Samy: Ain Shams University; pioneer of **shoulder arthroscopy**
- Topic: **Management of Anterior Shoulder Instability**
- Questions sent at the end
  - All answered

Objectives of the Lecture Ask AI

- Dr. Maged thanks Dr. Osama Abdel-Rashid; Dr. Atef El-Beltagy as **Chair**
- Objectives → exam points + daily practice
- **Shoulder instability** — must know
  - The **shoulder stabilizers**
  - **Soft tissue operations** vs bony operations
  - **How to make the decision**
  - **Tips and tricks**

Shoulder Stabilizers Ask AI

- Frequently asked: what maintains shoulder stability?
- As in **any joint** → static + dynamic factors

Static Factors Ask AI

- **Labrum and ligaments**
  - Superior, middle, inferior **glenohumeral ligaments**

  - Key: inferior glenohumeral ligaments → **hammock** → inferior support
- **Bony orientation**
- **Joint volume**
  - Larger, stretched capsule → greater volume → lower stability
  - Tighter capsule → greater stability
- **Adhesion–cohesion forces**
  - Two smooth surfaces + **thin film of fluid** — e.g., two glass slides, drop of fluid
  - Slide easily over each other; difficult to lift apart
- **Bony configuration**
  - Small glenoid, large humeral head → low **bony stability**
  - **Glenoid labrum** → deepens glenoid by more than 50%
    - Fibrocartilage attached to glenoid edge
  - Low bony stability ≠ unimportant
    - Glenoid size **very critical** → minimum acceptable
    - Any smaller → highly unstable joint
  - → **Bony defect** very critical in instability

Dynamic Factors Ask AI

- Muscles surrounding the shoulder
  - Most famous → **rotator cuff muscles**
  - **Subscapularis** → just anterior to anterior capsule
    - Torn, retracted → unstable shoulder
    - Treatment → subscapularis repair or **tendon transfer**

Classification of Shoulder Instability Ask AI

Classical Classification Ask AI

- **Voluntary** → habitual dislocators; dislocate by themselves
  - No operation
- **Involuntary** → dislocates against patient's will
  - Traumatic
  - **Atraumatic** → generalized ligament laxity; no trauma history
- By direction: anterior, posterior, **multidirectional**
- Each → acute, chronic or **recurrent**
- Practice → treatment-guided classification instead

Treatment-Guided Classification Ask AI

Anterior

- Anterior, involuntary, **non-habitual** instability → subdivide
- **Subluxation**
  - Certain movements → catching ("bone on bone", "muscle on muscle")
  - Fear → stops movement, goes back
  - No dislocation needing reduction; arm stays in place
- **Frank dislocation** — must know
  - **First-time** or recurrent dislocator?
  - Ordinary or **locked dislocation**?

    - Locked anterior → not common, but seen
    - Humeral head locked firmly on anterior glenoid rim
- **Subtle instability** → often missed

Subtle Instability Ask AI

Subtle

- Subtle = hidden → instability without instability symptoms
- No dislocation, no catching → presents with **pain**
- **Overhead athletes**: swimmers, handball, volleyball players
  - 17–18-year-old → shoulder pain with every overhead activity
- Exam → classic **painful arc** → misdiagnosed as impingement
- Cause: **overstretched capsule** → head migrates superiorly → rubs acromion
  - Not a narrow subacromial space; very lax capsule
- Treatment → the instability, not impingement

> **Pitfall**
> - No **subacromial impingement** in a boy of 17, 18 or 20 years
> - Painful arc in this age → subtle instability

Treatment Principles and Patient Problems Ask AI

Building the Algorithm Ask AI

- Algorithm slide left empty → built step by step, not ready-made
- Principles
  - Understand **where the problem is** → very simple, logical
  - **Target** the specific problem → give the best chance
    - No operation with high recurrence rate in his situation
  - Leave room for **revision surgery** — "do not burn all your boats"
    - Simple procedure with good success → start with it
    - Last option not first → if it fails, nothing left

Soft-Tissue Problems Ask AI

- Analyse each problem → know how to target it
- **Capsular laxity**
  - E.g., subtle instability, ligament laxity
  - Lax capsule only; labrum not avulsed from glenoid rim
  - Lax/redundant capsule → larger joint volume → unstable
- **Labral tear**
- **Inferior glenohumeral ligament tear** without labral tear
  - Tears from glenoid side or humeral side
- **Deficient capsulolabral complex**
  - Generalized ligament laxity
  - Failed **Bankart repair** once or twice → nothing left to repair
  - → soft tissues cannot be relied upon again
- **Rotator interval lesion**
  - **Rotator interval** → anterosuperior capsule between supraspinatus and subscapularis

  - Very wide → instability
  - Sign: **external rotation** more than 90° in abduction
    - → interval too wide → tighten; reduce external rotation range
- Soft-tissue scenarios
  - **Labral tear** only → repair; easy
  - **Capsular** problem → dislocates turning over in bed
    - Or when coughing; dislocations uncountable — "fifty, sixty"
    - Soft tissues unreliable → **even without bony defect**
  - **Torn subscapularis** → repair

Bony Problems Ask AI

- **Hill-Sachs defect**
  - Defect: posterolateral humeral head
  - External rotation → posterolateral head jams on anterior glenoid rim
  - Glenoid much harder → **indentation** of head
  - Indented head → catches on anterior glenoid rim → easy dislocation
- **Glenoid defect**
  - Fractured part of **anterior rim**
  - Glenoid inherently very small relative to head
  - Smaller still → bony problem; **soft-tissue solutions** cannot solve it
- **Bipolar defect**
  - Hill-Sachs + glenoid defect together
  - **Very deceptive** → each per se may not be large
  - **Additive defect** large → special handling
  - "Small + small" → together no longer small
- **Version defect** → not common
  - Humeral head normally retroverted about 30°
  - Less **retroversion** → greater anterior dislocation chance
- Images
  - Hill-Sachs defect
    - Defective posterolateral humeral head
  - **Defective glenoid**
    - Anteroinferior part eroded
    - **Probe** → distance from **bare area** (glenoid centre) to anterior and posterior edges
      - Normal → equal
      - Anterior erosion → anterior distance shorter
- **Bipolar lesion** image: Hill-Sachs (blue) + fractured anterior glenoid → much easier dislocation

Failed Surgery Ask AI

- Last group → **failed surgery**
  - Bankart repair → dislocated again
  - Bony procedure → dislocated again
- All options laid out → patient's problem → best choice

Clinical and Radiological Assessment Ask AI

Clinical Assessment Ask AI

- Full examination not covered
- **Apprehension test**
  - Seated → **crank test**
  - Supine (better) → **fulcrum test**
  - 90° abduction + external rotation → apprehensive or not
- **Apprehension in mid-abduction** → test at 45° abduction
  - Apprehensive at 45° → **grave instability**
    - High degree → significant bony defect or highly **insufficient soft tissues**
    - Either case → think hard before soft-tissue operation
    - → usually not suitable for **soft-tissue operations**
- **Generalized ligament laxity** → collagen disorder
  - Soft tissues unreliable → **bony** procedure even without bony defect

Plain X-ray Ask AI

X-ray

- Radiology follows examination
- "Anterior" dislocation = actually **anteroinferior**
  - Posterior dislocation → directly posterior
  - Reason: head must slide under **subscapularis**
  - Principle of operations fixing subscapularis → head cannot slip under → no dislocation
- **Bony Bankart lesion** → inferior edge of glenoid
- **Hill-Sachs lesion**
- Glenoid defects → **special views**

MRI and CT Ask AI

- MRI **axial cuts**
  - Hill-Sachs → posterolateral humeral head
  - **Bankart lesion** → small black triangle avulsed away from anterior rim
- MRI **oblique sagittal** → glenoid shape
  - Normal → **pear** ("comma-shaped"): narrow above, wide below
  - Anteroinferior erosion → same width above and below
  - Large erosion → wider above than below → **inverted pear**
- MRI **programs** → measure bony defect accurately
- **CT** → classically better than MRI for bone
  - Little relied on now → MRI suffices
- (Audio cut → X-ray and MRI repeated)
- X-ray in dislocated patient
  - Bankart lesion visible if **bony avulsion**
  - Hill-Sachs = **posterolateral impression fracture**
  - **Bernageau view** → glenoid defect; usually MRI enough
- MRI axial
  - Hill-Sachs posterolateral; avulsed anterior labrum (black triangle)
  - Oblique sagittal: pear → equal width (**anteroinferior erosion**) → inverted pear (superiorly wider)

- MRI: axial → Bankart + Hill-Sachs; oblique sagittal → **glenoid defect**
- Any bony defect → seen clearly on CT
  - CT axial → **Hill-Sachs lesion** clearly
  - **En face view** → glenoid end-on, like a circle → glenoid defect size
- MRI **software** measures bony defect → CT not necessarily needed

Modality	View	Shows
X-ray	Standard	Bony Bankart (inferior glenoid edge); Hill-Sachs
X-ray	Special (Bernageau)	Glenoid defect
MRI	Axial	Hill-Sachs; Bankart triangle
MRI	Oblique sagittal	Pear → equal width → inverted pear
CT	Axial	Hill-Sachs
CT	En face	Glenoid defect size

Operative Options and Scenarios Ask AI

Available Operations Ask AI

- Review → which operation suits which patient
- **Capsulolabral repair** → capsule, labrum, or both
  - Avulsed labrum sutured back to glenoid rim → **Bankart repair**
  - Overstretched, wide capsule tightened → **capsular plication**
  - Either alone, or both together
- **Subscapularis plication procedures** → formerly done
  - Subscapularis cut → sutured **double-breasted** (overlapped)
  - **Putti-Platt**
  - **Magnuson-Stack**
  - **Obsolete**
    - Sutured with arm in internal rotation → complete loss of external rotation
    - → **posterior subluxation** → arthritic changes
    - Later overstretching → subscapularis damaged
    - → no longer acceptable
- **Glenoid augmentation** (bony procedures)
  - Small glenoid → **bone graft** fixed anteriorly → larger glenoid
  - Graft on anteroinferior aspect
    - Anterior, lower half of glenoid
  - Most famous → **Latarjet**, Bristow

Matching Scenarios to Operations Ask AI

- Scenario 1: **subtle instability** → overstretched capsule on examination
- Scenario 2: **capsulolabral deficiency** → more than one failed soft-tissue operation
- Scenario 3: **simple Bankart lesion** → handball player, arm caught; no bony defect

- Scenario 4: **rotator interval lesion** → external rotation in abduction more than 90°

Scenario	Problem	Operation
Subtle instability	Large, too wide, overstretched capsule	Capsular plication
Capsulolabral deficiency	1–2 failed soft-tissue operations → no capsule	Bony operation
Simple Bankart lesion	Labral avulsion only	Bankart repair; high-contact sport → bony procedure
Rotator interval lesion	Interval too wide	Rotator interval closure

- High-contact sport → high Bankart failure chance

Labral Lesions and Terminology Ask AI

Bankart, ALPSA and GLAD Ask AI

Labrum

- **Bankart lesion** → labrum avulsed from anterior glenoid rim
  - Labrum deepens glenoid about 50% → as if 50% of glenoid lost
  - Occurs in **acute** injury
- **ALPSA** → Anterior Labral Periosteal Sleeve Avulsion
  - Recurrent dislocations → labrum slides medially on glenoid neck
  - Heals in medialized position → no longer deepens glenoid
  - Every patient starts as Bankart → later turns **ALPSA**
  - Surgery after 2–3 years of recurrence → labrum fixed medially, not loose
- MRI: Bankart → small **triangle** (arrow)
  - Labrum avulsed from anterior glenoid neck
- MRI: ALPSA → triangle slid medially → adherent to **periosteum** of glenoid neck
  - Terminology liked by examiners
- **GLAD lesion** → sometimes
  - Avulsing labrum takes piece of glenoid **articular cartilage**
  - Glenoid Labrum Articular Disruption
- All → glenoid labrum disruption → same treatment: labrum back to its place
- (Audio lag → important, difficult section repeated)
- Bankart → not remaining loose
  - Labrum slides medially, adheres to glenoid neck → no deepening
  - **ALPSA** = chronic form of Bankart lesion
- Bankart or ALPSA → labrum nonfunctional → patient **dislocates**
- GLAD → red arrows on image

Lesion	Timing	Pathology	MRI
Bankart	Acute	Labrum avulsed from anterior glenoid rim; loose	Small triangle separated from anterior rim

Lesion	Timing	Pathology	MRI
ALPSA	Chronic, after recurrence	Labrum slid medially; adherent to glenoid neck periosteum	Triangle displaced medially on rim/neck
GLAD	—	Labral avulsion + piece of articular cartilage	Red arrows

- Functionally identical → treated as **one entity**

Exam Pearl

- **ALPSA** = Anterior Labral Periosteal Sleeve Avulsion
- **GLAD** = Glenoid Labrum Articular Disruption
- ALPSA = chronic Bankart
- All three → return labrum to its correct place

Bankart, Reverse Bankart and SLAP Ask AI

Terms

- Two-minute break → heavy terminology
- Dr. Atef's questions
  - Bankart → exact site of labral tear?
  - Reverse Bankart?
  - **SLAP lesion**?
- (Cameras off → sound cutting out)
- (Server overload → all cameras off)
- Labrum attached all around glenoid; anterior part relevant here
- Labrum deepens glenoid about 50% → avulsion → advantage lost → **anterior shoulder instability**
- Anterior labral avulsion → chronic form: ALPSA
- (Sound check with Dr. Atef → fine)
- **SLAP** = Superior Labrum Anterior to Posterior
- **Long head of biceps** → attached mainly to superior labrum
  - Not mainly to supraglenoid tubercle, as taught in anatomy

Lesion	Site	Consequence
Bankart	Anterior glenoid labrum avulsion	Anterior instability
Reverse Bankart	Same injury, posterior labrum	Posterior instability
SLAP	Superior labrum tear with long head of biceps origin	SLAP injury

Bankart Repair and First-Time Dislocation Ask AI

Immobilization After First-Time Dislocation Ask AI

- Question (Dr. Atef): immobilization period after reduced **first-time dislocation**?
  - Great variability among surgeons
  - One school → period has no effect on recurrence

- **No general agreement**
- Somewhat agreed → younger patient → longer immobilization

Age	Immobilization
Under 20 years	One month
20–30 years	3 weeks
Over 30 years	2 weeks

- Position → **neutral rotation** (hand forward) better than sling in internal rotation
- Labral avulsion healing in good position → chance minimal
  - Trend → immobilization makes little difference
  - Bankart persists regardless of **conservative management**
  - Recurrence → depends on age + activity level
- More than one dislocation → immobilize first two days only, until pain subsides
  - Fixed Bankart → turned **ALPSA** → sling pointless

Patient Selection Ask AI

Bankart

- Labral injury → acute loose Bankart, or medialized adherent labrum
  - (= ALPSA) → **Bankart repair**
- Requires **proper patient selection**
  - No **significant bony defects**
  - Good soft tissues
    - No generalized ligament laxity
    - No previous Bankart
    - Not 30–40 dislocations
  - No **contact sport** — boxing, wrestling
    - → Bankart repair usually fails
- **Ideal patient**: 25-year-old, two dislocations, no bony defect
- Investigations → Bankart lesion; no large Hill-Sachs
  - MRI sagittal → **pear-shaped** glenoid; superior narrower than inferior
  - No obvious humeral head or glenoid defect
  - **Arthroscopy** → separation between labrum and glenoid
- → ideal patient for Bankart repair

Main Steps Ask AI

Bankart

- **Diagnostic arthroscopy**
- **Labral release**
  - Unless first-time dislocator → labrum adherent medial to normal position
  - Release (mobilize) until free → pull back to normal place
  - Stitch in current position → no movement; already attached
  - Freed from glenoid neck → pulled laterally to glenoid rim
- **Freshening of glenoid neck**
  - Glenoid rim + neck → **raw surface** for labral healing
- Release → freshening
- **Repair** → anchor in glenoid bone; sutures through labrum

- Three steps → release, freshening, repair
- Images
  - **Yellow instrument** → plane between labrum and glenoid → complete separation
  - **Rasp** (or burr) → abrade glenoid neck
  - Anchor in glenoid bone → sutures through labrum
  - Sutures tied
- Final view → labrum seated on anterior glenoid rim
  - **Bumper effect** → buffer against head dislocation

HAGL Lesion Ask AI

HAGL

- Last capsular lesion
- **HAGL** = Humeral Avulsion of the Glenohumeral Ligament
- Ligaments torn from humeral side, not glenoid side
- Anchoring capsule and labrum to glenoid → pointless
- **Very deceptive**
- Humeral-side repair → quite inferior results → usually **bony procedure**

First-Time Dislocation: Operate or Not? Ask AI

- Depends on two things → **patient's age** + type of sport
  - Younger → more likely **recurrent dislocator**
  - Contact sport → more likely recurrent dislocator
- Conservative treatment → recurrence approximately four times that of **surgery**
- **Young patient** under 20 + contact or overhead sport → operate
- Otherwise → wait until recurrent, then operate

Must-Know

- Under 20 + contact or overhead sport → operate after **first-time dislocation**
- Conservative recurrence ≈ four times surgical

Bony Defects: Hill-Sachs and Glenoid Ask AI

- Soft-tissue lesions done → **bony defects**

Engaging and Off-Track Ask AI

Hill-Sachs

- **Hill-Sachs** → posterolateral head impression/indentation; catches on anterior glenoid rim
- Arthroscopy: abduction + external rotation → defect caught on anterior rim
  - → head dislocates in front of us
  - Catching = **engagement** → engaging lesion
- "Engage" → little used now
- (Repeated at Dr. Atef's request)
  - External rotation → defect reaches anterior rim, catches → **engaging lesion**
  - Very small → never reaches anterior rim
    - → **non-engaging lesion**
- Terms replaced
  - Stays inside glenoid at any external rotation → **on-track** (on its glenoid track)
  - Reaches anterior glenoid border, catches, may dislocate

        - → **off-track lesion** → left glenoid track → dislocation
- Radiological method → exact on/off-track calculation; details not covered
    - Small → no difference; never catches
    - Off-track → catches → **dislocation**
- Engaging Hill-Sachs → goes off track, catches, dislocates
    - → **needs management**
    - Bankart repair + large off-track Hill-Sachs left → **Bankart fails**

Remplissage Ask AI

Hill-Sachs

- One option for Hill-Sachs
- **Remplissage** = filling a defect
- Posterolateral filler → **infraspinatus**
- Anchor in defect → muscle sutured into defect
- Significant Hill-Sachs + Bankart repair
    - → add remplissage in the **same operation**

Glenoid Defect Ask AI

Glenoid

- More serious
- **Glenoid defect**
    - Torn labrum + small glenoid + deficient anterior part → very small glenoid
    - → head dislocates with utmost ease
- **Acute fracture** → fix fractured fragment; easy
- **Recurrent dislocation** + eroded glenoid on imaging (previous image, far right)
    - Only solution → **enlarge the glenoid**: compensate anteroinferior deficiency
    - Bone graft → fixed anteriorly, inferior half of glenoid
- Main indication for **bony procedures** → glenoid defect
- Final appearance → enlarged glenoid → head never leaves it
- Engaging Hill-Sachs → anterior rim moved further away
    - Catches on original glenoid → cannot reach end of graft
    - → used **even for Hill-Sachs defects**
- Radiological measurement methods → X-ray and CT
- Agreed: defect more than 20% of glenoid width → **bony procedure**
    - Bankart repair → fails
- Measurement methods on X-ray
    - And on CT → details not covered
- → bony procedure; famous options follow

High Yield

- Glenoid defect more than 20% of width → **bony procedure**
- Bankart repair in this case → fails
- Main indication for bony procedures → glenoid defect

Latarjet Procedure and Decision-Making Ask AI

Latarjet Concept Ask AI

Latarjet

- In current guidelines → **Latarjet procedure**
- **Coracoid osteotomy** → junction of vertical and horizontal parts
  - Whole horizontal part taken, with **conjoined tendon** attached
- Bone + tendon → **subscapularis split** → inferior half of glenoid neck
- → glenoid enlarged
  - Anteroinferiorly
- (Dr. Osama: difference between the two operations)
- **Coracoid transfer** → classically two operations
  - Bristow → tip only; almost no longer performed
  - Latarjet → large piece, most of horizontal part

	Bristow	Latarjet
Coracoid taken	Tip only	Most of horizontal part
Current use	Almost no longer performed	In guidelines

Mechanisms of Action Ask AI

Latarjet

- Principle → enlarge the glenoid
  - Glenoid defect → compensated
  - **Engaging Hill-Sachs** → anterior rim moved further; never reached in external rotation
- Graft through subscapularis split → inferior half of glenoid
  - Larger glenoid → **functional glenoid arc**
- Graft passes through mid-subscapularis; **conjoined tendon** descends from it
  - Anteroinferior dislocation → head must slip under subscapularis
  - Conjoined tendon pulls inferior half of subscapularis down → head cannot slip under
  - → **sling effect**
- Latarjet → two mechanisms
  - Most important → larger **functional glenoid arc**
  - **Sling effect** → inferior subscapularis fixed → head cannot slide below

Off-Track Hill-Sachs: Remplissage or Latarjet Ask AI

- Hill-Sachs → catches on anterior glenoid rim (arthroscopic image, right)
  - Anterior rim lengthened (more anterior) → engaging lesion no longer engages
  - → **glenoid bone graft** treats glenoid defect + humeral head defect
- Engaging (now **off-track**) Hill-Sachs → two options
  - **Bankart repair** (three glenoid anchors) + infraspinatus fills defect (large humeral anchor)
  - Hill-Sachs left as is → glenoid enlarged → head cannot reach anterior rim
- → Bankart repair with **remplissage**, or Latarjet

Decision Rules Ask AI

Bipolar

- Bankart repair
  - Significant glenoid defect → **fails**

  - Hill-Sachs, Bankart alone → fails
  - Hill-Sachs, with remplissage → satisfactory; or Latarjet
- **Bipolar defect** → humeral head defect + glenoid defect
  - Low threshold for Latarjet → decided very easily
  - High threshold for Bankart → decided with great difficulty
    - Study very carefully, **even if both defects small**
  - Latarjet → chosen with utmost ease

Bipolar scenario	Decision
Both defects small	Latarjet may succeed; Bankart may succeed too
Glenoid defect significant	Latarjet — must
Hill-Sachs more significant	Bankart + remplissage possible; Latarjet favoured
Both significant	Latarjet — must

- Latarjet → works for **all bipolar lesions**; Bankart → very selected case

High Yield
- Bipolar defect → **high threshold for Bankart**, low threshold for Latarjet
- Bipolar defect → combined effect large even if each small

Locked Anterior Dislocation Ask AI

Locked

- Not common, but seen
- E.g., years of **tramadol** abuse → head locked on anterior glenoid rim
  - Significant impression fracture
- **No closed reduction** → glenoid digs further each attempt → head fractures
  - → open reduction
- Open reduction → **very large Hill-Sachs**
- Fill with **bone graft**, or same principle
  - Large Hill-Sachs left → **coracoid transfer** (Latarjet)

Generalized Ligament Laxity Ask AI

Laxity

- Hyperextended elbows + knees + **recurrent shoulder dislocation**, no bony defect
- Problem → very poor **soft tissues**
- Bankart repair → recurrence 75%
- Even without bony defect — usually none
  - Head needn't engage on glenoid to dislocate
  - Very lax capsule → dislocates without bones catching
- Fail soft-tissue operations → need **bony operation** (Latarjet)

Latarjet: Technique, Indications and Failure Ask AI

Graft Orientation and Surgical Steps Ask AI

Latarjet

- (Dr. Osama asked for details)
- Graft → most of horizontal coracoid + conjoined tendon

- **Classic Latarjet** → inferior surface (border) of coracoid on glenoid neck
- **Congruent arc** (most surgeons now) → medial border of coracoid on glenoid neck
- **Approach** → small incision centred on coracoid
- Exposed: coracoid, conjoined tendon, **coracoacromial ligament**
- Medial border → **pectoralis minor**
  - Cut, with coracoacromial ligament, before osteotomy
- **Coracoid osteotomy** → fixed on glenoid neck after subscapularis split
- Subscapularis **split**, not detached (elevated) → preserves sling effect
- Very small incision ≈ a **saw blade's length**
- Medial border chosen → before osteotomy
  - Pectoralis minor cut off medial border
  - Saw → **freshening** of medial border → raw surface → healing
  - Medial border → laid on glenoid neck
  - Room for **two screws** — 4 mm
  - Drill holes far apart → holes must not join
- Subscapularis split → glenoid neck exposed → **rasped** for healing
- Bone graft fixed with two screws
- Graft **flush** with glenoid surface; never lateral
  - Not a buttress blocking the head
  - Buttress → high friction → **arthritis** very soon
  - Never more protruding laterally
- X-ray → graft at lower glenoid; two screws
- Medial border fixation → graft takes glenoid **curvature**
  - Better than inferior border fixation

Pitfall

- Lateral graft → **buttress** → friction → early arthritis
- Graft flush with glenoid surface, inferior half

Operative Video Ask AI

Latarjet

- Four-minute video
- Incision directly over coracoid → not classic **deltopectoral approach**
- Coracoid palpated → **4 cm approach**; proximal end on coracoid
- Dissection → **conjoined tendon**: attached at coracoid tip, runs inferiorly
- Muscle detached from medial border → **pectoralis minor**
- Pectoralis minor detached → **osteotomy** begins
- Significant coracoid piece + conjoined tendon
- Room for two holes → two 4 mm screws
  - Two screws → no **rotation**; fixed well on glenoid neck
- **Subscapularis split**, not detached
  - Middle, or two-thirds above / one-third below
  - Not higher → greatly affects sling effect
- Split → joint seen → glenoid neck prepared (dark field)
- Holes drilled in glenoid neck → graft in inferior half
  - Flush with glenoid surface; not **overriding laterally** the glenoid edge

- Usually one hole → one screw → then second screw
  - Second drilling **easier** after first screw
- Working through split → subscapularis intact → **sling effect** working
- **Screw length** measured → screw fixed → Latarjet done

Indications for Latarjet Ask AI

Latarjet

- **Glenoid defect** → classic indication
- **Bipolar defect** → glenoid defect + Hill-Sachs
- **Failed Bankart** (or failed plication) → very poor soft tissues
- **HAGL lesion** → avulsed from humeral side; suturing to glenoid pointless
- **Hill-Sachs defect** → Bankart with remplissage, or Latarjet
- **Locked dislocation**
- **Generalized ligament laxity** → defective soft tissues; repair illogical
- **ISIS** higher than 3

Instability Severity Index Score Ask AI

ISIS

- **Instability Severity Index Score** → points for
  - Patient's age
  - Contact sport
  - Generalized ligament laxity
  - Hill-Sachs defect size
  - Glenoid defect size
- Previously → score more than 6 → **Latarjet**
- Now → more than 3 → Latarjet
  - E.g., one factor + contact sport → already needs it
- More factors → further **away from Bankart**
- **Modification** → measured by CT instead of X-ray
- **Arthroscopic Latarjet** → long learning curve

Failed Bankart and Failed Latarjet Ask AI

- **Failed Bankart** → soft tissues unreliable → Latarjet
- **Failed Latarjet** → a problem
  - Straightforward patient → no "Latarjet for everyone"
  - Leave a **line of retreat**
  - No bony defect, no laxity, 3–4 dislocations → Bankart
- Failed Latarjet case, dislocating
  - **Coracoid graft** completely resorbed; screws standing alone
  - No place for soft-tissue operation → another bony operation
- **Eden-Hybinette** → tricortical iliac crest graft
  - Larger than coracoid → functional glenoid arc increased considerably
  - No **sling effect** (no tendon attached) → not important
  - Stability results → excellent
  - High **arthritis** rate → 25% of patients
  - Done only if forced
- Failed Latarjet → **iliac crest graft**

Glenoid bony augmentation	Status
Coracoid — Latarjet	Gold standard
Bristow	No longer done
Iliac crest graft (Eden-Hybinette)	Salvage for failed Latarjet; high arthritis rate

Take-Home Messages Ask AI

- **Bankart repair ± capsular plication** → properly selected patients
  - No significant bony defects
  - No deficient soft tissues
- Sizable Hill-Sachs (engaging → now **off-track**) → Bankart + remplissage, or Latarjet
- **Latarjet**
  - Classic → glenoid defect more than 20%
  - Bipolar defect, **even if both small** → combined effect likely large
  - Deficient soft tissue → generalized ligament laxity or failed soft-tissue operation
  - **High-contact sports** — wrestling, judo, boxing → Latarjet much better than Bankart
  - Score higher than 3
- Why not Latarjet for all patients? (some schools abroad do)
  - Salvage of failed Latarjet → iliac crest → **poor results** (arthritis)
  - Good-Bankart category → no need to exhaust patient's chances
  - Properly selected patients → Bankart
- Everything fails → **iliac crest graft** as last option
- (End of lecture)

Questions and Answers: Diagnosis and Decision-Making Ask AI

Announcements Ask AI

- Dr. Atef: valuable lecture; difficult topic simplified well → questions begin
- (Dr. Osama asks Dr. Maged to switch on video)
- Saturday (day after tomorrow), 9 o'clock Cairo time → **Prof. Dr. Ahmed Naeem**
  - Professor of Hand Surgery, Ain Shams University
  - Topic: **scaphoid fractures**
- Lectures **recorded** → YouTube or Facebook page
- (Dr. Atef begins while Dr. Maged's audio connects)

Knee versus Shoulder Arthroscopy Ask AI

- Questions → reflect many colleagues' minds
- Question (Dr. Osama → Dr. Atef): **knee arthroscopy** or shoulder — which more difficult?
- Answer → depends on **learning curve**
  - 1000 knee + 100 shoulder arthroscopies → knee much easier
  - Sometimes shoulder seems easier to work in
- (Dr. Maged's video on)

Primary Bankart Repair After First Dislocation Ask AI

- (Audio muted, then restored)
- Question: **primary repair** of Bankart after first dislocation in young patients?
- Yes → prefers repair, especially in sports players
  - Sports → recurrence → worse results → later ALPSA and bony defects
  - No need to wait for all this
  - Especially contact or overhead sports
  - Younger than 20 → first-time dislocation → **Bankart repair**

Subtle Instability versus Impingement Ask AI

Subtle

- Question: how to differentiate?
- Young patient labelled **subacromial impingement** → wrong diagnosis; never at this age
- Painful arc cause → head **rides proximally** on arm raising (very lax capsule)
  - Not a narrow subacromial space
- **Push-pull test**
  - One hand at **wrist** → pulls arm forward
  - Other hand fixed on **acromion** → pushes opposite
  - Same complained pain → subtle instability

> **Clinical Pearl**
> - Swimmer, volleyball player, repeated overhead activity + painful arc → **subtle instability**
> - No impingement patients at this young age
> - Push-pull test reproduces the pain

Traumatic versus Atraumatic: TUBS and AMBRII Ask AI

- (Thanks from attendees relayed; Dr. Maged thanks Dr. Atef)
- Question (Dr. Atef): classification of traumatic vs atraumatic dislocation
- Traumatic/atraumatic terms (and acronyms) → superseded
- Atraumatic → now **generalized ligament laxity** patients
  - Overstretched capsule
  - Dislocate without significant trauma, with certain movement
  - Usual problem → **deficient soft tissues**
- Author → **Matsen classification**

	TUBS	AMBRII
Mechanism	Traumatic	Atraumatic
Pattern	Unilateral	Multidirectional, bilateral
Lesion	Bankart lesion	Ligament laxity
Treatment	Surgery	Rehabilitation; surgery → inferior capsular shift

- AMBRII patient → the **ligament laxity** patient
- Overhead athlete → **rehabilitation** usually fails / not enough
  - → **capsular plication**

  - Still a risk → capsule **restretches** over years

Risk Factors for Recurrence Ask AI

Recurrence

- Question summarizing the lecture: **risk factors for recurrence**?
- **Age** → younger → higher risk
  - Under 20 years → over 80%; some say over 90%
  - Older → risk decreases
- **Bony defect** from first traumatic dislocation → very high risk
  - Off-track Hill-Sachs
  - Significant glenoid defect
- **Generalized ligament laxity** → very high risk
- **Contact sport** → high risk
  - Boxers, judo players (opponent pulls arm to throw)

Questions and Answers: Measurement and Technique Ask AI

Treatment of Subtle Instability Ask AI

Subtle

- Complaint → pain, not subluxation or dislocation
- First → **rehabilitation**, two main focuses
  - **Posterior capsular stretches**
    - → prevents the **SLAP injury**
  - **Subscapularis** strengthening → anterior support
- Rehabilitation fails, pain persists → large capsular volume, overstretched capsule
  - → **arthroscopic capsular plication**

Intraoperative Measurement of Glenoid Defect Ask AI

Glenoid

- Question: **glenoid defect** measured intraoperatively?
- Radiology → many methods
- Intraoperatively → view from **posterior portal**
- **Bare area** → centre of glenoid; not covered by cartilage
- Normal, inferior half → bare area to posterior edge
  - ≈ bare area to anterior edge → exactly in the middle
- **Probe** measurement → longer posteriorly than anteriorly → **anterior erosion**
- Described method → in real practice, likely unreliable
  - Posterior distance usually appears longer anyway

HAGL Repair versus Latarjet Ask AI

HAGL

- Question: HAGL repair before Latarjet?
- **HAGL repair** → two problems
  - Technically much more difficult than Bankart repair
  - Even in experienced hands → **high recurrence rate**
- Advice → Latarjet; much better results

Order of Remplissage and Bankart Ask AI

- Question: remplissage + Bankart → which first?
- Technical point

  - Anchor in **Hill-Sachs defect** → sutures passed, not tied
  - Bankart completed → remplissage sutures tied **blindly** at the end
- Infraspinatus tied first → space tightened → Bankart **more difficult**
  - And remplissage spoiled
- Easiest while joint still **spacious**
  - Anchor in defect → sutures through infraspinatus, untied
  - Whole Bankart repair → then sutures tied blindly from outside

Diagnosis of HAGL Ask AI

HAGL

- Question: HAGL diagnosis?
- MRI **coronal cuts**
  - Sign related to **U-shaped axillary pouch** → pouch seen descending
  - Torn from humeral side → clear on coronal cuts

On-Track versus Off-Track: Radiological Assessment Ask AI

Track

- Important; many questions
- Somewhat difficult → simplified
- **Glenoid track** → contact area between humeral head and glenoid
- Range of motion to full external rotation → head contacts 83% of glenoid
- Measure **inferior glenoid width**
  - Sagittal en face cuts → circle drawn → width measured
  - E.g., 30 mm width → 83% ≈ 25 mm *[C1 · transcript: "83% of 30 mm = 26 mm"]*
  - → glenoid track ≈ 25 mm *[C2 · transcript: "glenoid track 26 mm"]*
- Axial cuts → **Hill-Sachs** width in mm
  - Whole impression fracture + **bone bridge** to rotator cuff insertion
  - E.g., 22 mm → smaller than track → on-track
  - E.g., 28 mm vs 25 mm track
    - Hill-Sachs 28–29 mm → larger than track → off-track
- Measured by **software**, not on printed images
- **Radiologist** → measures width, takes 83%, measures Hill-Sachs, reports

	On-track	Off-track
Hill-Sachs vs glenoid track	Smaller	Larger
Example (track 25 mm)	22 mm	28–29 mm
Behaviour	Does not catch	Catches → dislocates
Management	Bankart enough	Remplissage or Latarjet

High Yield

- **Glenoid track** = 83% of inferior glenoid width
- Hill-Sachs larger than track → off-track → remplissage or Latarjet

- In practice, borderline Hill-Sachs, not measured → two options
  - Proceed to **Latarjet** directly → preferred
  - Arthroscopy → maximum external rotation → see if defect catches

- Catches → **remplissage**
- No catch → Bankart alone

Questions and Answers: Special Situations and Rehabilitation Ask AI

Epileptic Patient with Recurrent Dislocation Ask AI

- Frequently asked: **epileptic patient** with recurrent dislocation
  - Management?
- Must be **fit-free** before surgery
  - Newer recommendations → at least six months

Habitual Dislocators Ask AI

Habitual

- Question (Dr. Atef): **habitual dislocation** → dislocate and reduce as if normal
- Answer → leave them
  - Described management → likely home exercises only

Goalkeepers and Handball Players Ask AI

- Question: football goalkeeper, handball or volleyball player → chance after treatment and rehabilitation?
- Better chance
- **Football goalkeeper** → higher chance with Latarjet
- **Handball player** → depends on age
  - Under 20, no bony defect → **Bankart**
    - Latarjet might fail later → all chances exhausted
  - Late twenties or early thirties → **Latarjet** directly

MRI After First-Time Dislocation Ask AI

- Question: young first-time dislocator → MRI work-up, or wait
  - Content with X-ray, **sling**?
- Depends → athletic or not
  - **Contact sport** → imaging, certainly
    - Bony defect → must operate
    - No bony defect, contact or overhead sport → better to operate; less mandatory
  - Non-athlete, no **violent trauma** in history
    - → **conservative** management as usual
- Investigation → must build a **decision** on it
  - No operation decision anyway → no imaging first time

Proximal Humeral Osteotomy for Hill-Sachs Ask AI

- (Many questions; colleagues very interactive)
- Question: role of **proximal humeral osteotomy** in Hill-Sachs?
- Described mainly → **decreased humeral retroversion** (classically)
- More retroversion → Hill-Sachs further from anterior rim
  - But may cause **posterior subluxation**
- Hill-Sachs options → remplissage, Latarjet, or **bone graft** to fill defect
  - Not version change
- Exception → very large Hill-Sachs, more than 40% of head

  - **Hemiarthroplasty** decided → version changed with it
  - Very odd indication
- Otherwise → not done

Postoperative Rehabilitation Ask AI

Rehab

- Last question: **postoperative rehabilitation** → can spoil good operations
- Avoid **shoulder stiffness**
- Protect the repair
- Latarjet → much more **accelerated rehabilitation** than Bankart
  - Many Bankart concerns absent in Latarjet
- **After Bankart repair**
  - **Sling** for four weeks
  - From second day → **pendulum exercises**: forward flexion, circular movements
  - First month → passive and assisted-active range of motion
    - **Extension** forbidden
    - External rotation in abduction forbidden
  - From 3–4 weeks → **deltoid strengthening**
  - Afterwards → **coordination exercises**
- **After Latarjet**
  - Same protocol; much less concern about extension and abduction–external rotation
  - Bankart → movement may cut **anchor sutures**
  - Latarjet → movement won't break graft screws
  - → range of motion much faster

	Bankart repair	Latarjet
Pace	Standard	Much more accelerated
Extension, abduction–external rotation	Forbidden first month	Much less concern
Movement risk	Cuts anchor sutures	Won't break graft screws

Neglected Dislocation Ask AI

Neglected

- Final question: **neglected dislocation** → management?
- (Colleagues invited to suggest future topics in chat)
- **Neglected anterior dislocation** → very uncommon; easy to diagnose
- Neglected non-locked → arm hanging → usually **reduces by itself**
- → neglected non-locked anterior dislocation highly uncommon
- Management → two things
  - **Open reduction**
  - **Bony procedure** + **capsular shift**
    - Capsule overstretched → long time anteroinferior
    - Cannot rely on capsule alone → **augment** with bony procedure

Closing Ask AI

- Dr. Atef: questions covered the topic; topics already arriving
- Dr. Maged → upcoming lectures

  - Colleagues ask about related topics → he will be informed
- Thanks → Dr. Maged, Dr. Osama, **Elixir** company
- Dr. Maged: an honour
- Reminder: **Dr. Ahmed Naeem** → Saturday, 9 o'clock Cairo time
  - Room opens before start
  - Lecture ≈ 45 minutes to one hour; chat ≈ half an hour
  - No overlap with **Nasr City (Medinet Nasr) Hospital** lectures at 10:30
- Final thanks → Prof. Dr. Atef El-Beltagy, Prof. Dr. Maged Samy, attendees
- See you Saturday; happy new year

Knee Menisci

(All You Need to Know)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 97) As spoken; classic Apley: pain on distraction suggests ligament (collateral) injury, not chondral injury.

Objectives and Meniscal Function Ask AI

- Lecture objectives
  - **Meniscectomy**
  - **Meniscal repair**
  - **Degenerative meniscal tears** — among commonest in practice; much malpractice
  - Newer topics → **meniscal root tears** and ramp lesions
  - **Discoid meniscus** — brief

Anatomy Ask AI

- Two menisci → medial and lateral
- Lateral meniscus → anterior and posterior horns **closer together**
- **Ligament of Wrisberg** → arises from posterior horn of lateral meniscus
  - Close association with PCL
  - Injury → **hypermobility** of posterior horn of lateral meniscus

Functions Ask AI

- **Load transmission** and shock absorption
  - Femoral condyle = ball; **tibial plateau** = flat surface
  - Ball on flat floor → **point contact**
  - Point shifts with flexion → still a point → high cartilage wear
  - Meniscus → ↑ **surface area** of contact
- → **Load distribution** on cartilage
- Second role → **joint stability**
  - **Meniscectomized knee** → stability affected
  - Instability + meniscal injury with ligament injury → repair meniscus → **adds to stability**
  - Not merely load distribution and **OA prevention**

High Yield

- Main concept → **menisci contribute to joint stability**
- Instability with meniscal + ligament injury → consider meniscal repair

Microstructure, Vascular Zones, and Tear Patterns Ask AI

Collagen Orientation Ask AI

- Periphery (**meniscosynovial junction**) → inwards: fibre orientation changes
  - Peripheral → **circumferential** fibres
  - Central → **radial** fibres
- **Blood vessels** → all from periphery
  - Central part → no blood supply
- Picture **collagen orientation** when planning any repair
- Aim → **anti-rip** stitch: resists cutting through and pulling out
- Stitch direction vs **fibre orientation** at that site → decides ripping risk

Vascular Zones Ask AI

- More peripheral → more blood supply

Zone	Location	Blood supply	Healing potential
Red-red	Outermost	Most	Greatest
Red-white	Slightly inwards	Less	Lower
White-white	Central	None	Much lower

Tear Patterns Ask AI

- **Longitudinal tear** → runs along length of meniscus
- **Radial tear** → cuts across the meniscal circle
  - May extend all the way to periphery
  - Central radial tear → wholly avascular → **very poor healing**
- Plane of tear
  - **Vertical tear** → superior → inferior
  - **Horizontal tear** → upper + lower leaflet; opens as **fish-mouth**
- Combined terms
  - **Longitudinal vertical**
  - **Radial vertical** — still cuts superior → inferior
  - **Horizontal cleavage** → upper and lower leaflets
- Whole fragment displaced into knee → **bucket handle tear**

MRI Diagnosis of Meniscal Tears Ask AI

Feature	Medial	Lateral
Plateau shape	Concave (more or less)	Convex
Triangular plateau in any cut	Definitely medial	—
Fibular head visible	—	Lateral

- Sagittal → anterior horn and posterior horn
- Intrasubstance signal (whiter) → **meniscal degeneration** only; no significance
  - **Not** an indication for surgery
  - Not a meniscal tear
  - Posterior intrasubstance signal crossing → still not a tear
- **Meniscal tear** → bright signal cuts upper or lower articular surface, or both
- Line cutting top → bottom → tear (example: medial)
- **Double PCL sign** → second PCL-like structure anterior to PCL
  - → **displaced bucket handle** flipped into notch
- Sagittal → look for **interruption** of meniscus: upper, lower, or both surfaces
- Meniscus absent over successive cuts
  - Posterior horn entirely absent → two possibilities
    - Previous **meniscectomy** → evident from surgical history

        - **Bucket handle** completely displaced → flipped into notch
    - Meniscus cannot simply be absent → tear flipped out of place
    - **Coronal** → meniscus found in notch → displaced bucket handle tear

High Yield

- Tear only if signal **cuts an articular surface** — upper, lower, or both
- Intrasubstance signal → degeneration; no surgery
- Double PCL sign → displaced bucket handle in notch
- Absent posterior horn, no prior surgery → search **coronal** images for notch fragment

Clinical Presentation and Examination Ask AI

Symptoms Ask AI

- Symptoms vary → isolated tear vs **ligament injury**
    - Associated ligament injury → **instability**, giving way may dominate
    - Isolated tear → **pain**
- Pain site → **medial** or lateral joint line
- **Recurrent effusion**
- **Mechanical symptoms** → something catches, then releases
- **Locking** → frank bucket handle tear
    - Requires careful patient description; "catching" ≠ locking

	OA stiffness	True locking
Pattern	Stiff after sitting; keeps straightening until it straightens	Cannot extend beyond 10–20° short of full extension
Release	Gradual	Back to full flexion + rotation → releases displaced fragment

- Clear history required before labelling **locking**

Examination: McMurray and Apley Tests Ask AI

- Two best-known tests → **McMurray** and Apley grinding
- **McMurray test**
    - Knee flexed
    - **External rotation** → medial meniscus
    - **Internal rotation** → lateral meniscus
    - Fingers on **joint line**
    - Extend knee → feel meniscus under hand
    - Click / popping → **unstable tear**
        - Usually large **flap tear** or bucket handle
    - Not the rule in **medial tears** → most have negative McMurray
    - Positive finding → **medial joint line tenderness** only
- **Apley grinding test**
    - Prone; knee flexed 90°
    - Rotation with **compression**, then with distraction from ankle

Pain on rotation	Diagnosis
With compression and distraction	Chondral injury *[Q1 · As spoken; classic Apley: pain on distraction suggests ligament (collateral) injury, not chondral injury.]*
With compression only	Meniscal tear

Exam Pearl

- McMurray click → **unstable tear**: large flap or bucket handle
- Most medial meniscal tears → McMurray **negative**
- Medial tear → joint line tenderness is the positive sign
- Apley pain on compression only → meniscal tear

Meniscectomy vs Repair: Decision Principles Ask AI

- Tear confirmed → choose **meniscectomy** or repair

Principles of Meniscectomy Ask AI

- Only in **unfavourable repair situations**
- **Partial**, not total → avoids accelerated OA
- Do **not injure cartilage**
  - Avoid large instruments, repeated in-out passes
  - Scraping **femoral condyle** cartilage → destroyed by procedure end
- Whole aim of meniscal surgery → **preserve cartilage**
- Meniscectomy → only partial, only when unfavourable for repair
- Defining favourable situations first → identifies unfavourable ones

Favourable Repair Situations Ask AI

- **Young** patient; acute tear
- **Chronic repair** sometimes justified
  - ACL tear + large bucket handle, months old, good tissue → repair
  - Alternative → almost **subtotal meniscectomy** if very long bucket handle
- Ideal situations → not repairing is unacceptable
- Young, acute, **vertical**, red zone
  - Vertical + red zone → **longitudinal** = bucket handle
  - **Radial tear** always starts centrally (white zone) → never in red zone
- **Stable** tear
  - Reduced bucket handle jumps back into notch on removing instrument → **highly unstable**
- Longer tear (posterior → anterior) → ↑ **instability**
- Rule → **opposite of fractures**
  - Fractures → larger surface area (**spike fractures**) → better healing
  - Longitudinal vertical / bucket handle → **longer tear** → worse healing
  - Long tear → unstable → displaces → ↑ repair failure
- Blood supply better in **lateral meniscus**
- Concomitant **ACL reconstruction** → ↑ stability + ↑ blood supply

  - Tunnel drilling → **mesenchymal cells** → growth factors → better healing
- Best scenario → stable **bucket handle** + concomitant ACL reconstruction + young patient
- These conditions → meniscectomy = **crime** in current practice
- Best bucket handle → **peripheral**, red-red or red-white zone
  - Far commoner than **white-white** bucket handles
- Short → stable; no flipping; easy **reduction**; good tissue
- **Locking** → no reason to delay surgery
  - Repeated flipping out and back → **macerated** tissue → stitches tear through
- Lateral meniscus heals better → better **blood supply**
- Medial → definitely repaired, especially with concomitant **ACL reconstruction**

Must-Know

- Ideal repair → young, acute, **vertical longitudinal**, red zone, stable
- Longer tear → more unstable → worse healing — opposite of fractures
- Concomitant ACL reconstruction → mesenchymal cells, growth factors, stability
- Locking → operate early; delay → macerated tissue

Partial Meniscectomy: Complex and Radial Tears Ask AI

- Case 1 → right knee; **anterolateral** portal view
- **Macerated complex tear** → posterior horn medial meniscus; extends to white-white zone
- Patient 40+ (or 30+, 20+) → **not for repair**, even in red-red zone → remove
- Access without scraping cartilage
  - **Pie-crusting** → repeated needle punctures + valgus → medial compartment opens
- **Basket** → remove tissue → reach **stable rim** (fundamental principle) → test
- Remnant may open as horizontal split → **thermal welding** seals it
- Case 2 → lateral meniscus; right knee; anterolateral portal
- **Radial vertical** tear → white-white zone; entirely central → not for repair
- Radial tear reaching **meniscocapsular junction**
  - Young patient → obliged to attempt repair
  - **Saucerization** → edge made smooth with meniscal body instead of leaving radial tear
  - Saucerization to the end → **non-functional** meniscus = total meniscectomy functionally
  - Repair → peripheral part may heal → **residual meniscal function**
- Radial, complex, macerated tears → **partial meniscectomy**; not for repair
- Goal → leave enough **residual meniscal tissue**
- Each removal leads to more removal → **stop**, even without better tissue left

Unstable Bucket Handle Tear Ask AI

- **Displaced bucket handle** → right knee
- Anterolateral portal view
- Reduce with **blunt instrument** → suture once reduced
- **Very long**, from far posterior → jumps out once blunt instrument withdrawn
  - Long + high **tension** → stitching very difficult; feasible only if very small
- Jumps back + high tension → **failure** chance very high → lean towards meniscectomy

Meniscal Repair: Stitch Configuration and Techniques Ask AI

Horizontal vs Vertical Mattress Ask AI

- Any repair technique → **two stitch types**
- Vertical stitch across fibres → enters **perpendicular** to collagen → no cut-through

	Horizontal mattress	Vertical mattress
Path	Across tear, transversely	Lengthwise; one limb peripheral outside tear, other through tear
Coverage	Larger area	—
Pull-out strength	Weaker	Stronger
Number needed	Fewer	More stitches
Best site	Central tears → anti-rip	Peripheral tears

- **Central** vertical stitch → tears straight through → **ripping**
  - Central → stitch across → horizontal → anti-rip

Three Techniques Ask AI

- **Inside-out** → needles loaded with sutures
  - Portal → meniscus → skin → exit outside
- **Outside-in** (out-in) → needle through skin → meniscus → suture retrieved inside knee
- **All-inside** → implant introduced and **fired** inside knee
  - All sutures inside; nothing exits skin

Inside-Out Repair Ask AI

- Special needles inside a **tube**
  - Suture loaded into needles
  - Tubes **straight** or curved → chosen repair direction
- Enter from inside knee via **portal** towards tear
- Needles placed lengthwise → **vertical mattress**; across → horizontal mattress
- Both needles threaded → pulled through → exit **skin**
- Suture forms **loop** around meniscus → tied onto capsule from outside
- Advantages
  - **Cheap** → needle reusable
  - Needle push **reduces** bucket handle (outside-in pushes it displaced)
  - **Ideal for body tears**
- Disadvantages
  - **Anterior horn** → impossible; angle never allows from portal
  - **Posterior horn** → very risky; needle exit unknown → neurovascular bundle injury
    - **Safety incision** for needle exit → no longer common; or other technique
- Current practice → inside-out **ideal for body tears**
- Steps
  - **Freshen tear** with shaver → raw area for healing

  - Shave **meniscocapsular junction** of peripheral part before reduction
  - Tube with two needles → exit other side → stitch
  - Tightening → meniscus **flips** upwards
    - Next stitch from **under surface** → corrects flipping
  - Horizontal or vertical → as wished
  - Sutures exit through skin
  - Small skin incision → tie on **capsule** from outside
- Good in body; very posterior part → fear of **needle exit**

Clinical Pearl

- Every repair starts by **freshening** the tear with a shaver
- Tightening flips meniscus upwards → next stitches from **under surface**

Outside-In Repair Ask AI

- No special **instruments** needed
- **IV cannula** → green preferred over grey → less traumatizing to meniscus
- Choose entry site for each stitch
- Not all stitches through **same portal**
  - Example → ACL reconstruction case; bucket handle; stitches in the middle
  - Anterior body ≠ posterior body ≠ **posterior horn**
  - Posterior horn of **medial meniscus** reachable
- **Separate portals** per region; horizontal or vertical as wished
- Needle enters from outside
- **Prolene** passed first
  - Prolene = **monofilament** → feeds through needle
- Prolene → **shuttle** (eyelet) → pulls final suture
  - **FiberWire** preferred; Ethibond if unavailable
- Sutures exit outside → tied

Case: Chronic Bucket Handle with ACL Tear Ask AI

- Young man, 24 years
- Repeated **locking** and giving way
- MRI → meniscus absent; **double PCL sign**
- ACL tear + **bucket handle** medial meniscus tear
- Chronic, but age 24 + concomitant ACL + long bucket handle
  - Removal = whole meniscus → **try repair**
- Out-in steps
  - **Freshen** → shaving
  - **Reduce**
  - Suture; stitches from above flip meniscus → subsequent stitches from **below**
- Structure in notch → not ACL → displaced **bucket handle** of medial meniscus
  - Very long; whole meniscus; reduced → did not jump back → start
- **Medial condyle** changes → from prolonged displacement
- Meniscal changes too → still repair at this age with **concomitant ACL** injury

- Ordinary needle → skin → meniscus
  - Both through meniscal tissue transversely → **horizontal mattress**
  - One peripheral in synovium, other inside → **vertical mattress**
- Feed **Prolene**; Ethibond / FiberWire not easily fed through needle
- Retrieve via another portal → **eyelet** → drop in Ethibond or FiberWire
- Second pass → Prolene → replaced by Ethibond → two limbs **lock** → one stitch
- First stitch → **body**
- Tighten first body stitch → confirm meniscus pulled back
- More posterior → needle **peripheral to tear** → vertical suture
  - First **horizontal** stitch did not pull firmly
  - **Vertical**, peripheral → pulls more strongly
  - Vertical needs more stitches; chosen when horizontal fails to pull
- Further back → more **additional portals**
- All stitches pull upwards → meniscus **flipping**
  - More posterior stitches from **under surface** → pull down → stable
- **Final tightening** → after ACL reconstruction
  - → little knee movement after tightening
- After all stitches → **test stability** by pulling
  - Part comes → **additional stitch**; costs only a suture

Combining Repair with Partial Meniscectomy Ask AI

- Displaced **bucket handle**; whole meniscus flipped out → reduced
- Inner part → small **flap tear**
- Young + concomitant **ACL**
  - Suture bucket handle as normal
  - **Meniscectomy** of small flap
  - → combine **partial meniscectomy** with bucket handle repair → most of meniscus saved

Entry Sites and Safety Ask AI

- Displaced meniscus → pulled back repeatedly with out-in
- **Highly unstable** → high repair failure; jumps out before pulled into place
- Multiple **entry points** → cover whole meniscus
- Safety study → 2010, **Arthroscopy** journal
  - Safe to midline of **semitendinosus** tendon without neurovascular bundle injury

Sequencing with ACL Reconstruction Ask AI

- Pass stitches → finish **ACL reconstruction** before tying → no repeated pull on repair
- **ACL stump** → not removed unless causing impingement
  - Torn ACL displaced posteriorly → drill through its centre
  - Preserve **tibial fibres** and femoral fibres
  - Fibres stuck onto **posterior capsule** → femoral-side injury
- New graft through centre of torn ACL → **graft-in-graft**
  - ↑ **Collagen fill**
  - ↑ Proprioception
  - ↑ Blood supply
- ACL fixed → less tension on

sutures → start **tightening**

- Assistant keeps remaining stitches taut → meniscus stays **reduced** while tying
  - Full extension; scope not in use → risk of suturing while displaced
- Out-in or inside-out → sutures finally exit the same way

Lateral Meniscus Ask AI

Out-in

- Basketball player; **ACL tear**; bucket handle lateral meniscus tear
- Problem → **common peroneal nerve** behind fibula
  - Cannot reach whole posterior horn of **lateral meniscus**
- **Marker line** → needle entry never posterior to it
- Most posterior lateral part
  - Never **inside-out** → needle exit unknown
  - → **all-inside**
- Body → inside-out, or out-in
- **Anterior horn** → only out-in; inside-out cannot reach
- **Posterior horn** → all-inside safest, especially lateral
  - Medial posterior horn → out-in possible
- Displaced lateral meniscus → needle passed, never posterior to drawn line
  - Line → **posterior border of fibula**
- **Prolene** shuttle → exchanged for Ethibond → stitch
- Both limbs in meniscus → **horizontal mattress**
- Tightening → meniscus starts **flipping**
- **Popliteus tendon** → out-in reaches it or its midline
  - Further towards midline → unreachable with out-in in lateral meniscus
  - Hand restricted → cannot angle entry posteriorly
- Towards **root** (pointed with hook) → needs all-inside
- Stitches tied at the end

All-Inside Repair Ask AI

- Many device types
- **Freshen** tear → reduce meniscus
  - From inside knee → helps reduce **bucket handle**, like inside-out
- No stitches exit to outside
- Device → **two parts**
  - One anchors **behind capsule**
  - Stitch or bar onto meniscus, per design
- Position on tear → **fire** → withdraw → repeat per design
- Advantage → nothing exits → no risk to **blood vessels** or from exiting needles
- Disadvantages
  - Cost → about 4,000–5,000 **Egyptian pounds** per stitch
    - Usual 4–6 stitches → about 20,000 pounds
  - **Anterior horn** → cannot be reached from inside
- Reserved → **posterior horn tears** not doable inside-out
  - Medial → out-in technically difficult; needs training
  - Lateral, most posterior → out-in **not safe**, not accepted

- Classic, undisputed indication → **posterior horn lateral meniscal tears** → justify price
- Freshen → reduce → **fire** implant
- Lucrative field → many designs: one, two, or three stitches per **implant**
- Needs **training** → wrong firing loses 4–5 thousand pounds

Take-Home Message Ask AI

Repair

- **Always attempt repair** of repairable tears → red zone, not macerated
- **Longitudinal vertical** → best indication, especially with concomitant ACL injury
- **Out-in** → easiest; no instruments; all sites except posterior horn lateral meniscus
- **In-out** → easy with needles; ideal for body tears
- **All-inside** → posterior horn tears, especially lateral meniscus

Site	Inside-out	Outside-in	All-inside
Anterior horn	Impossible	Only option	Impossible
Body	Ideal	Yes	—
Posterior horn, medial	Very risky (neurovascular bundle)	Possible; needs training	Reserved for these
Posterior horn, lateral (most posterior)	Never	Not safe; not accepted (common peroneal nerve)	Classic indication

Degenerative Meniscal Tears Ask AI

- "**Most common real practice dilemma**"
  - Patient 40–50 years → knee pain
  - Pain → medial, lateral, several sites, front and back
- Reflex **MRI** → "meniscal tear" → arthroscopy → very common practice
- **Not a tear**
  - Degeneration within or crossing substance → tear only if it **cuts** upper, lower, or both surfaces
- Example → cuts lower surface
  - Undisputed tear → **horizontal cleavage**
- Horizontal cleavage → very small **cyst**
- Horizontal cleavage → upper + lower leaflets with **large cyst** → clearly troublesome
- Questions before deciding surgery
  - **Age** → young, middle-aged (35 upwards), elderly
  - **Osteoarthritis** evidence → present? which grade?
  - **Trauma** history → traumatic vs degenerative tear
- Options → **conservative**, arthroscopic partial meniscectomy (APM), meniscal repair
- Degenerative tear findings
  - **Calcifications** → calcium pyrophosphate dihydrate or calcium hydroxyapatite
  - **Horizontal cleavage**
  - **Radial tears**
  - **Complex tears**

Work-Up: 2016 Consensus Ask AI

- Patient **over 35** + knee pain + no traumatic event
- Pain on joint line or **diffuse** → 4 X-ray views
- **No MRI** initially
  - Meniscal tear very common in 40s–50s → no operative meaning
  - MRI → not justified first
- Views → detect **arthritis**
  - **Weight-bearing AP**
  - **Lateral**
  - **Patellofemoral**
  - **Schuss view**
    - **Weight-bearing**, flexed 30°, beam posterior → anterior
    - Shows **joint line** and any narrowing
- Aim → arthritis? any other **pain cause** besides meniscal injury
  - Starting chondral ulcers, narrowing, osteoarthritis
- No improvement after about 3 months of **conservative management** → MRI justified
  - Applies to 35- or 40-plus patient, no traumatic event
- MRI + X-ray → **osteoarthritis** evidence?
- No OA + no obvious cause + **joint line** pain, not diffuse → APM justified
- OA present → APM vs nothing → **no outcome difference**
- Many studies → no additional benefit of APM vs **non-operative** treatment in degenerative tears
- Conclusion → APM **not first-line** treatment
- "Analgesics failed" → **not** a surgical justification
- Only justification → 3 months + **specific joint line pain** (not diffuse) + no arthritic changes

Horizontal Cleavage and Meniscal Cyst Ask AI

- No trauma + degenerative-looking tear → **repair** possible only for horizontal cleavage
- Always **progresses to periphery** → functionally total meniscectomy
- May form large **parameniscal cyst** → indication for repair
- Horizontal cleavage with **meniscal cyst** → depends
  - No arthritic changes → **arthroscopic repair**; decompress cyst from inside tear first
  - Not repairable → **partial meniscectomy**
  - Cyst → evacuate from inside, or **excise** (usually palpable under skin)

Summary Ask AI

Degen tear

- **Degenerative meniscal tears**
  - Middle and **old age**
  - **No trauma** history
  - ± **Arthritic changes**
  - **Conservative** → first line
- **APM** → only after failed conservative management, with
  - Significant **mechanical symptoms** → catching and releasing; or
  - **Specific joint line pain**, not improving, no arthritic changes
- MRI → only **after 3 months**
- Definite arthritic changes on X-ray (mild–moderate, sometimes severe) → **MRI not justified**
  - Tear certainly present → **decision unchanged**

  - Exception → **mechanical symptoms**

Must-Know

- Over 35, no trauma, knee pain → **4 X-ray views**, not MRI
- MRI → only after 3 months of failed **conservative** treatment
- APM → specific joint line pain + **no arthritic changes**, or mechanical symptoms
- OA present → APM = doing nothing
- Horizontal cleavage → only degenerative tear considered for **repair**

Meniscal Root Tears Ask AI

Significance Ask AI

- Relatively new topic; **popular** in recent years
- Root tear → **whole meniscus** non-functional
- Consequences → same as **total meniscectomy**
- → **Rapid** OA progression
- Young patient + obvious arthritic changes + severe **chondral injuries** → think root tear
- Root tear = total meniscectomy in function
  - Compartment **contact pressure** ↑ by 235%

Classification Ask AI

- **5 types** of meniscal root tear
  - **Partial** (small piece)
  - **Complete** (type 2)
  - Complete with **bucket handle**
  - **Oblique**
  - **Bony avulsion**
- Differentiate **two kinds**

	Acute	Chronic degenerative
Cause	Acute trauma; many with associated ACL injury	Degeneration in elderly
Side	Commoner lateral; also medial	Commoner medial; the one seen most
Indication	Best indication	Less indicated

- Root tear "**fashion**" → women in 40s–50s imaged → told surgery needed → need questioned

MRI Diagnosis Ask AI

Root

- **Complete root tear**
  - → **type 2**
- **Coronal** → **meniscal extrusion**: meniscus out, not in line with tibial plateau
- **Posterior coronal** cuts (fibula visible) → medial meniscus not reaching midline
  - **Fluid signal** at that site
- **Sagittal** → last cut before notch: meniscus absent → **ghost sign**

- **Axial** → meniscus not reaching midline
- **Anterior root tears**
  - Less common
- Root tear followed up → definite **rapid progression** to OA, as after total meniscectomy

Indications for Root Repair Ask AI

- First → **surgical candidate**?
- Candidates → divided into acute and chronic
- **Acute root tear**, with or without ACL injury (commonly with) → best indication
- **Acute traumatic** → young patient
  - Elderly → degenerative, not traumatic
  - Acute traumatic → best indication for **meniscal root repair**
- **Chronic**, 40s–50s → assess
  - **Arthritis**?
    - None on X-ray + **no cartilage changes** at arthroscopy → good indication for repair
    - Confirmed only at arthroscopy; X-rays first
  - Arthritis present → **very cautious**
    - Current view → opposite of a 2014 paper: no benefit from **partial meniscectomy**
    - Trimming loose root piece → **subjective pain** improvement only
    - **OA progression** → unchanged
    - No definite mechanical symptoms from **loose flap** + changes → don't operate; = conservative management

Root Repair Technique Ask AI

Transtibial Root Repair Ask AI

- Formerly **open**; no longer done
- Probe → confirm torn root
- **Two steps**
  - Pass **stitches** through meniscal root
  - **Tibial tunnel** (unless anchors — not commonly used)
- Pull stitches through tunnel → tie over anterior **bone bridge**
- **Abrade** root footprint (should lack cartilage) towards eminence
  - Towards **tibial eminence** → root not malpositioned; proper tension
- Abrade **tibial tunnel** site → pull root into it
- Suture-passing instruments → e.g. **lasso**: curved; suture fed through
  - **Prolene** → shuttle → definitive Ethibond or FiberWire
- One stitch, then a second → both through root
- Tibial tunnel exits at footprint → root descends into tunnel
- Stitches pulled down → tied on
  - **Anterior cortex** of tibia
- Lasso stitches through root → reduce root into place
- **Anatomic insertion site** → guide wire must exit there
  - **ACL tibial guide** → drawback: scrapes femoral condyle
  - **Special root guide** → avoids scraping femoral condyle

- Guide wire as for ACL tibial tunnel → ream with **4.5 drill**
  - Cannulated, or **EndoButton** (suspensory fixation) drill
- Grasper / **hook** through tunnel → pull root stitches down tunnel
- Pulling stitches → root returns to place
- Stitches exit **anterior tibial cortex** → tied over bone bridge or **washer**
- After repair → **gap** sign lost; root back in place
- **Acute** → good healing; healed and fixed on second look
- **Chronic** → lower healing potential; weak hold → function restoration **questionable**

Case: Missed Root Tear with ACL Tear Ask AI

- **ACL tear** patient
- Fibres present but all gone **posteriorly** → lax ACL
  - **Partial**, unstable → indicated
  - No need for whole ACL "lying on the floor"
- Meniscus looked good; **medial femoral condyle** changes at age 29
  - Cause not investigated → such changes **must be explained**
- **Mosaicplasty** prepared in case needed
- Cartilage sometimes better than expected at **arthroscopy**
- Not exposed bone → **very soft cartilage** → not indicated for mosaicplasty
- Cause → missed **type 2 complete root tear**
  - **Posterior coronal** cuts not reviewed
  - Acute traumatic → **must be repaired**

> **Pitfall**
> - Young patient + condylar chondral changes → **explain the cause**
> - Missed root tear → **posterior coronal** cuts not reviewed
> - Acute traumatic complete root tear → must be repaired

Root Repair Without Special Instruments Ask AI

- No special instruments → **ACL tibial guide** at chosen root tunnel site
  - Guide wire → **4.5 drill**
- No suture passers → no **lasso**, no meniscal Scorpion
- Ordinary needles (**cannulas**) through portal → pierce meniscus
  - Not curved → won't exit like **lasso**
  - Once pierced → withdraw slightly → feed suture → emerges from **under surface**
- Feasible, **very easy**, not traumatizing
- **Prolene** → two stitches near root via simple needles
- ACL **guide wire**, back end first, through tibial tunnel
  - Prolene fed through its centre
- Prolene exits tunnel → retrieved
- Prolene enters from **tibial tunnel**
  - Pulled out via **portal** → tied to root stitches
- Pulling → Ethibond stitches drawn down **tibial tunnel** → root seated
- Only **two steps** → root stitches + correctly placed tibial tunnel

Combined ACL Reconstruction and Fixation Ask AI

Root

- **ACL reconstruction** as usual
- ACL **remnant** kept → collagen fill + proprioception
  - Bare bone in femur and tibia → not a goal
- Original torn ACL kept unless **impingement** → graft through its centre
- **Final tightening** → very last, after ACL, as in any meniscal repair
- **Two tunnels**
  - Root tunnel → exits anteriorly at **tibial crest**
  - ACL tunnel → exits **medially**
- Root stitches from **same hole** → no bone bridge possible
- **Washer** → one thread through it, other outside → acts as bone bridge → final tightening

Take-Home Message Ask AI

Root

- **Complete root tear** → functionally a total meniscectomy
- **Acute traumatic** → definitely indicated for repair
- **Chronic** → needs evaluation
  - Arthritic changes → **poor prognosis**
    - Clear mechanical symptoms → consider **APM**; otherwise no value
  - No arthritic changes → **meniscus repair** → prevents otherwise definite OA
- Usual → **transosseous repair**
- Alternative → **anchors**
  - No tibial tunnels; inserted through **posterior portals**
- **Healing potential** → acute good; chronic lower — problem in chronic root repair

Ramp Lesions Ask AI

Definition Ask AI

- Emerging topic → **ramp lesions**
- Ramp lesion → **vertical longitudinal tear**
  - **Vertical** → superior → inferior
  - **Longitudinal** → along meniscal length
- Site → most medial part of meniscus, near **posterior root**
- **Vertical longitudinal meniscocapsular** tear at meniscocapsular junction
  - Like very small **bucket handle**
  - At posterior root of **medial meniscus**
- Occurs **only in ACL deficiency**
- **Incidence** high → about 10% → thousands missed
- **Pivoting** during giving way → cause
- Present → **deficient ACL**
  - ACL tear, or **graft failure** after reconstruction (absent initially)

MRI Diagnosis Ask AI

Ramp

- **Unstable** ramp → ↑ OA risk
- Most posterior cuts, approaching **notch** → fluid signal line cutting top → bottom
  - Present → not necessarily ramp

    - Absent → **no ramp lesion**
- **Sensitivity** → high
- **Specificity** → not high
- **Bone marrow contusion** / oedema → ↑ specificity
    - Old; **most posterior** medial tibial plateau

High Yield

- No fluid signal → **no ramp lesion**
- Fluid signal alone → ramp not certain
- Fluid signal + posterior medial plateau **bone marrow contusion** → ramp likely
- MRI → excellent to **rule out**; misses some

Classification and Indications Ask AI

Ramp

- **5 grades** by stability → displaces into joint on pulling?
    - Very small, upper or lower surface only, little movement on **hook** → stable
    - Stable → usually no effect on joint stability
- Menisci and root → **add to joint stability**
- Any ramp → **ACL tear** present → reconstruction planned
    - **Ramp repair** → adds stability
    - Unstable ramps → mainly **types 4 and 5**
- Stable, small ramp → **can ignore**
- MRI → excellent to **rule out** ramp; need not show all
- ACL + ramp injury → higher **rotational instability**

Arthroscopic Case Ask AI

Ramp

- MRI uncertain for ramp; **torn ACL**
- Right knee → always **probe**
    - Pulled well → nothing; at **root** → starts to come
- Far back at root → **meniscocapsular**, vertical longitudinal ramp
    - Comes with probe → **unstable**
- Ramp never occurs alone → **ACL tear** present
- Cuts top → bottom + unstable on probing → **type 4**
- Unstable on probing → type 4 → repair like any **meniscal repair**
- Repair techniques
    - **Posteromedial portal** → sutures or anchors
    - **Out-in** suture far posterior → safe zone from published Arthroscopy study
        - Technically difficult
- Ramp seen + **unstable** → peripheral tear
- **ACL** → turned back posteriorly to posterior capsule
    - Need not fall to the floor
    - Instead of **lateral femoral condyle** → turned posteriorly
    - → **partial / tricky** ACL tears
- Needs
    - **ACL reconstruction** → preferably stump kept, graft through centre
    - **Ramp repair**

- Ideal → **posteromedial portal** + anchors
- Done here → out-in via **safe portal** from behind
- First ramp stitch → through meniscus; **Prolene** fed
  - Out via **anteromedial portal** → replaced by permanent FiberWire or Ethibond
  - Here → Ethibond
- One limb only → no **loop** → second needle needed
- **Out-in** → reaches nearly to end of root

Ramp Repair: Completion Ask AI

- Prolene out via **anteromedial portal** → tied → shuttle → Ethibond pulled back → loop
- **Ramp repair** with out-in suture → like any meniscal repair
- Stitch not tied until **ACL reconstruction** finished
- Existing **stump** left; graft through centre
- Last step → tighten **horizontal mattress** from outside → good reduction of small ramp
- Ramp → usually **small piece**
  - Very long, reaching front → ordinary **bucket handle**, not ramp
- Held well by stitch

Take-Home Message Ask AI

Ramp

- **Stable**, small → nothing, or needle passed 2–3 times
  - Needle injury → **bleeding** → stimulates healing → should heal
- Mobile, **displacing** → no healing unless sutured

Ramp	Management
Stable	Nothing, or needle injury to stimulate healing
Unstable	Ramp repair + ACL reconstruction

Discoid Meniscus Ask AI

Definition and Types Ask AI

- **Discoid medial meniscus** → very rare → discussion = discoid lateral meniscus
- C-shape → complete or near-complete **disc** reaching tibial eminence
- **Bilateral** → 20–80%
- **Very thick** → may exceed fully opened basket jaw
- Large size → **more prone to tearing** → main problem

Type	Size	Feature
Complete	Reaches eminence	—
Incomplete	Larger than normal	Does not reach eminence
Wrisberg type	Normal	Wrisberg ligament deficient; posterior part displaces repeatedly

- **Wrisberg type** → close to hypermobile lateral meniscus
  - Pulling posterior horn → flips into joint

- Cause → **posterior attachment** torn / absent, not a bucket handle tear

Clinical Picture Ask AI

Discoid

- **Asymptomatic** → incidental at arthroscopy
- **Common complaint** → child aged 10, 12, or 14 (boy or girl)
  - Parents hear loud **squeaking** on bending and straightening

	Popping	Snapping
Sound	Continuous throughout bending and straightening	Loud clunk
Meaning	—	Unstable meniscus moving out of place
Treatment	Not needed	Needed

- Child 10–12 + worried parents → **not too enthusiastic** to operate
  - **Partial meniscectomy** / saucerization to reduce size → reason below
- Other presentations
  - **Giving way** → knee slips, catches; tear displacing repeatedly
  - **Recurrent effusion**
  - **Quadriceps atrophy**
- **Abnormal collagen** + very thick
  - Partial meniscectomy → removing **central part** → upper and lower leaflets
  - Closed at most central part → opens → **horizontal cleavage**
  - Chasing cleavage to closed part → whole meniscus removed to **meniscocapsular junction**
  - Cleavage extends right to the end
- Partial meniscectomy in small child, no tear
  - Must be able to **repair** horizontal cleavage tear
  - Otherwise → **total meniscectomy** → tragedy at age ten
- **Examination**
  - **Joint line tenderness**
  - **McMurray** → may be positive
  - **Effusion** possible

Pitfall

- Discoid partial meniscectomy → central removal opens **horizontal cleavage**
- Chasing cleavage → reaches meniscocapsular junction → **total meniscectomy**
- Cannot repair → do not operate on child

Imaging Ask AI

Discoid

- **X-ray** → thickness causes
  - Wide **lateral joint space**
  - **Flattened** lateral femoral condyle
  - Tibial plateau → possibly **cupped**
- **MRI** → 3 cuts
  - Anterior horn continuous with posterior horn → **bow-tie sign**

Treatment Ask AI

Discoid

- Treatment → **depends on symptoms**
  - Sound only → **do nothing**
  - Incidental, no tear → do nothing
    - Especially MRI **horizontal cleavage** → opening it → torn
- Created cleavage → followed → **total meniscectomy**
  - → Severe degeneration of **lateral compartment** + OCD
- Case → girl, 15 years; **open lateral meniscectomy** at age 8
  - Huge **OCD** of lateral femoral condyle at 15 → mosaicplasty
  - Mosaicplasty → will fail within few years; **cause persists**
  - No **valgus** knee → no compartment unloading → disaster
- Partial lateral meniscectomy → **horizontal cleavage** appears → be ready to suture
  - **Out-in** → stitch passed below → upwards; tied as usual
  - **All-inside** → also acceptable
- **Pain**, locking, giving way → operate
- **No total meniscectomy** in children
  - Adults → acceptable if repair impossible
- Cannot do **meniscal repair** in small child → no arthroscopy; leave, refer later
- Ideal → symptomatic (unstable), no tear → **saucerization** → normal size
- **Torn discoid** → saucerize, then suture
  - Suturing discoid alone → technically very difficult; may not reach
  - **Re-tear** incidence → very high
  - → Size must be **reduced**
- Goal → return discoid to **normal size**
- **Partial meniscectomy** → not easy
  - Needs **baskets curved** right and left
  - Straight basket only → parts unreachable (left knee case)
- Aims
  - **Saucerization** → near-normal size
  - **Stable rim**
  - **Probe** → pull → no missed peripheral tear
  - No **horizontal cleavage** left → if opened → suture
  - Rim comes with pull → unnoticed **peripheral tear** → suture

Discoid	Management
Not symptomatic	No surgery
Symptomatic, no tear	Partial meniscectomy (saucerization)
Symptomatic, with tear	Partial meniscectomy + repair

Take-Home Message Ask AI

Discoid

- Operate **only if symptomatic** → giving way, locking, persistent pain, effusion
  - Sound alone → not "symptomatic"
- **Saucerization** (partial meniscectomy) → normal size ± repair if torn
- **Avoid total meniscectomy** at all costs, especially young children

ACL: A to Z

(From Graft Choice to Rehabilitation)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 123) “fiber discontinuity listed as indirect sign” → “fiber discontinuity listed as direct sign”
- **C2** (p. 125) “less pain than quadriceps tendon” → “less pain than patellar tendon”
- **C3** (p. 128) “drilling at 22 degrees” → “drilling at 120 degrees”
- **C4** (p. 131) “9 mm tunnel: screw 10 or 15” → “9 mm tunnel: screw 10 or 11”

ACL Anatomy and Goals of Reconstruction Ask AI

Lecture Scope Ask AI

- Scope: applied anatomy → function → injury mode → diagnosis → technique → tricks → **rehabilitation**
  - Rehabilitation added at Prof. Osama's request → course after surgery

Goals of Reconstruction Ask AI

- Goal: **restoration to native dimensions**
  - Tibial attachment
  - Femoral attachment
  - Mid-substance collagen orientation → original fiber direction
  - Insertion sites at anatomical sites
- Not all achievable → lecture identifies which are
- Achievable items → must be done well; only ones truly restorable
  - Remainder → cannot be brought back

Femoral Attachment Ask AI

Femur

- Posterior femoral cortex → continues as **lateral intercondylar ridge**
  - Entire ACL femoral attachment → posterior to ridge
  - Nothing attached anterior to ridge
  - "Posterior" = extended knee; arthroscopy, knee flexed → **inferior**
- Area inferior to ridge → **direct ACL fibers** attach
- Footprint ≈ 18 mm × 8 mm
  - Rectangular graft 18 mm (≈ 2 cm) × 8 mm impossible → **footprint not fillable**
  - Achievable: **proper anatomic position**, not anatomic dimensions
- Posterior notch cleaning → **radiofrequency** → bony landmarks incl. ridge preserved
  - Front → back: cartilage end → ridge → area inferior to ridge = **femoral tunnel** site
- Ridge identified clearly; no tunnel part superior to it
  - Tunnel anatomically **inferior to lateral intercondylar ridge**

Tibial Attachment Ask AI

Tibia

- Landmarks: anterior and posterior horns of lateral meniscus
  - **Oval area on tibial eminence**
  - Start: anterior horn of lateral meniscus → end: just anterior to posterior horn
  - Size ≈ 17 mm × 10 mm → no 17 mm graft → dimensions not restorable
  - Target: **center** of footprint
- Center: **posterior edge of anterior horn** of lateral meniscus → followed to tibial footprint
  - Meets footprint at its middle, slightly medial
  - Lands on **lateral border of medial tibial spine** → tibial tunnel exit point
  - → femoral and tibial points defined

Mid-substance Ask AI

- Large at attachments → **ribbon-like** in mid-substance
  - Flattened: ≈ 12 mm wide, ≈ 4 mm thick
- Closest mid-substance match: **patellar tendon** > hamstrings

  - Patellar tendon flat, ribbon-like; hamstrings rounded, circular
  - Raises question: ACL as two bundles? (asked in exams)

Functional Bundles Ask AI

Bundles

- Not two bundles: **multiple bundles twisted** around each other
  - Tens of different bundle directions
  - Simplified → **two functional bundles**
- AM / PL = **functional, not anatomic** bundles
  - Named by tibial attachment
- Orientation and function:

	Anteromedial (AM)	Posterolateral (PL)
Tibial start	Anteromedial	Posterolateral
Femoral position	Most posterior	—
Orientation	Vertical, upright	More horizontal
Function	Anterior tibial translation on femur	Rotational stability

- Vertical graft only (AM direction) → anterior drawer stable
  - Complaint: **giving way on turning** / direction change → no horizontal component
- Part of graft must be **horizontally directed**
  - Functional description only; true double bundle does not exist
  - → oversimplification of anatomy
- True anatomy: different fibers **twisted** around each other, tibia → femur
  - Graft should reproduce this
- Graft preparation (optional): **strands twisted** around each other
  - Arthroscopically → looks like two bundles → belief in two bundles
- After passage: apparent anterior bundle = strands twisted on each other
  - → origin of term **double bundle**; anatomy more complex

High Yield

- Femoral tunnel: entirely inferior (posterior) to lateral intercondylar ridge
- Tibial point: posterior edge of anterior horn of LM → lateral border of medial tibial spine
- Footprints not fillable → aim at anatomic position, not dimensions
- AM bundle → anterior translation; PL bundle → rotational stability

Footprint Fibers, Innervation and Restorable Features Ask AI

Femoral Direct and Indirect Fibers Ask AI

Femur

- **Direct fibers**: just inferior to lateral intercondylar ridge
  - Indirect fibers: fan-like extensions → posterior articular surface
- Reconstruction target: **direct fibers**
  - Case: right knee, anterolateral portal
  - Tear in **mid-substance**, not femoral attachment (most common site) → femoral stump present

  - Front → back: ridge → normal ACL attachment inferior / posterior to it
- Direct fibers lifted → **fan-like extensions** behind → posterior articular surface

Tibial Attachment Shape Ask AI

Tibia

- **C-shaped**, not oval; middle of C empty
  - Start: anterior part of anterior horn of lateral meniscus → end: posterior horn of lateral meniscus
  - No fibers behind posterior horn of lateral meniscus
  - Most fibers **more medial** on eminence → closer to medial tibial spine
- Closer to medial than lateral tibial spine → target **lateral border of medial tibial spine**
  - → central tibial tunnel
  - Posterior edge of anterior horn → followed to lateral border of medial tibial spine
- Meeting point with tibial footprint → **tibial tunnel**
  - Variations, irrelevant to reconstruction: C-shape, J-shape, C within smaller C on tibial eminence
- Shape not reconstructable → **center of tibial footprint** only
  - Tibial footprint larger than expected

Tibial Indirect Fibers Ask AI

Tibia

- Landmarks: anterior horn, posterior horn of lateral meniscus, **anterior cortex**
  - Beyond direct fibers → fan-like extensions forward, almost to anterior cortex
  - Mirror of femur: posterior on femur, anterior on tibia
  - Tibia also has direct and indirect fibers
- Extensions far forward → **anterior horn of lateral meniscus**

Mid-substance Revisited Ask AI

- **Ribbon-like**: ≈ 12 mm wide, flattened, ≈ 3.5 mm thick
- **Round grafts** cannot simulate mid-substance → closest = patellar tendon

Blood and Nerve Supply Ask AI

- **Blood supply**: middle genicular artery + lateral and medial inferior genicular arteries
- ACL has **no pain receptors** → pain after tear not from ACL
- Nerve supply → **vasomotor** function; more important → proprioceptors
  - Types: **Ruffini**, Pacini, Golgi-like receptors
  - Location: mainly femoral and tibial attachments
  - Role: constant **position sense** → brain, without conscious thought
- Incorrect movement imminent → **reflex arc** → muscle contraction → position corrected
  - With every normal movement
  - Removed receptors → no regeneration → permanent position-sense deficit
  - → attachment proprioceptors **preserved** unless removal unavoidable
  - Responsible for **ACL reflex** → works continuously
- ACL absent → feedback lost → **quadriceps weakness**
  - Normally quadriceps contracts continuously with ACL reflex

What Reconstruction Can Restore Ask AI

- Only achievable target: **center of anatomic attachment site** → must not be missed

Feature	Native	Restorable by reconstruction
Femoral attachment	≈ 18 × 8 mm, rectangular	No
Tibial attachment	≈ 17 × 10 mm, C-shaped	No
Mid-substance	Ribbon-like, ≈ 12 mm wide	No
Center of attachment site	—	Yes
Proprioceptors	At attachments	Not by graft → preserve existing

Function, Mechanism and Clinical Diagnosis Ask AI

Function Ask AI

- **Primary restraint** against anterior tibial translation relative to femur
- Secondary: resists **internal tibial rotation**
  - Weak secondary restraint to varus and valgus
  - → collaterals examined in full extension, then 30° flexion → ACL role abolished

Mechanism of Injury Ask AI

- Main: **dynamic valgus load**
  - Fixed foot (e.g. winger kicking with other leg) → fixed flexed leg → valgus
  - Valgus + tibial external rotation + femoral internal (medial) rotation → **ACL tear**
- Contact and **non-contact** injuries
- Other activities:
  - Landing (basketball, jump and land)
  - **Rapid direction change**
  - Dangerous tackle (football, hit on knee from behind → red card)

Clinical Picture Ask AI

- Presentation: **pain**
- Pain source (ACL has no pain fibers):
  - **Haemarthrosis**, capsular stretch
  - Associated injuries: muscular, chondral, meniscal, bone bruises
- **Pop** sensation in knee
- Later: **giving way** with movement → "my knee betrays me"
- **Giving way ≠ ACL tear**; causes:
  - Recurrent patellar dislocation → leg "dislocates" in certain movement
  - Quadriceps weakness
  - Acute sharp knee pain → **reflex quadriceps inhibition** → sudden relaxation → giving way

Examination Ask AI

- **Lachman test** feasible in acute case despite pain and swelling
  - = anterior drawer at 30° flexion

  - Anterior translation + end point assessed; compared with other side

	Lachman	Anterior drawer
Flexion	30°	90°
Grading	Soft vs hard end point	+1, +2, +3
Acute swollen knee	Feasible	Marked swelling, locked knee → not positive

- Anterior drawer → thumb check first:
  - **Medial tibial plateau** ≈ 5 mm anterior to medial femoral condyle
  - → neutral start, not **posteriorly sagged tibia**
  - Sagged start → pulled forward → false-positive drawer → actually **PCL injury**
- Start at 90°; fingers confirm **hamstrings relaxed**
  - Tense patient → test fails
  - Marked swelling + locked knee → not positive
- **Pivot shift test**:
  - Internal rotation + axial loading + valgus → tibia **subluxed anteriorly**
  - Flexion → reduction via **iliotibial band**
  - Depends on **MCL**: complete MCL tear → pivot shift not positive → collaterals examined
- Examination in clinic + again **under anesthesia** before surgery
  - Missed ligament injury + isolated ACL reconstruction → early failure
  - Collaterals (varus / valgus stress) + **posterolateral corner** examined
- Case: Lachman +, anterior drawer +
  - **LCL opens in full extension** → because ACL torn
  - Intact ACL → LCL tested at 30°
  - Missed LCL + isolated ACL → opens up → **failure**

Pitfall

- Missed ligament injury → one of commonest causes of failed ACL reconstruction
- Giving way ≠ ACL tear
- Drawer from posteriorly sagged tibia → false positive; actually PCL injury
- Complete MCL tear → pivot shift not positive

Exam Pearl

- Soft / hard end point → Lachman test only
- Anterior drawer → graded +1, +2, +3

Imaging Ask AI

X-ray Ask AI

- X-ray → little
  - **Segond fracture** (lateral rim avulsion) → always associated with ACL tear
  - Tibial eminence avulsion

MRI Signs Ask AI

- **Direct** and indirect signs
- Direct signs: *[C1 · transcript: "fiber discontinuity listed as indirect sign"]*
  - **Fiber discontinuity**: tibia → femur not continuous
  - Signal irregularities; "fuzzy ACL" / "ACL sprain" in reports → **normal ACL not fuzzy**
  - Widened appearance
  - Changed angle instead of attaching to femur
- Indirect: **bone bruises**
- Normal ACL: not fuzzy, straight tibia → femur
  - **PCL** taut → not buckled, not comma-shaped
  - Assessed on ≥ 2 cuts, sometimes 3; never one cut
- Tear criteria:
  - Not reaching femur, fallen down → **definite tear**
  - Fuzzy, inclined backward → **posterior capsule** instead of femur → ACL injury
- **Anterior tibial translation** on femur → sometimes, uncommon
- **Empty notch sign** (coronal): lateral wall lacks ACL fibers, fluid instead → femoral avulsion
- Classic **bone bruises** → lateral compartment:
  - Mid-femoral condyle
  - Posterior tibial plateau (mid and posterior)
- Small part attached, rest → posterior capsule; angle changed, fallen → **ACL tear**
  - Small stretched fiber visible → rest still torn
- **Buckled PCL**: ACL torn → tibia forward → pushes PCL → comma-shaped
- Only small part attached, fuzzy remainder posterior → **partial ACL injury**

Case: Single-cut Pitfall Ask AI

- ACL intact-looking in one cut only; adjacent cuts → clearly torn → **> 1 cut** required
  - Symptomatic → **partial tear**
- Arthroscopy: ACL lax, fully posterior → attached at **indirect fiber** region
  - Far behind ridge, not just behind it → far posterior to normal place
  - MRI appearance looked good

Indications and Timing of Surgery Ask AI

Indications Ask AI

- Depends on:
  - **Symptoms**
  - Age: 20-year-old, complete tear, sports, asymptomatic → **high risk** of later meniscal + chondral injury → reconstruction advised
    - Older (≈ forties), asymptomatic, not bothered → increased meniscal tear risk explained; non-operative acceptable
  - **Activity level**:
    - Fifties, athlete, non-arthritic knee, instability → operate
  - Associated ligament injuries or **meniscal tears** in non-arthritic knee → indicated → rapid deterioration

Timing Ask AI

- Prerequisites:
  - **No effusion** (quiet knee) → otherwise difficult rehab → stiffness
  - Good ROM; most important → **full extension**
  - Full flexion not mandatory
  - No **wasted quadriceps** → exercises first → good quadriceps power
- Exception: **locked bucket-handle** tear → immediate surgery → full extension otherwise unattainable
- Conditions met → **as early as possible** → best results, avoids secondary chondral / meniscal injury
  - ≈ within 3 months → anatomic reconstruction

Graft Choice Ask AI

Graft Requirements Ask AI

- Principle: **strong graft** + anatomic points + secure tibial and femoral fixation
  - Strong
  - Easy to pass
  - Length: ≥ 8 cm; 9 cm more comfortable; < 8 cm unworkable
  - Minimal **donor site morbidity**
  - Easy to fix

Patellar Tendon Ask AI

BTB

- Oldest, most famous: **bone–tendon–bone**: patella piece + patellar tendon + tibia piece
  - Middle ribbon-like → like ACL mid-substance
  - Bone blocks → **bone-to-bone healing** in tunnels (theoretical)
  - Drawback: **high donor site morbidity** → anterior knee pain, esp. kneeling
- Patellar cut: saw at **45° slant**; straight entry → patella fracture

Hamstrings Ask AI

Hamstrings

- Faster donor site recovery, easier harvest → **preferred**
- **Semitendinosus + gracilis**, each doubled → quadruple graft
  - Semitendinosus alone if long enough → tripled
  - Very small scar ≈ finger phalanx length
- Allows single or double bundle
- Small incision → may be vertical; larger → not vertical → **infrapatellar branch of saphenous nerve** at risk
  - **Sartorius fascia** opened
  - "SGS" ≠ sartorius over gracilis over semitendinosus → gracilis + semitendinosus enveloped in superficial sartorius layer
  - Both tendons exposed; **both freed** before harvester → neither cut
- Harvester on **semitendinosus first**
  - Gracilis first → interconnections → semitendinosus slips distally → hard to retrieve
- Semitendinosus connections → usually **medial head of gastrocnemius** → exposed, cut → stripper introduced
- Graft hung on finger; stripper **push-and-pull**, no force
  - Resistance → back off → advance → full length
  - Order: semitendinosus → gracilis
  - Tibial detachment with **periosteal strip** → length + improved tunnel healing

Quadriceps Tendon Ask AI

Quadriceps

- **Very good graft**, underused
- Patella piece + **quadriceps tendon**
  - Better length and thickness than patellar tendon
  - Anterior knee pain much less than patellar tendon *[C2 · transcript: "less pain than quadriceps tendon"]*
  - Split in two → double bundle reconstruction

Allografts Ask AI

Allograft

- Cadaveric, used abroad: BTB, quadriceps with bone, **calcaneus + Achilles tendon**
  - Achilles graft splittable → double bundle
  - Readily available, quick, no donor site morbidity
  - Not available here
- Main advantage: **multiligament** injuries → autografts insufficient → large allografts
  - Return to sports less reliable; not for young athletes
  - **Patellar tendon allograft** → poor results
  - Theoretical **disease transmission** risk

Synthetics and Internal Brace Ask AI

- **Synthetics** (e.g. LARS ligament) → bad reputation → synovitis
- Newer generations better
  - Better in collaterals / **extra-articular** than intra-articular (ACL)
  - Uses: ACL repair; reconstruction needing temporary support
- **Internal brace**: very thin FiberTape, not large tape like LARS
  - Augments ACL repair
  - Supports reconstruction until graft matures and vascularizes

Graft Comparison Ask AI

- Differences small → aim = good result
- Outcomes:

Parameter	Finding
Stability: PT vs hamstrings	Similar post-op; early studies favored PT
Muscle strength	PT → quadriceps weakened initially; hamstring graft → hamstrings affected
Sprinters	Hamstrings avoided (speed) → PT
Function	No difference
Osteoarthritis	Slightly higher with PT
Quadriceps vs PT	Quadriceps thicker, longer, less donor morbidity; harder rehab first 3 months; same stability
Autograft vs allograft	Same stability; allografts worse overall; autografts used here, no allografts

Graft Selection Summary Ask AI

- **No clear best graft**; all have a place
- PT → possibly better **return to sports**
  - PT + quadriceps → professional athletes, revision
- **Hamstrings** → easy harvest, easy pass, excellent results
- **Allografts** → multiligament injuries, ACL repair

Femoral Tunnel Technique Ask AI

Portals and Transportal Drilling Ask AI

- Anatomic reconstruction → **accessory anteromedial portal**
  - Femoral tunnel drilled via AM portal, not tibial tunnel → **transportal femoral drilling**
  - → horizontally directed graft, not vertical
- Portals: classic lateral + medial
- **Accessory medial portal**: just medial to medial portal → femoral drilling
  - Medial portal → view of femoral point → choice confirmed
  - Large drill bit (not only guide wire) via accessory portal → **femoral condyle cartilage** protected
- First step: **femoral footprint prepared**
  - Torn ACL stuck far posterior on capsule; no femoral attachment → proprioceptors lost → removed
  - **Tibial attachment kept**

Notch Preparation Ask AI

- **Radiofrequency minimized** → thermal cartilage injury
  - Shaver → up to most posterior part → then radiofrequency
  - Shaver posteriorly → **bleeding** obscures view, even under tourniquet
  - Shaver continued → **lateral intercondylar ridge erased**
- Cleaned to far posterior end; ridge identified → stay posterior (inferior, knee flexed)
  - Area exposed well, esp. early in learning
  - Orientation: **right knee**, anterolateral portal

Marking the Femoral Point Ask AI

- Palpation: far posterior **over-the-top point** → ridge → between them
  - Via accessory medial portal → femoral tunnel point marked
- With experience optional: **portals switched** → view from anteromedial portal
  - Instrument via anterolateral portal → point confirmed vs ridge and over-the-top
- Back to anterolateral view → **guide wire**
- Suspensory fixation: **4 mm drill bit** → complete tunnel → length measured
  - Then final reamer = graft diameter
- **Posterior wall** ≈ 2 mm
  - Result: horizontal, posterior to ridge, ≈ 2 mm posterior wall
  - **Blunt obturator** via accessory portal → horizontal graft direction shown

Tibial Tunnel, Graft Passage and Tunnel Geometry Ask AI

Tibial Point and Tibial Guide Ask AI

- Exit at **center** of large tibial footprint
  - Posterior edge of anterior horn of lateral meniscus → lateral border of medial tibial spine
- **Tibial guide** types:

Guide	Pin exit	Placement
Elbow aimer	Anterior to tip	Slightly posterior to target
Tip aimer	Exactly at tip	Exactly at target

- Tibial tunnel reamed to **graft diameter**

Transportal vs Transtibial Ask AI

- Demonstration: **two blunt obturators**
  - Medial portal → femoral tunnel → horizontal
  - Tibial tunnel → femur → vertical
  - Reference 9 and 10–11 o'clock → horizontal between 9 and 10; vertical ≈ 11 o'clock
- **Transtibial drilling** → vertically oriented graft → hence transportal

Passing and Checking the Graft Ask AI

- Graft pulled through tibial tunnel → tension + relation to **PCL** checked
- Knee 90° → ACL and PCL **crossing** at ≈ 90°
  - Synovium on PCL shaved → difference shown
- Graft horizontal, 9–10 o'clock, **no impingement**
- Femoral tunnel drilling → knee flexed ≈ 120°
- Full extension → no **impingement** between graft and notch

Posterior Wall and Tunnel Blowout Ask AI

- Femoral tunnel = graft size; behind → posterior wall → **popliteal fossa**
- Just behind → **over-the-top** position → must not break
- Breach → tunnel opens into popliteal fossa → **tunnel blowout**

Fixation	Minimum posterior wall	Reason
Suspensory	≈ 1 mm acceptable	—
Interference screw	≥ 2 mm	Screw compression → wall breaks → blowout → no fixation

- Interference screw → thicker wall, **whole tunnel inferior to ridge**

Clock-face Orientation Ask AI

- Transportal drilling at 120° flexion → step back → notch **clock face**
- Right knee: 9, 10, 11, 12 o'clock
  - 9–10 o'clock → **horizontal** (target)
  - Higher → vertical
  - Old terms: vertical = **anteromedial graft**; horizontal = posterolateral graft

  - Vertical graft → AM bundle reconstructed → AP stable, **rotationally unstable**
- Good anterior drawer; complaint → **instability on direction change**

Step Summary Ask AI

- Front → back to **lateral intercondylar ridge**
- **Tibial C-guide cannula** ("bullet") fixed posterior / inferior to ridge
  - Drill guide through it → no slipping from point
- Drilling at 120° → guide wire → **final size** *[C3 · transcript: "drilling at 22 degrees"]*
- Tibial point: posterior edge of anterior horn of LM → lateral border of medial tibial spine → exit
- **Suture** pulled through tibial tunnel first
- Graft pulled through middle of **tibial stump**, unshaved
  - Femoral stump completely torn → cleaned (unattached)
  - **Tibial stump always left**
  - Stump in the way → pulling side chosen → graft enters with it; stump fibers adhere to graft
  - Pull until **button flips**
- Flip confirmed → pull does not bring graft down

High Yield

- Follow anatomic femoral and tibial ACL attachment points
- Center of footprint
- No tunnel part anterior (superior, knee flexed) to lateral intercondylar ridge
- Avoid vertically oriented grafts

Partial ACL Injury and Augmentation Ask AI

ACL Augmentation Ask AI

- Old term "single bundle injury" → now **partial ACL injury**
  - Large part torn; small part attached, usually very posterior
- Management: **ACL augmentation**
- Case: left knee, ACL present but far posterior
  - Tunnel just anterior to it, still posterior (inferior) to ridge
- Remnant ACL **left intact**: no trimming, no shaving
  - Tunnel ends anteriorly at original ACL; tibial footprint palpated; same technique
- Graft passed **through middle of remnant** ACL
  - Patient: partial injury + symptomatic
  - Conditions: anatomic, inferior to ridge → = ACL augmentation

Why Preserve the Stumps Ask AI

- Both stumps in augmentation; usually tibial only
  - **Collagen fill**: native attachment ≈ 2 cm × 1 cm → unmatched by graft → native collagen kept
  - Added vascularity
  - Preserved **proprioceptors** at femoral and tibial sites
  - Partial injury → **residual function** of injured ACL + reconstruction

Graft Fixation Ask AI

General Points Ask AI

- Two ways: **suspensory** (button) or screw (many types)
- **Nitinol (elastic) wire** placed before pulling graft
  - Wire after graft → ACL fibers trapped between wire and tunnel → screw cuts fibers
- Graft **marked at 25 mm** → true length in tunnel known
- Screw via **accessory medial portal**, same angle as tunnel drilling
  - Otherwise → does not seat
  - Twisting under graft → unseen → presses **posterior cortex** → may break it

Femoral Fixation Options Ask AI

- Differences small

Type	Device
Aperture	Interference screw → plugs tunnel opening
Suspensory	Buttons with loops → graft hangs

- **Interference screws**:

	PEEK	Bioabsorbable
Appearance	Off-white, not transparent	Transparent: white, green or blue
Material	PEEK	Polylactic acid
Current use	Standard	Rarely used
Problem	Does not dissolve → stays permanently, not a problem (patient informed)	Reaction, tunnel widening

- Suspensory: **button** on cortex → loop → graft

Loop	Length	Use
Fixed	15, 20, 25 mm	Slightly longer than needed → room for button flip
Adjustable	≈ 60 mm, shortened as needed	Tightened until graft at desired length → much easier

- Example: tunnel 45 mm →
  - 30 mm graft inside → **15 mm loop**
- Drilling: 4 mm drill full length; reamer (8 or 9 mm) only to 25–30 mm, per graft length
- 4 mm drill → reaches end
- **Pull-out strength**: suspensory > interference screw
- Drawback: uncompressed graft → side-to-side swing → **windshield wiper effect**
- **Tunnel widening** causes:
  - Suspensory → windshield wiper effect

  - Interference → synovial fluid enters tunnel → dilatation
  - Polylactic acid (bioabsorbable) screws
- Results equal; **revision rates higher** with suspensory fixation
  - Possible cause: technical error → button not properly flipped
  - Flip confirmed after pulling →
    - **Sutures** running in expected direction

Tibial Fixation Ask AI

- **Cycling** → tension → tibial side at 0–20° flexion → pull hard → interference screw
- Extra length → additional fixation: **staple** or screw and washer

All-inside Reconstruction Ask AI

- **All-inside**: suspensory fixation on femur and tibia
  - Incomplete tunnels (sockets) in femur and tibia
  - Tibial interference screw → requires complete tunnel of graft size → short graft → all-inside advantage
  - **Adjustable loop** mandatory; fixed-length loop not usable

Double Bundle Reconstruction Ask AI

- Trend ≥ 10 years ago
- Rationale: ACL = two bundles → **double bundle** reconstruction = "anatomy"; now understood otherwise
  - AM bundle: femur more posterior, vertical; tibia anteromedial
  - PL bundle: femur slightly more anterior, horizontal; tibia posterolateral
  - → 2 tibial tunnels ± 2 femoral tunnels
- Lecturer's technique (~10 years ago): 2 tibial tunnels, 1 femoral tunnel, second strand via **over-the-top** position
  - Graft 1: tibia posterolateral → femur more anterior → = PL bundle
  - Graft 2: tibia more anterior and medial → femur far more posterior → = **AM bundle**
- Studies: double bundle much better stability, esp. rotation
  - Comparator: **vertical single bundle** from transtibial femoral drilling
- Horizontal **transportal drilling** → double bundle advantage questionable

Causes of Failure Ask AI

Causes Ask AI

- **Wrong tunnel position**
  - Correct example: tibia middle, femur posterior
  - Imaged at 8–9 months → excellent direction and tension → **ligamentized**
- **Missed ligament injury**
  - Excluded: MCL, LCL, posterolateral corner, PCL injury
  - Revision → focus on missed injury behind original failure
- **Weak graft**
  - E.g. hamstrings cut early in harvest → other leg or patellar tendon
  - No 6 mm graft → **< 8 mm in adults** usually fails
  - Required: diameter 8–10 mm, length 8–10 cm
- **Poor fixation**: blowout, interference screw not holding, small interference screw

  - Femoral interference screw = tunnel size (9 → 9)
  - Tibial screw 1–2 mm larger (tunnel 9 → 10 or 11) → **failure commoner at tibia** *[C4 · transcript: "9 mm tunnel: screw 10 or 15"]*
- **Rehabilitation**

Must-Know

- Graft diameter 8–10 mm, length 8–10 cm
- < 8 mm graft in adults → usually fails
- Femoral interference screw = tunnel size; tibial screw 1–2 mm larger
- Revision → exclude missed MCL, LCL, PLC, PCL injury

Imaging Examples Ask AI

- Post-op MRI, incidental
- **Femoral tunnel horizontal** → not vertical at 12 o'clock
  - 2–3 o'clock (left knee), posterior in femur
- Tibia in thirds → **tibial tunnel** in middle
- External case: **tibial tunnel very anterior**; femoral tunnel also anterior

Case: Multiple Errors Ask AI

- Original injury: **PCL**, not ACL
  - Drawer from posterior sag → misdiagnosed as ACL → ACL reconstructed
  - Tibial tunnel very anterior; femoral tunnel very anterior → **failed**
  - Revision; careful examination
- No missed ligament injury confirmed → **old graft removed**
  - Left knee, anterolateral portal; old tunnel displayed
- Old tunnel: horizontal (acceptable) but **very anterior**
  - Entire tunnel anterior to ridge, just behind articular cartilage
  - Horizontal, crossing PCL
  - Behind ridge → area untouched by any instrument
- That area → **proper femoral tunnel** site
  - Failure cause: very anterior, not vertical

Case: Two Failed Grafts Ask AI

- ACL reconstructed twice, both failed
  - Second graft posterior but **very vertical** (≈ 12 o'clock)
- **Instability** with any movement
  - Debridement → correct site intact: horizontal + posterior; untouched in both surgeries
  - First graft: horizontal but very anterior
  - Second graft: posterior but **very vertical**

Two Commonest Femoral Errors Ask AI

- **Vertical tunnel** → usually transtibial drilling
- Cartilage seen → tunnel just behind it → ridge mistaken for **over-the-top**
  - Scope not moved back to true posterior end
- Correct tunnel site → further posterior, behind ridge

Tunnel	Orientation	Position	Outcome
Correct	Horizontal	Posterior	—
First wrong	Horizontal	Anterior	Failed
Second wrong	Vertical	Posterior	Failed

Anterolateral Ligament and Lateral Extra-articular Tenodesis Ask AI

- **Anterolateral ligament**: capsular thickening
  - Origin: just posterior to lateral epicondyle
  - Insertion: proximal and anterior to Gerdy's tubercle
- ALL reconstruction with tendon graft alongside ACL → **overdoing**
- Return → **Lemaire modification** of old operation
  - ITB strip (attached to Gerdy's tubercle) → behind fibular (lateral) collateral ligament → tenodesis
  - Fixed at lateral epicondyle or just posterior and proximal to it
  - = **iliotibial band tenodesis**
- Indications:
  - **ACL revision**
  - High-grade **pivot shift** → knee shifts and drops hard
  - Professional athletes → very **debatable**
- Claim: addition → improved **rotational stability** → high-grade pivot shift, rotational instability

ACL in the Skeletally Immature Ask AI

- Detailed lecture by another colleague
- **Partial injury** → activity modification
- **Complete injury** → reconstruction required
  - Untreated → chondral + meniscal injuries → ruined knee
- Reluctance in young → **sub-optimal reconstruction**
  - Must last to full growth without re-injury
  - Later re-graft possible → not large proportion
  - Growth → graft grows in length, **not in thickness** at same rate
  - 5–6 mm graft in small child → not 10 mm at adulthood → long, thin → weak

Age	Technique	Growth plate
< 11 (F) / < 12 (M)	All-epiphyseal	Not crossed in tibia or femur
Up to 14 (F) / 15 (M)	Partial transphyseal	Crossed in tibia only
> 15	Classic (complete) transphyseal	Crossed in femur and tibia

- Partial transphyseal: tibia **central** → crossed; femur eccentric → not crossed

All-epiphyseal Technique Ask AI

- Tunnels in **epiphysis** only: femoral tunnel + short tibial tunnel, no crossing
- Alternative: **hamstrings left attached**, no tibial tunnel

  - Over anterior tibial cortex → under **intermeniscal ligament** → into knee
  - Far behind lateral femoral condyle (over-the-top point) → tenodesis
  - Back outside → fixed at **Gerdy's tubercle**
  - = ACL + anterolateral ligament equivalent
- Short tibial tunnel → verified with **image intensifier** + tibial guide
- Femur: epiphysis not crossed?
  - Reverse: **guide inside**, bullet outside → intensifier → below growth plate → drill out-to-in

Partial and Complete Transphyseal Ask AI

- Older → tibia crossed normally
- Femur still below growth plate → **partial transphyseal**
- > 15 years → femur + tibia crossed → **complete transphyseal**

ACL Deficiency with Varus Knee: High Tibial Osteotomy Ask AI

- Varus knee + torn ACL → **HTO** + ACL reconstruction

Decision Factors Ask AI

- **Thrust** present?
- Medial compartment **degeneration** starting?
  - Lifelong varus not always corrected with ACL
- **Posterior tibial slope**: higher → more ACL failure
- Aim: HTO **reduces slope** → lateral closing wedge > medial opening wedge
- Wedge effect on slope:

Wedge	Effect on slope
Lateral closing	Decreases by default
Medial opening	Increases by default → unless opened much more posteriorly than anteriorly

- More slope reduction → **better ACL results**
- **Tibial slope** = angle on lateral image; high slope → higher failure
  - Revision → **slope change** considered → lateral closing wedge
  - **PCL injury** → increased slope advantageous
  - Slope not an issue; varus with instability or thrust →
    - or medial degeneration starting → medial compartment saved → **medial opening wedge** easier + ACL

Staging Ask AI

- Preferably **single stage**
- Two stages → **osteotomy first**, never ACL first
  - Non-athletic: slope reduced → stability improved → ACL may be unnecessary
  - ACL with persisting malalignment → **progressive graft laxity**
- Alignment correction → likely **graft protection**, better stability, medial compartment unloaded

Combined HTO + ACL Technique Ask AI

- Harvest incision = **osteotomy incision** (little slope change needed)
- Sequence:
  - **Short tibial tunnel**: tibial guide at 50 instead of 55 → higher → osteotomy entirely distal to tunnel
  - ACL completed except final **tibial interference screw** tightening
  - Osteotomy distal to tibial tunnel
  - Osteotomy fixed → graft tensioned → **tibial interference screw**
- Recap: harvest → anatomic tunnels (tibial higher) → femoral fixation (interference or suspensory) → pulled through tibia, unfixed
- Osteotomy distal to tibial tunnel → aimed at **tip of fibular head** → stop ≈ 1 cm before lateral cortex
  - Fixation: **locked plates** better than Puddu plates; Puddu quicker
  - Then graft pulled
  - Tip: object in tibial tunnel before **proximal screws** → drill contact felt → redirected
  - Then: ACL tensioned → cycling → pull in full extension or 20° flexion
  - Tightening with tibial interference screw → **brace**

Rehabilitation Ask AI

Goals Ask AI

- At Dr. Osama's request
- Goals: decreased pain and swelling; good ROM; **muscle strength**
- Old ≈ 1-year return-to-sport programs abandoned → **4–6 months**

Range of Motion Ask AI

- Started **early**; first priority → full extension
- Worst outcome: **fixed flexion** deformity
  - Back of knee touching mattress
  - Flexion gradual, passive, not pushed
- Other leg extends operated leg fully; flexion < 90° at first
  - **Passive ROM** initially, even if able to lift
  - Goal: full extension + flexion short of 90°
  - **Long immobilizer** rather than hinged brace → worn constantly; off only for flexion / extension exercises → easier, safer

Weight Bearing Ask AI

- Weight bearing on reconstructed ACL allowed → **knee fully extended**
  - No bending during stepping → graft stress avoided
  - Requires **good quadriceps power** → straight leg raise
  - Taught: muscle tightened on stepping → full extension
  - In immobilizer

Return to Sport Ask AI

- Depends on sport and circumstances; usually **≥ 4 months**

Alarming Signs: Infection Ask AI

- Early post-op fine (tightening, lifting, full extension) → after 2–4 days or 1 week: **full extension lost** + no SLR
- + quadriceps weakness + **effusion** → suspect infection
  - Not "just swelling": e.g. extension lost at 3 days, or weak quadriceps + effusion at 10 days
- Management:
  - **Aspiration** → cytology
  - **Antibiotics** once bacteria shown, before culture
  - **Arthroscopic lavage**

> **Clinical Pearl**
> • Good early quadriceps → later loss of extension, weak quadriceps, effusion → suspect infection
> • Aspirate → cytology → antibiotics before culture
> • Arthroscopic lavage

Rehabilitation Phases Ask AI

- **First 2 weeks**
  - Passive knee extension: towel under heel; support from other leg or wall
  - Flexion active, leg supported, not in air
  - **SLR** in immobilizer locked in full extension (or hinged brace) → no bending during lift
  - Unable to tighten and lift → **electric stimulation**; able → no role
- **2–6 weeks**
  - Stiff, poor flexion → **patellar mobilization**
    - Patella moved medially and laterally by hand → no adhesion to femur
  - Strengthening: **closed chain** + balancing exercises
  - Flexion > 100° → **stationary bicycle**
    - Reaching 100°: healthy leg in front of operated leg → pushes it back
    - Then bicycle
- **Up to 3 months**: normal walking + balancing
- **From 3 months**: jogging (slow running) on even ground
- **4 months**: uneven ground, direction change, crosses and turns, sport-specific activities

Exercise Types Ask AI

- Joint movement: closed vs open chain
- Strengthening: isometric,
  - isotonic, **isokinetic**
- Chain comparison:

	Closed chain	Open chain
Foot	Fixed on something	Free
Joint force	Compression → more stability	Shearing

	Closed chain	Open chain
Proprioception	Stimulated	Not stimulated
Timing	Start with	Avoided early

- **Isotonic**: dumbbell, slow flexion (biceps shortening) and slow extension (biceps lengthening) → contracting both ways
- Squat with bar on shoulders:
  - Rising from squat → **concentric** contraction
  - Slow descent, contracting while lengthening → **eccentric** contraction
- **Isometric**: dumbbell held, elbow at 90° → no lengthening or shortening
  - Squat held still, neither rising nor descending
- **Isokinetic**: contraction at fixed speed
  - E.g. bicycle at constant speed; isokinetic machines

Questions and Answers Ask AI

Tunnel Blowout Ask AI

- Blowout → **interference screw impossible**
  - Causes: too posterior, poor view, ≈ 1 mm wall left → screw compression → posterior cortex broken
- Suspensory fixation → no problem → hangs on cortex
- No suspensory device → **non-absorbable sutures** (FiberWire bundle)
  - Out through femoral cortex → tied over **screw and washer**
- **New femoral tunnel**:
  - Not horizontal again → tunnels converge, or exit too anterior → useless
  - Via tibial tunnel → posterior vertical femoral tunnel
  - **Vertical posterior** >> horizontal anterior

Short Graft Ask AI

- **All-inside** → one option
- **Suspensory fixation** saves length: 1 cm in tibia sufficient vs interference screw 2 cm
- Native femoral / tibial dimensions **larger than center** reconstructed

Isometry Check Ask AI

- Graft fixed in femur, free in tibia → flex / extend → **length unchanged**
- Lengthens markedly in extension, slackens in flexion → **points wrong**
- Anatomic point → **isometric**:
  - Constant length and tension through ROM
  - Length change with flexion / extension → tunnel placement wrong

Long Graft Ask AI

- Long graft after pulling → **staples**
- Femur → different fixation + interference screw added → no problem

Summary Ask AI

- Associated injuries not missed:
  - **Ligament** and meniscal injuries
- **Anatomic reconstruction** → center of tibial and femoral footprints
- Complications handled → **blowout**, short graft
- **Supervised rehab**
  - First 2 weeks → exercises taught by surgeon; physiotherapy referral after
  - Follow-up → no "creative" therapist returning problems
  - **Program written** for them + follow-up

Anatomic ACL Points

(Locating Them Arthroscopically)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Anatomic ACL Reconstruction — Principles Ask AI

Objectives Ask AI

- Describe the **anatomic bony attachment points** of the ACL
- Locate them **arthroscopically**
- → crucial for successful ACL reconstruction

Definition Ask AI

- **Anatomic ACL reconstruction** = restoration of the ACL to its:
  - Native dimensions
  - Collagen orientation
  - Insertion sites
- Native dimensions → **not restorable**
  - Femoral + tibial attachments larger, more complex than a rounded graft covers
- Collagen orientation → **not restorable**
  - Different ACL collagen orientations → tendon graft cannot mimic
- Insertion sites → the **only restorable** element
  - Graft exactly in **center** of proper anatomic insertion sites
  - Fundamental to successful reconstruction

Must-Know

- Dimensions + collagen orientation → cannot be restored by a graft
- What remains → **center of anatomic insertion sites**

Functional Bundles of the ACL Ask AI

- Two functional bundles: **anteromedial (AM)** + posterolateral (PL)
- Naming (AM vs PL) → based on **tibial attachment**

	AM bundle	PL bundle
Orientation	More vertical	More horizontal
Femoral attachment	More posterior	A little more anterior
Main role	Main restraint against anterior translation of tibia on femur	More important in rotatory stability

Exam Pearl

- Bundle names → from **tibial attachment**
- AM → **anterior translation** restraint
- PL → **rotatory stability**

Tibial Attachment Ask AI

- Site: area **between medial and lateral tibial spines**
- **Not** on the tibial spines themselves
- Shape: broad **oval / C-shaped** area
  - 11 mm in coronal plane
  - 17 mm in sagittal plane
- Rounded graft → cannot cover → no complete **collagen fill**
- Therefore:
  - Graft in **center** of the area
  - Leave old (remnant) ACL fibers intact → proper **collagen fill**

Femoral Attachment Ask AI

Direct Insertion Ask AI

- Main component: **direct insertion**, located in a depression:
  - Just **posterior** to lateral intercondylar ridge
  - **Anterior** to posterior articular margin of condyle
- Large **rectangular** area: ~18 mm length × 8 mm width
- Rounded graft → cannot completely fill
- Graft in **center**; leave still-attached fibers, especially **posteriorly**

Key Point — Lateral Intercondylar Ridge Ask AI

- **No ACL fibers** attach anterior to lateral intercondylar ridge
- Identify the ridge → move **posterior** to it
- Whole graft lies between:
  - **Lateral intercondylar ridge**
  - **Over-the-top position** (posterior articular margin)
- Graft orientation → **horizontal**

Indirect Fibers Ask AI

- **Indirect ACL fibers** → expand from main direct fibers
- **Fan-shaped** spread → all the way to posterior articular margin of posterior condyle

High Yield

- **No ACL fibers** anterior to lateral intercondylar ridge
- Graft: between ridge and **over-the-top position**
- Femoral graft → **horizontally** oriented

Tibial Footprint — Extent Ask AI

- Inserts **between medial and lateral tibial spines**
- Anteriorly → far, almost to **anterior rim of tibia**
- Posteriorly → just to **anterior edge of posterior horn** of lateral meniscus
  - No fibers extend far enough to overlap **posterior horn of lateral meniscus**

- Fibers **more anterior** than once anticipated

Footprint Comparison Ask AI

	Tibial footprint	Femoral footprint (direct)
Location	Between medial and lateral tibial spines	Depression posterior to lateral intercondylar ridge, anterior to posterior articular margin
Shape	Oval / C-shaped	Rectangular
Size	11 mm coronal × 17 mm sagittal	~18 mm length × 8 mm width
Rounded graft	No complete collagen fill	Cannot completely fill
Strategy	Center + leave remnant fibers	Center + leave attached fibers, especially posterior

Arthroscopic Landmarks — Femoral Side Ask AI

- **Right knee**; view through standard anterolateral portal
- Landmarks:
  - **Lateral intercondylar ridge** (the arch)
  - Far posteriorly: **over-the-top position** (point) = posterior articular margin
  - **Popliteal fossa**
- Graft placement zone:
  - **Horizontal** orientation
  - **Posterior** to lateral intercondylar ridge (whole arch)
  - **Anterior** to over-the-top point
- Achieved with **transportal drilling**

Transportal Femoral Drilling Ask AI

- Route: **accessory anteromedial portal**
- Knee flexed → graft almost **perpendicular to PCL** (~90°)
- Results:
  - **Posterior wall** intact
  - **Horizontal** tunnel
- Clock face (right knee) → almost **10 o'clock** position

Arthroscopic Landmarks — Tibial Side Ask AI

- **Tibial footprint** = whole large area → far anterior, almost to anterior rim of tibia
- Do **not remove** it → graft alone cannot achieve collagen fill
- Landmark: **posterior edge of anterior horn** of lateral meniscus
- Steps (right knee, anterolateral portal view):
  - Follow posterior edge of anterior horn of lateral meniscus → **center of footprint**
  - Palpate **medial tibial spine** → stay just lateral to it
  - → place **tibial ACL guide** here

- Sequence: anterior horn posterior edge → medial tibial spine → just lateral → **tibial ACL guide**
- → exactly the target position

Clinical Pearl

- Tibial guide → line of **posterior edge of anterior horn** of lateral meniscus
- → just lateral to **medial tibial spine**
- Keep footprint fibers → **collagen fill**

Case Example: Mid-substance Femoral Avulsion Ask AI

Femoral Side Ask AI

Case

- Right knee; **mid-substance femoral attachment** tear → native attachment site visible:
  - Far **posteriorly**
  - Just posterior to **lateral intercondylar ridge**
  - **Horizontally** oriented

Tibial Side Ask AI

Case

- Identify **posterior edge of anterior horn** of lateral meniscus
- Tibial attachment → far anterior, almost to **anterior rim of tibia**
  - **More anterior** than we think
- **Tibial guide** placement:
  - Center of footprint
  - Same **coronal line** as posterior edge of anterior horn of lateral meniscus
  - Just lateral to **medial tibial spine**
- Really large attachment area → **leave fibers** → collagen fill needed

Femoral Drilling: Transportal vs Transtibial Ask AI

- Alternative to transportal (**accessory anteromedial portal**): drilling through anatomic tibial tunnel
  - = **transtibial drilling**

Transportal Drilling Ask AI

- **Horizontally** oriented tunnel
- Proper **posterior wall** → no blowout
- Graft almost 90° with **PCL**

Transtibial Test Ask AI

- Create **anatomic tibial tunnel**:
  - Posterior edge of anterior horn of lateral meniscus
  - → center of footprint, just lateral to **medial tibial spine**
- Drill femur through it → where does it lead on femoral side?

	Transportal	Transtibial
Route	Accessory anteromedial portal	Anatomic tibial tunnel

	Transportal	Transtibial
Orientation	Horizontal	More vertical
Clock face (right knee)	Almost between 9 and 10 o'clock	Almost between 11 and 12 o'clock
Posterior wall	Proper, no blowout	—
Angle with PCL	~90°	—
Rotation control	Better (horizontal graft)	—

- **Horizontally oriented grafts** → control rotation better

Pitfall

- **Transtibial drilling** → vertical tunnel, 11–12 o'clock
- Transportal → horizontal, 9–10 o'clock → better **rotation** control

Graft Passage and Final Check Ask AI

- Transportal graft (horizontal) → **pulled through**
- Good **tension** at 90° knee flexion
- PCL vs graft → proper **orientation** confirmed: 90° between them

Summary — Keep It Simple and Safe Ask AI

Femoral Side Ask AI

Summary

- Right knee: palpate **lateral intercondylar ridge** (the arch)
- Drill **posterior** to it → no part of fibers anterior to it
- **Transportal drilling** → horizontally oriented graft
- → proper **anatomic femoral point** for ACL

Tibial Side Ask AI

Summary

- Right knee, view from **anterolateral portal**
- Identify **posterior edge of anterior horn** of lateral meniscus
- Palpate **medial tibial spine** → stay just lateral
- Preserve native ACL collagen fibers → proper **collagen fill**

Take-home Message Ask AI

- Follow **anatomic femoral and tibial** ACL attachment points
- Be in **center of footprint**
- **No part of ACL** anterior to lateral intercondylar ridge of femur
- Avoid **vertically oriented grafts** → transportal, not transtibial, femoral drilling

Patellofemoral Instability

(Stabilizers, Assessment and Surgical Options)

Prof Dr Maged Samy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 160) As spoken; usual pathological TT-PCL threshold is about 24 mm, not 20 mm.

Introduction and Diagnostic Strategy Ask AI

Opening Remarks Ask AI

- Thanks → Orthopaedic Department, **Nile Health Insurance Hospital**
  - Dr. Osama's team → well-known, no longer unknown soldiers
- Thanks → sponsoring company, supporting the series since day one
- Topic rarely addressed as a lecture or subject (chairman's view)
- Lecturer → **Prof. Dr. Maged Samy**
- Topic → **patellofemoral instability**: diagnosis and management
  - With tips and tricks
- Host → Nile Hospital, headed by **Dr. Osama Abdel-Rashid**
- Chairman → **Prof. Dr. Atef El-Beltagy**, the lecturer's professor
- Topic → not very common

Why a Structured Approach Is Needed Ask AI

- **Patellofemoral instability**: not uncommon in practice, uncommon in teaching
  - Clear guidelines → only in the last few years
- Sequence → know the **patellofemoral stabilizers** → assess each one → defect or not
- Management → **every defective stabilizer** addressed
  - Some addressed, others left → failure
- **No standard operation** for every patient
  - Identify source of pathology (defective stabilizer) → address it
- Tool → **clinical and radiologic assessment** of each stabilizer
  - Some stabilizers → clinical; some → radiological
- Endpoint → **treatment algorithm** → healthy, functioning patellofemoral joint

Must-Know

- No standard operation for patellofemoral instability
- Every defective stabilizer addressed → otherwise failure

Patellofemoral Stabilizers and Kinematics Ask AI

Requirements of a Healthy Patellofemoral Joint Ask AI

- Patella **located in the trochlear groove**; no instability
- Intact **extensor mechanism**
- **No pain**
  - Severe patellofemoral pain → **quadriceps inhibition**
  - Quadriceps → essential part of patellar stabilization

Dynamic Stabilizers Ask AI

- Stabilizers → **static** and dynamic
- **Dynamic stabilizers** = muscles keeping the patella in the trochlear groove
  - **Quadriceps** → most famous, most important

  - **Hip abductors** and hip extensors → balance the pelvis → proper patellar tracking
- **Balanced pelvis** → required for proper patellar tracking
- "Strengthen your muscles" → **quadriceps**, hip abductors, hip extensors
- Instability case → assess every stabilizer
  - One by one → intact or not → only then the **management plan**

Static Stabilizers Ask AI

- **Medial retinaculum**
  - Lateral patellar dislocation → requires medial retinacular injury
- **Lateral retinaculum** → surprising inclusion
  - Not always a **deforming force**
  - Normal role → proper patellar tracking
  - **Lateral retinacular release** ≠ first, easiest step in instability
  - Non-indicated release → increased instability → **medial instability**
    - Seen only in this setting; never occurs on its own
- **Trochlear groove depth**
  - Patella seated in a **deep trochlear groove**
  - Very shallow groove → patellar instability
- **Patellar height**
  - Normal relationship with trochlea vs more proximal position
  - Proximal position = **patella alta** → role in stability
- **Patellar shape**
- **Limb alignment** → increased instability
  - **Valgus knees**
  - **Pes planovalgus** → valgus knees
  - **Exaggerated femoral anteversion**

> **Pitfall**
> - Lateral retinaculum → not routinely a deforming force
> - Non-indicated lateral release → medial instability

Patellar Kinematics with Knee Flexion Ask AI

- Imagine patellar motion during knee bending
- Knee flexion → patella moves **distally**
- Highest patellar position → **extension**
- Beyond 90° → patella **fully engaged** in trochlear groove
- **Patella alta** → very proximal starting position
  - Early flexion → patella not yet reaching the groove
  - Not engaged → **prone to dislocation** with flexion
- Why it dislocates → **no early engagement**; needs much more flexion
- Beyond 90° → patella engaged with almost any pathology
- Imaging the patellotrochlear relation at 90° → **no value**
  - Shallow trochlea or patella alta → patella still appears centred, engaged
  - **Skyline view** at 90° → no useful information

Causes of Patellar Instability Ask AI

- Causes to look for:
  - Shallow trochlea = **trochlear dysplasia**
  - **Q angle**
  - **Patella alta** → very proximal patella
  - **Lower limb deformities**

Patellar Tracking and Presentation Ask AI

- **Patellar tracking** → proper track in groove; not even subluxation, let alone dislocation
  - **Repeated subluxation** → accelerated wear → patellofemoral arthrosis
- Instability ≠ only **frank dislocation**; presentations:
  - **Pain**
  - **Subluxation** → patella moves and returns
  - **Frank lateral dislocation**
- **First-time dislocation** vs recurrent → different management
- Arthroscopic example → patella lateral to **lateral femoral condyle** (most lateral trochlea)
  - → clear **lateral subluxation**

Clinical Assessment Ask AI

Stabilizers to Keep in Mind Ask AI

- Subluxation or dislocation → consider all stabilizers:
  - Direct restraints → **medial and lateral retinacula**
  - **Patellar height** → alta or not
  - **Trochlear shape** → dysplasia or not
  - **Overall limb alignment** → valgus knees, femoral anteversion, pes planovalgus → valgus knees

Q Angle Ask AI

- **Q angle**: ASIS → patellar centre → tibial tuberosity
  - Normal → not exceeding 15°
- **Lateralized tibial tuberosity** → 20–25°
  - Patellar tendon → reflects **quadriceps pull**
  - Pull directed laterally → patella dislocates laterally
- **High Q angle** / lateralized tuberosity → lateral subluxation or dislocation tendency
- Rarely relied on in instability → patella **already subluxed laterally**
  - ASIS line reaches laterally subluxed patella → angle not increased
  - Error → patella assumed central, actually laterally displaced
- Correct technique
  - Push patella → **centred in trochlear groove** → then measure
  - Never measure with existing lateral displacement

Clinical Pearl

- Laterally subluxed patella → falsely normal Q angle
- Centre the patella in the groove before measuring

Elements of the Clinical Examination Ask AI

- **Overall limb alignment**
- **Exaggerated femoral anteversion** → in-toeing; Craig test (as heard)
- **Patellar position**
- **Q angle** → lateralized tibial tuberosity or not
- **Quadriceps wasting**
- **Hip extensor and abductor power**
- **Gait**
- Dislocation → **in extension or in flexion**

Instability in Extension vs Flexion Ask AI

- **Extension dislocation** → most common
  - More flexion → more engagement → fewer dislocations
  - Extension → patella rises, leaves groove → dislocates
  - Flexion → enters centrally; extension → exits = **J sign**
- Classic **type 1** = dislocation in extension → target of most management
- **Type 2** = dislocation in flexion → rare
  - Main treatment → bony malalignment, e.g. **malrotation of the femur**

Medial Retinaculum: Patellar Glide Ask AI

Med. retin

- Most famous stabilizers to assess → **medial and lateral retinacula**
- **Patellar glide** → medial retinaculum
  - Quadriceps relaxed → push patella medial → lateral
  - Normal → moves
    - No more than **half the patellar width** laterally
  - More than half → **deficient medial retinaculum**
- **Apprehension test** → lateral push, knee flexed → gradual extension
  - Patient apprehensive of dislocation
- Medial → lateral push = medial retinaculum test

Lateral Retinaculum: Patellar Tilt Test Ask AI

Lat. retin

- **Patellar tilt test** → patella between two fingers; lift lateral border
  - Lifted toward the ceiling → lateral retinaculum **not tight**
  - No movement → **tight lateral retinaculum**
- Case → patella fully subluxed outside **lateral femoral condyle**
  - Glide on medial border → whole patella displaced
  - → **medial retinaculum deficient**
  - Up to half → acceptable; more than half → deficiency

Summary of Clinical Assessment Ask AI

Element	Clinical assessment
Medial retinaculum	Patellar glide
Lateral retinaculum	Patellar tilt
Patellar height	Rough only
Q angle	Rough idea → tibial tuberosity lateralized or not
Trochlear dysplasia	No idea clinically
Overall limb alignment	Assessable

Plain Radiographs: Patellar Height and Trochlear Dysplasia Ask AI

What Imaging Must Show Ask AI

- Before requesting → what must I see → which images

Target	Question	Significance
Patellar tracking	Lateral displacement?	Medial retinaculum deficient
Patellar tracking	Lateral tilt?	Lateral retinaculum tight
Patellar height	Patella alta?	Clinical → rough two-knee comparison; accurate → radiology
Trochlear groove	Depth; dysplasia?	Unknown clinically
Tibial tuberosity	Lateralized?	—
Patella	Shape	—
Retinacula	Medial (= MPFL), lateral	—
Overall limb	Alignment	—

- Each point → intact or not → needs addressing or not → case-specific **algorithm**

AP View Ask AI

- **AP view** → little information
  - Patella rarely subluxed at rest; dislocates and reduces

Lateral View Ask AI

- **Lateral view** → much information in instability
  - Dislocated patella → abnormal appearance; uncommon → usually reduced
  - First value → **patellar height**
- Patella alta or not → very important; many methods, two most famous

Insall–Salvati Index Ask AI

- **Insall–Salvati index** → whole patellar length : patellar tendon length
  - Tendon length → inferior pole → tibial tuberosity
  - Normal → 1:1; maximum 1:1.2
  - Above 1.2 (e.g. 1:1.4) → long tendon → proximal patella → **patella alta**

Caton–Deschamps Index Ask AI

- **Caton–Deschamps index** → lecturer's recommendation
  - Uses **articular surface** length, not whole patella
  - Rationale:
    - Articular part → engages trochlea; functional
    - Non-articular part → no role in engagement
    - **Long-nose patella** (large inferior pole)
      - → false normal index; functional articular part short
  - Second line → articular surface end → **plateau–metaphysis junction**
  - Landmark clearly seen; tibial tuberosity → exact point uncertain
    - **Osgood–Schlatter** with small ossicles → tuberosity unidentifiable
- Postoperative check after **tibial tuberosity distalization** for alta
  - Target → 1:1, maximum 1:1.2
  - **Insall–Salvati** → unchanged; patellar and tendon lengths fixed
    - Postoperative value = preoperative despite correction
  - **Caton–Deschamps** → changes
    - → accurate confirmation of correction

	Insall–Salvati	Caton–Deschamps
Patellar measurement	Whole patellar length	Articular surface length
Distal landmark	Tibial tuberosity	Plateau–metaphysis junction
Long-nose patella	False normal	Functional length measured
Osgood–Schlatter	Tuberosity ambiguous	Landmark clear
After tuberosity distalization	Unchanged	Changes → confirms correction

High Yield
- Caton–Deschamps preferred → articular length, clear landmark
- Insall–Salvati unchanged after tuberosity distalization

Trochlear Dysplasia: Lateral X-Ray Lines Ask AI

Trochlea

- Normal trochlear groove lines:
  - Black line → **floor of the trochlear groove**
  - Red line → highest point, **medial trochlear facet**
  - Green line → highest point, **lateral trochlear facet**
- Normal order → floor most evident → medial facet → lateral facet
- Lateral X-ray:
  - Groove floor = **Blumensaat's line**

  - More anterior → medial facet line
  - Most anterior → lateral facet line
  - Medial and lateral lines meet high at the **shaft**, not at condyles
- Shallow trochlea → floor ≈ level of both facets
  - Blumensaat's line meets both facet lines → single low point
  - = **crossing sign**
- **Severe trochlear dysplasia** → common confusion
  - Misconception → dysplasia = **small lateral facet** acting as buttress
  - Fact → key is **trochlear groove depth**
  - Lateral facet → normal size
  - Hypoplastic facet → **medial**, not lateral
  - Concave → **convex** trochlea; lateral facet still anterior to medial
  - Medial facet lower; floor ≈ lateral facet level → floor anterior to medial facet
    - Blumensaat's line crosses red line → lies anterior to it
  - Two lines instead of one = **double contour sign**

Radiographic Examples Ask AI

Lateral X-ray sign	Meaning
Crossing sign → lines meet at condyles, not shaft	Mild dysplasia → no dysplasia-specific treatment
Crossing sign + supratrochlear spur	Increasing dysplasia
Double contour sign → Blumensaat's line anterior to medial facet line	Severe dysplasia

- Lateral view → **patellar height** + trochlear dysplasia
- **Trochlear depth** → measurable on lateral view; calculations not covered
- Recap → non-articular part counted in **Insall–Salvati**
  - Long nose / long inferior pole → apparently normal height; functional alta

Exam Pearl

- Severe dysplasia → hypoplastic medial facet, not lateral
- Crossing sign → mild; double contour → severe

Axial Imaging Ask AI

Purpose and Knee Position Ask AI

- **Axial images** → very important
  - Trochlear groove shape and depth
  - Patellar position in the groove:
    - Whole patella shifts laterally → **lateral patellar displacement** → medial retinaculum deficient
    - Lateral border inclined downward → **patellar tilt** → lateral retinaculum tight
- **90° flexion** → patella engaged
  - Even with alta, even with instability → wrong position for assessment
- Aim → **maximal extension**; more flexion → more misleading

- Full extension → not feasible on X-ray; feasible on **CT**
- X-ray → **30° flexion**
  - Patient position and beam direction as shown
  - Request → axial X-ray in 30° flexion
  - **Skyline view** → avoid
- 30° axial view = **Merchant view**

	Merchant view	Skyline view
Knee flexion	30°	90°
Position / beam	As shown	Prone; beam at 45°
Value	Useful	Very misleading → everything looks fine

Pitfall

- Never request a skyline view in patellar instability
- Request axial X-ray in 30° flexion (Merchant view)

Sulcus Angle Ask AI

- First measurement on 30° view → **sulcus angle** = trochlear depth
  - Normal → about 140°
  - Higher → toward straight line (180°) → **shallow trochlea**

Congruence Angle Ask AI

- **Congruence angle**
  - Sulcus angle (e.g. 140°) → bisector (70°)
  - Second line from same point → **lowest patellar point** (medial–lateral facet junction)
  - Normal → patellar line **medial** to bisector
  - Patellar line lateral → whole patella lateral → **lateral patellar displacement**
    - = translational shift → **medial retinaculum deficient**

Patellar Tilt Angle Ask AI

- Lines → lateral patellar facet and **lateral trochlear facet**
  - Lines parallel → **tilt** of lateral border
  - Tilt → **lateral retinaculum tight**; not medial deficiency

Axial finding	Measurement	Structure
Shallow trochlea	Sulcus angle above 140°	Trochlear dysplasia
Lateral displacement (shift)	Patellar line lateral to bisector (congruence angle)	Medial retinaculum deficient
Lateral tilt	Lateral facet parallel to lateral trochlear facet	Lateral retinaculum tight

Other Axial Findings Ask AI

- **Patellar shape** → predominantly lateral facet → usually dislocates
  - Possible treatment → **patellar osteotomy** → medial + lateral facets
- **Avulsion** → medial patellar border, MPFL attachment
- **Loose body**
  - Sources → **medial patellar facet** or lateral femoral condyle
    - One strikes the other as the patella reduces
  - Medial facet defect → usually clearly visible
  - Source not visible → usually **lateral femoral condyle**

CT Assessment Ask AI

Advantages of CT Ask AI

- **Axial image in full extension**
  - No knee bending → no false engagement
- **Fixed reference line** → posterior femoral condylar axis
- **Overall limb alignment**
  - **CT scanogram** → knee valgus angle
  - Valgus knee deformity → predisposes to instability
- Other predisposing alignment factors:
  - **Exaggerated femoral anteversion** → posterior femoral condylar line vs head–neck line; superimposed images
  - **Exaggerated external tibial torsion** → posterior tibial condylar line vs bimalleolar line
- **Trochlear depth** and dysplasia on axial view:
  - Concave but shallow
  - Completely flat
  - **Convex** → severe
  - Convex with a **cliff** (descending step) → hypoplastic medial femoral condylar facet
- → **severe trochlear dysplasia**; grading easy on CT

Roman Arch: Choosing the Correct Cut Ask AI CT/MRI

- CT and MRI → trochlear depth judged only on the **Roman arch** cut
  - Cut must be distal enough; proximal cut → no groove visible
  - Roman arch = **intercondylar notch** appearance at trochlear level
  - No Roman arch → cut too high → cannot judge
- Roman arch cut + very small patellar portion → patella above groove → **indirect patella alta**
- **Patellar tilt** on CT
  - Transverse patellar axis vs posterior femoral condylar line
  - Effect of **quadriceps contraction** visible

Bisect Offset Ask AI

- **Bisect offset**
  - Deepest trochlear groove point → perpendicular from posterior condylar axis
  - Proportion of patella lateral to this line
  - Above 60% → **lateral patellar displacement** → medial retinaculum deficient

TT-TG Distance Ask AI

- Very important
- **Lateralized tibial tuberosity** → quadriceps pull → patella lateral → deforming force → dislocation
- **Q angle** → rough only; misleading with lateral displacement
- Definition → transverse distance: groove floor → **most anterior tibial tuberosity**
- CT technique:
  - Axial cuts → trochlear groove; lower → tibial tuberosity
  - **Superimpose** both → measure transverse distance
  - = **tibial tuberosity–trochlear groove (TT-TG) distance**
  - Adults → normal below 15 mm; above 20 mm → **abnormal** → lateralized tuberosity → deforming force
- **Dynamic CT** → flexion–extension in scanner → displacement at every degree
  - Determines dislocation **in extension or in flexion**

MRI and Refined Tuberosity Measures Ask AI

Role of MRI Ask AI

- **Soft tissues** → torn MPFL
  - Broken piece of **medial facet** → cartilage well seen
- **Patellar height** → patellar or articular length; patellar tendon length
- **Bisect offset** → posterior condylar line; perpendicular at deepest groove point
  - Proportion of patella lateral → displacement or not
- **Trochlear dysplasia** → axial cuts, as with CT
- **TT-TG distance** → two points; superimposed images, as with CT
- **MRI values differ** → same patient: pathological above 20 mm on CT = above 15 mm on MRI
- Relevant groove depth → **cartilage**, not bone (functional surface)
  - CT → bone; MRI → cartilage → more accurate

TT-TG distance	CT	MRI
Normal (adults)	Below 15 mm	—
Pathological	Above 20 mm	Above 15 mm
Groove reference	Bone	Cartilage → more accurate

TT-TG Index Ask AI

- Fixed values (20 mm) → ignore patient height (1.5 m vs 2 m)
- **TT-TG index** → relates TT-TG to patient size
  - Numerator → transverse tuberosity–groove distance
  - Denominator → vertical distance: highest groove point → most proximal tibial tuberosity
    - Reflects patient height
  - Above 0.23 → **exaggerated**

TT-PCL Distance Ask AI

- Problem → **severe dysplasia**: convex or flat trochlea → deepest point unidentifiable
- Alternative → **tibial tuberosity–PCL (TT-PCL) distance**

  - Point 1 → **PCL attachment** on tibial cut
  - Point 2 → most prominent tibial tuberosity
  - Transverse distance measured
  - Both points on tibia → **femoral rotation** irrelevant
  - Above 20 mm → pathologically lateralized tuberosity *[Q1 · As spoken; usual pathological TT-PCL threshold is about 24 mm, not 20 mm.]*

Measure	Reference points	Pathological	Advantage
TT-TG distance	Groove floor ↔ tuberosity	CT above 20 mm; MRI above 15 mm	—
TT-TG index	TT-TG ÷ vertical groove–tuberosity distance	Above 0.23	Adjusts for patient size
TT-PCL distance	PCL attachment ↔ tuberosity	Above 20 mm	Usable in severe dysplasia; femoral rotation irrelevant

Patellofemoral Engagement Ask AI

- **Patellofemoral engagement** → axial: lateral displacement; sagittal: patella alta

MRI vs CT in Children Ask AI

- **Birmingham** group → MRI rather than CT, especially young children
  - **CT radiation exposure** → very high → undesirable in children
- MRI covers → **patellar height**
  - **MPFL**
  - Trochlear dysplasia
  - TT-TG distance
  - → CT no longer needed

Imaging Protocol Ask AI

- Clear list → what to assess **clinically** and radiologically
- Protocol → **X-ray + CT** or X-ray + MRI
- Not all imaging types → know what each adds

Treatment Principles Ask AI

Building the Treatment Algorithm Ask AI

- Algorithm → not derived from a single picture (slide chosen deliberately)
- Reasoning:
  - All causative **pathologies** known?
  - Each assessable?
  - Each defect → which operation?
- Example → deficient medial retinaculum + patella alta + **trochlear dysplasia**
  - Retinaculum and alta treated, dysplasia ignored → failure
- **All pathologies** treated → otherwise high failure rate

Operation for Each Pathology Ask AI

Pathology	Operation
Severe patellar tilt + tight lateral retinaculum	Lateral patellar release or lengthening
Trochlear dysplasia	Trochleoplasty (deepening)
Patella alta	Tibial tuberosity distalization
Increased TT-TG (lateralized tuberosity)	Tibial tuberosity medialization
Malrotation (increased femoral anteversion)	Osteotomy
Deficient medial retinaculum	MPFL surgery → by default, ± one or more of the five

- **Lateral release** → very limited indications
  - Never in children
  - Mainly **type 2 instability** (dislocation in flexion) → uncommon
- Deficient medial retinaculum → **common to all**
  - Dislocating patella → medial restraint deficient → operated on by default

Must-Know
- Treat all pathologies → otherwise high failure rate
- Medial restraint deficient in every dislocating patella → addressed by default

Rehabilitation and Conservative Measures Ask AI

- Indicated → **first-time dislocators**
  - **Muscle strengthening** → quadriceps, hip abductors, hip extensors
  - **Braces**
  - **Taping**

Proximal vs Distal Realignment (Old Terminology) Ask AI

- Old division → **proximal and distal realignment** procedures
- **Key stabilizer** = medial retinaculum; lateral patella → medial restraint failed
  - Beyond dispute → then measure remaining factors
- Proximal → MPFL work, **lateral release**
- Distal → distal to inferior pole, e.g. **tibial tuberosity osteotomies**
- Old term → reference = **inferior pole of the patella**
  - Retinacula → proximal; tibial tuberosity → distal

Recap of the Assessment Ask AI

- Clinical → limb alignment, Q angle, **patellar glide** (medial), patellar tilt (lateral)
- Radiological → height, dysplasia, tilt, lateral subluxation
- CT → most important: **TT-TG** → not to be forgotten
- MRI → everything: height, dysplasia, **MPFL**, TT-TG

First-Time Dislocation Ask AI

- Rule → **conservative**; two exceptions:
  - **Bony avulsion** of medial patellar border by MPFL → reattach, fix

  - **Haemarthrosis + large loose fragment**
    - → remove or refix fragment
    - + address **deficient medial retinaculum** → medial facet struck lateral condyle → retinaculum torn
- Otherwise → **bracing** + muscle rehabilitation

Surgical Options and Lateral Release Ask AI

Indications for Surgery Ask AI

- First-time dislocator with **osteochondral fragment** or avulsed bony MPFL
- **Failed conservative** treatment → continued complaints, redislocation
- Presentation already **recurrent**

The Surgical Menu Ask AI

- **Lateral release** or lengthening
- **Medial plication** → historical; no longer done for dislocation
  - Over-stretches again with time
- **Tibial tubercle osteotomies**
  - Distalization → patella alta
  - Medialization → lateralized tibial tuberosity
- **Trochleoplasty** → trochlear dysplasia
- **Osteotomies** → limb malalignment
- **Patellar osteotomy** → abnormal shape with lateral facet only; rare
- Item done for everyone → **MPFL reconstruction**
  - Lateral dislocation → medial restraint torn by default → ± other procedures

Lateral Release Ask AI

- Complete cut or **lengthening**
- Rarely indicated; **never in children**
- Shown → **arthroscopic lateral release**
- Only indications:
  - **Lateral patellar hypercompression syndrome** → pain, no frank dislocation
  - **Type 2** (instability in flexion) + clear tilt on X-ray + positive tilt test
    - Lateral border not lifting → **tight lateral retinaculum**

Medial Plication Ask AI

- Old claim → medial plication + lateral release for all → no dislocations
- **Redislocation rate** in recurrent dislocation → close to 90%
- Technique → cut medial retinaculum, re-suture over-tight
  - Arthroscopic (over 10 years ago) → **shaving** of retinaculum → three wide sutures
  - Tied → **arthroscopic medial plication** → always overstretches
- Plication + release → possible for **subluxation**; never for frank recurrent dislocation

Pitfall

- Medial plication + lateral release in recurrent dislocation → ~90% redislocation
- Never for frank recurrent dislocation

MPFL Reconstruction Ask AI

Anatomy of the MPFL Ask AI

MPFL

- Rule → **MPFL reconstruction**
- Patellar attachment → **medial border of the patella**
- Femoral attachment → midway, **adductor tubercle** ↔ medial epicondyle
- **Adductor tubercle** → proximal and posterior to medial epicondyle

Reconstruction Technique Ask AI

MPFL

- **Tendon graft** → medial patellar border + femoral midpoint
- Layer → **deep to medial retinaculum**, superficial to knee synovium
- All **recurrent patellar dislocation** → adults and children
  - Skeletally mature or immature
- **Gold standard** → alone or combined
- Grafts → **hamstrings** (free); quadriceps tendon
  - Quadriceps → left attached to patella → under medial retinaculum → femoral midpoint

Schöttle Point Ask AI

- **Schöttle point** → very critical in young children → avoid physeal injury
  - Line along **posterior femoral cortex**
  - Two perpendiculars → most posterior **Blumensaat's line**; highest point of posterior condyle
  - Point → just anterior to posterior cortex, between the two lines
  - Graft **tunnel** here → physis-safe, anatomic

Indications and Intraoperative Confirmation Ask AI

MPFL

- All **recurrent dislocation**, mature or immature, with deficient medial retinaculum
- On the table → **patellar glide** confirms deficiency
  - Patella not coming out → suspect **diagnostic error**
  - Increased glide → deficient → MPFL reconstruction
- Graft fixed at **anatomic** patellar and femoral points
  - Patella → **medial proximal border**
  - Femur → midway, medial epicondyle–adductor tubercle

Graft Choices Ask AI

MPFL

Graft	Attachment kept	Fixation needed	Notes
Hamstrings	None (free graft)	Patella + femur	Disadvantage → double fixation
Medial third of quadriceps tendon	Patella; cut proximally	Femur only	Already attached to patella
Adductor magnus tendon	Femur at adductor tubercle; cut proximally	Patella only	Lecturer's current favourite, especially young children

- **Adductor magnus** → adductor tubercle near anatomic point
  - No drilling near **femoral physis** → much concern removed

- Graft passage → **anatomical layer**: under medial retinaculum, superficial to synovium
- Quadriceps tendon → fixed at **Schöttle point**, 30° knee flexion

Intraoperative and Radiographic Illustration Ask AI

- **Medial epicondyle** → adductor tubercle proximal and posterior → fix exactly midway
- Image → posterior cortex, posterior Blumensaat's line, highest condylar point → **Schöttle point** between
- **Quadriceps tendon** → medial third harvested
- Whip-stitched → turned toward medial femur
  - Passed under **medial retinaculum**
- Pulled out → fixed midway epicondyle–**adductor tubercle**
  - In 30° flexion
- **Free hamstring graft** → three incisions:
  - Hamstring harvest
  - Femoral fixation point
  - Patella
  - Graft passed subcutaneously between → **plane** under medial retinaculum, superficial to synovium
- Patient at one year → **patellar drill holes**
  - Femoral fixation, patellar fixation, reconstructed **MPFL** appearance

Combined MPFL and MPTL Reconstruction Ask AI

- MPFL + **medial patellotibial ligament** reconstruction
  - Hamstring left attached distally
  - Brought up along patella → exits patella → fixed in femur

Tibial Tuberosity Osteotomy and Trochleoplasty Ask AI

Medialization of the Tibial Tuberosity Ask AI

TT

- Lateralized tuberosity → **medialization**
  - Threshold → above 20 mm on CT; above 15 mm on MRI
- Technique:
  - Any tuberosity osteotomy → **long** → heals
  - Left attached distally; proximal part rotated medially → **2 screws**
- Case shown

Case: Trochlear Dysplasia and Patella Alta Ask AI

- **Recurrent patellar dislocation**; trochlea looks shallow
  - Cut too high → cannot judge → use **Roman arch** cut
  - Roman arch cut → shallow, not severe dysplasia appearance
    - e.g. **Dejour type B**
    - Higher cut → misleading; too proximal
  - Roman arch visible, patella absent → patella **proximal** to groove level
  - Patella : tendon → 1:1.5
    - → **patella alta**
- Treatment → **tibial tuberosity osteotomy** with distalization
  - Amount → restore ratio to maximum 1:1.2, not more

Trochleoplasty Ask AI

- **Severe trochlear dysplasia** → trochleoplasty
  - Not elevation of lateral facet (old practice); lateral facet not flat
  - Aim → **deepen the groove**: evacuate cancellous bone, press floor down
  - Convex → concave

Management Algorithm Ask AI

The Skeletally Immature Patient Ask AI

Immature

- **Open physes** → no bony operations:
  - No tibial tuberosity distalization
  - No tibial tuberosity medialization
  - No trochleoplasty
- Lateral release usually avoided → remaining: **MPFL reconstruction**
  - **Physeal-sparing** → femoral physis protected
- Procedures with poor results:
  - Patellar tendon **plication** for alta → very poor range of motion
  - **Roux–Goldthwait** → part of patellar tendon (not tuberosity) transferred medially → poor long-term results
- Main operation in children → MPFL reconstruction
- **Approaching skeletal maturity** + severe dysplasia or very high alta
  - → **brace**, wait → all procedures together later
  - Isolated MPFL → fails → redislocation → reconstruction lost → redo with bony operation
  - Age 9–10 → waiting likely not feasible (as heard)

Osteotomy for Malalignment Ask AI

- **Type 2** patient → standing → squinting of the patella
  - Cause → **femoral malrotation** → correction = treatment
  - Not the rule

Summary of Management Ask AI

- **First-time dislocation** → conservative
  - Unless bony patellar avulsion or osteochondral fragment in joint
- **Recurrent dislocation** → fixed operation
  - **MPFL reconstruction**
  - Very rare exception → type 2, flexion dislocation, clear malalignment, glide negative on table → medial retinaculum intact
- Rule → **MPFL reconstruction plus** → osseous anatomy check:
  - Trochlear dysplasia?
  - Patella alta?
  - Lateralized tibial tuberosity?
- None → check excessive **patellar tilt**

Finding (all with MPFL reconstruction)	Addition
Normal osseous anatomy, no excessive tilt	MPFL alone

Finding (all with MPFL reconstruction)	Addition
Normal osseous anatomy, significant tilt, adult (not child)	+ lateral release
Trochlear dysplasia	+ trochleoplasty
Increased TT-TG, normal patellar height	+ medialization
Increased TT-TG + patella alta	+ medialization and distalization
Patella alta, tuberosity not lateralized	+ distalization only

Take-Home Messages Ask AI

High Yield

- Proper preoperative clinical + radiologic assessment in recurrent dislocation
- All causative pathologies addressed when possible → not always in children
- MPFL = cornerstone, alone or combined
- MPFL in all age groups; exception → rare type 2
- Lateral release or lengthening → rarely indicated; overdone

Discussion: Technique and Controversies Ask AI

Opening of the Discussion Ask AI

- Chairman → praise; questions via Q&A and Facebook
- **Dr. Roshdy Abu Odeh** → praise: nothing left out
- Request → simple explanation of **trochleoplasty**

Trochleoplasty Technique Ask AI

- Not to be performed lightly → aim: **deepen the trochlea**
  - **Lateral incision**
  - **Burr** → curette bone under cartilage
    - → cavity under groove cartilage
  - Manual pressure → convex becomes **concave**
  - Cartilage must remain uninjured
- **Mini-open** and arthroscopic techniques → technically demanding
- Adult + **severe trochlear dysplasia** → failure without it

Femoral Graft Fixation Ask AI

- Question → **interference screw**?
- Lecturer → **interference screws**, 5 mm

Fixation method	Strength
Medial border curetted + anchors; graft sutured onto anchors	Weaker than tunnel

Fixation method	Strength
Two tunnels; suture passed, tied transosseously on lateral border	—
Bone tunnels + interference or biotenodesis screws	Strongest (described method)

- **Biotenodesis screw** vs interference screw
  - Interference → graft pulled out the other side to stay in tunnel
  - Biotenodesis → carries graft into tunnel on insertion

Postoperative Stiffness Ask AI

- Question → fear of **postoperative stiffness**
- Should not occur; two causes:
  - **Over-tightening** → a quarter of the patella must still move
    - Patella not moving laterally at all → wrong
  - **Non-anatomic fixation** → not midway
    - Medial epicondyle–adductor tubercle missed → too tight
- **Over-tight medial plication** → stiffness + high patellofemoral wear → arthrosis

Medial and Multidirectional Instability Ask AI

- Chairman → "deceptively simple", enjoyable lecture
- Question → **medial or multidirectional** instability?
- Medial instability → almost always **iatrogenic**
  - From non-indicated excessive lateral release
- Usually no tilt, no lateral tightness → routine release → iatrogenic **medial instability**
- Lateral release indications limited → clear **patellar tilt** + tight lateral retinaculum
- **Multidirectional instability** (dislocation in all directions) → no

Chairman's Emphasis Points Ask AI

- **MPFL reconstruction** → key cornerstone of treatment
- No role for **isolated lateral retinacular release** in instability
  - Done too easily; now even arthroscopically

Must-Know

- MPFL reconstruction = cornerstone of treatment
- No isolated lateral retinacular release for instability
- Medial instability → almost always iatrogenic

Tibial Tuberosity Medialization: Two Schools Ask AI

- Chairman's point of difference
  - **Medialization** → almost abandoned by current trends? Problems or narrower indication?
- **American school** → medialization for increased TT-TG
- **European school** → much less often
- **Distalization** for patella alta → beyond dispute
- Classic teaching → **pathology-specific treatment** retained

Questions and Answers Ask AI

Yamamoto Procedure Ask AI

- **Yamamoto procedure** → lecturer not aware

Muscle Imbalance and Imaging Choice Ask AI

- **Muscle imbalance** → important
- Imaging → **either CT or MRI**, not both
  - Children → avoid CT radiation; MRI provides everything
- **Weak hip abductors** → valgus attitude at knee
- Balance → hip abductors and extensors ↔ **quadriceps**

Weight Bearing and Range of Motion Ask AI

- **Full weight bearing** → immediately after surgery
- Full range → usually 6–8 weeks; depends on **type of repair**

Neglected Patellar Tendon Injury Ask AI

- Case → 15-year-old male; over 1-year-old **tibial tuberosity avulsion** / patellar ligament injury
  - Treated as patellar fracture → above-knee cast
  - Reinsertion impossible → **atrophied, shortened quadriceps**
  - **Quadriceps lengthening** done; flexion limited
- = **neglected patellar tendon** injury
- Avoid **V-Y lengthening** → marked quadriceps weakening
- Prefer **coronal splitting** of quadriceps tendon
  - Three layers → rectus, intermedius, vasti
  - **Z-lengthening** in coronal plane
- Chairman → **semi-T graft** for atrophied patellar tendon
  - Agreed → **patellar tendon reconstruction** with semitendinosus

Postoperative Plan (Questioner's Protocol) Ask AI

- **Knee brace** with open patella hole → patella centralized
- Passive ranging → 0–90°
- **Closed-chain exercises**
- Brace discontinued → after 4 weeks
- Immediate **weight bearing** in brace

Generalized Hyperlaxity Ask AI

- Question → specific instructions in **generalized hyperlaxity**?
- Always **deficient medial retinaculum** → MPFL reconstruction, definitely
- **Lateral release** → not indicated; condemned
- Additional pathology → must be addressed
  - Frequent **patella alta** → distalization required

Medialization vs Anteromedialization Ask AI

	Medialization	Anteromedialization
Example	—	Fulkerson procedure
Osteotomy	Medial shift	Oblique → slides anteriorly as moved medially
Use	Lateralized tuberosity	Medialization needed + patellofemoral pain
Effect	Patellar tracking	Patella moved away from trochlea → reduced contact forces; like anteriorization

- **Anteromedialization** → reduces patellofemoral pain; nothing to do with patellar tracking

Adductor Magnus Tendon Ask AI

- Question → extra fixation sites needed with **adductor magnus**? → No
  - Left attached at **adductor tubercle**
  - Anatomic MPFL point → **not exactly midway**; slightly nearer adductor tubercle → accepted
  - Advantage → attached to femur; fixed only in **patella**
- Length sufficient → patellar **tunnel** about 2 cm in adults; no more than 2 cm needed

Associated Deformities: Genu Valgum Ask AI

- Dr. Mohamed → **genu valgum** corrected in same setting?
  - Was correction planned from the start?
  - **Type of instability**:
    - Type 2 (in flexion) → start with **bony malalignment** correction
    - Adult, extension type (most common), no cosmetic correction
      - → deformity irrelevant locally

Tibial Tuberosity Medialization: Technical Details Ask AI TT

- **Dr. Hisham El-Sonbaty**: amount, thickness, length, screws?

Parameter	Recommendation
Screws	Bicortical → good grip; not prominent beyond posterior cortex
Osteotomy length	8–10 cm in adults → long healing surface
Thickness	Less than 1 cm → about 1 cm; screw grip without fracture
Medialization amount	From measured lateralization, e.g. above 20 mm → target no more than 15
Distalization amount	Restore normal ratio → measure before and after

- Chairman → many questions reflect lecture quality

"Miserable Triad": Order of Management Ask AI

- **Dr. Farag** (Libya) → which managed first?
- Term unknown; if alta + **trochlear dysplasia** + MPFL → all in the same session
  - Partial treatment → failure; work lost

Sole Indication of Isolated Lateral Release Ask AI

- Chairman → lateral release widely done; lecturer allows few indications
- Sole isolated indication → **lateral patellar hypercompression syndrome**
  - Lateral pain, no dislocation
  - High **patellar tilt**
  - **Wear** beginning in lateral facet
- Instability + confirmed lateral tightness → isolated release
  - → universally **condemned**
- Combined with other procedures → no problem

Exam Pearl

- Sole indication for isolated lateral release → lateral patellar hypercompression syndrome
- Lateral pain, high tilt, lateral facet wear; no dislocation

The Questioner's Triad Ask AI

- Dr. Farag's triad → **coxa valga**, genu valgum, tibial external rotation
- Likely **type 2 instability** → correct bony deformity
  - **External tibial torsion** → minimal role; last considered
  - Significant → **exaggerated femoral anteversion** and valgus
- Combination uncommon → three rarely together
- **Miserable patient** = exaggerated femoral anteversion + exaggerated external tibial torsion

Closing Remarks Ask AI

- This week's sessions → **Monday and Thursday** (Al-Ahly–Zamalek match)
- Thanks → Dr. Atef, chairman
- Thanks → sponsor; **Dr. Maged**, busy yet generous with the lecture
- Upload → **YouTube channel**
- Dr. Roshdy → lecture for **beginners in arthroscopy** agreed
- Dr. Maged → **mock exam** lecture for Master's and MD examinations
  - His own idea; in preparation, coming soon
- Prof. Atef → everything from **Dr. Maged** valued (Dr. Roshdy's summary)
- Personal benefit from every lecture; more awaited
- Lecturer → honoured by **Prof. Dr. Atef** leading the discussion
- Closing thanks; next meeting → Monday